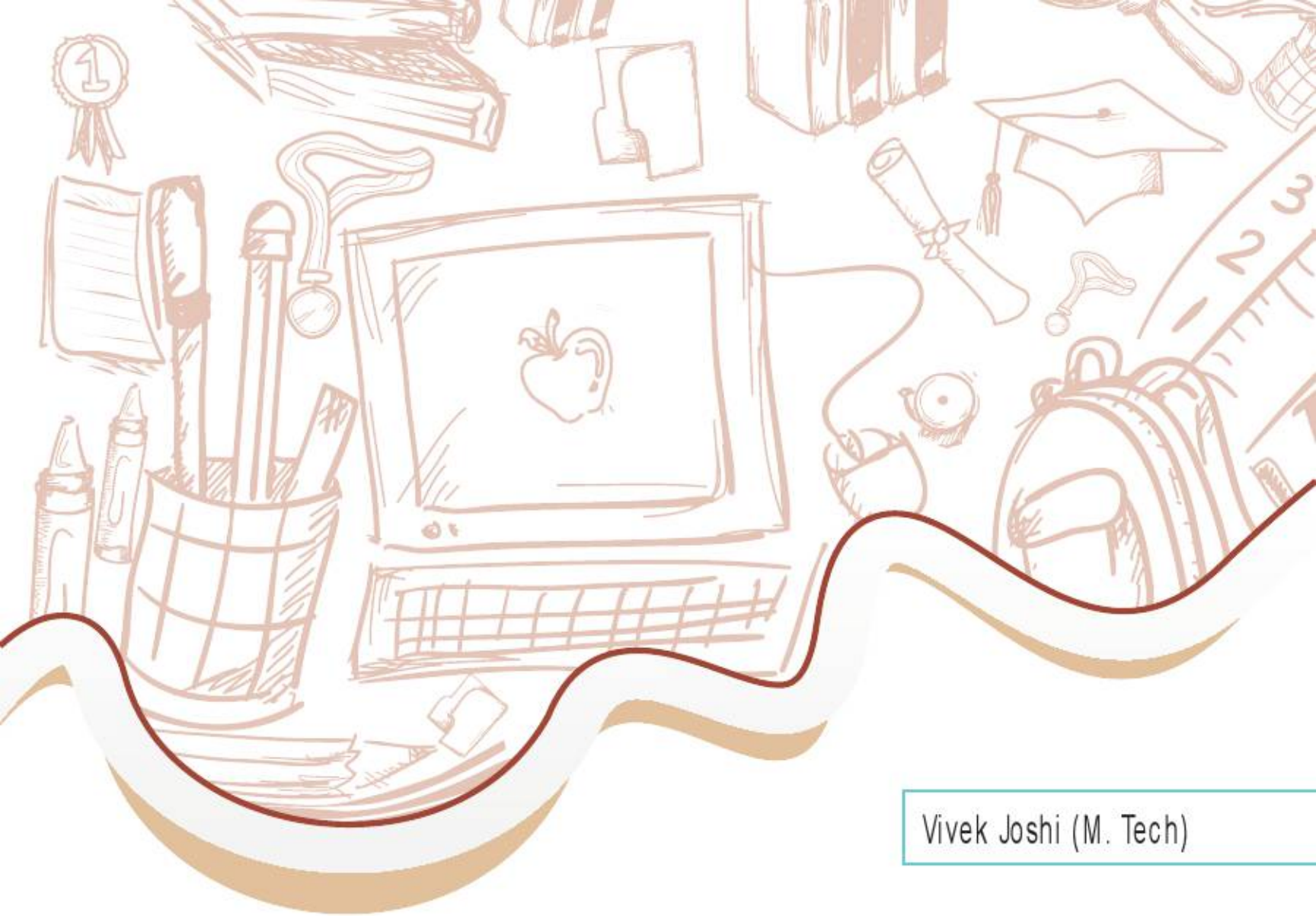
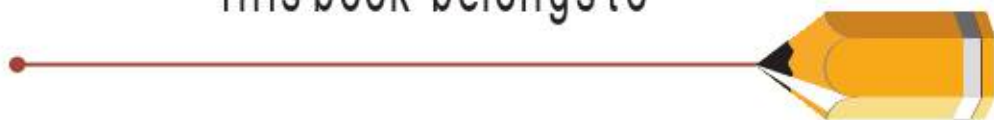




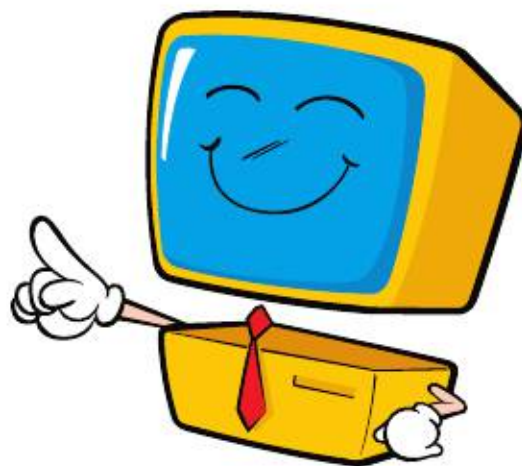
Vivek Joshi (M. Tech)

This book belongs to



Computer Interface

8



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PREFACE

In this modern age, computer plays an important role in our daily life. Computers are widely used everywhere for advancement and progress in our life. Mostly every field of our surrounding is depended on computers. So, all the schools and colleges are teaching computer as one of the main subject and preparing students to meet the challenges of the coming time.

This series **Computer Interface** of 8 book (I to VIII) written for the students studying in middle classes following **National Education Policy 2020** and covers every aspect which is related to computer according to the recent development in computer technology. This series is strictly written according to the latest syllabus based on the guidelines given by CBSE. Moreover the language used in this series is very easy to understand by the students, teachers and parents.

This series is designed to provide a complete knowledge with the latest information with lab exercises and more over.

We hope that students will enjoy learning computer with this series. Any suggestions for the improvement of these books will be highly appreciated.



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Computer Networking

Lesson at a Glance

Classification of Computer Network
Types of Network
Networking Hardware

Network Applications
Network Topologies
Transmission Media

A computer network, often simply referred to as a network, is a collection of computers and devices interconnected by communication channels that facilitate communication and allows sharing of resources and information among interconnected devices.

Classification of Computer Networks

Computer networks are classified based on various factors. They includes:

Geographical span

Inter-connectivity

Administration

Architecture

Geographical Span : Geographically a network can be seen in one of the following categories:

It may be spanned across a whole city.

It may be one network covering the whole world.

It may be spanned across a whole building, including intermediate devices to connect all floors.

Inter-Connectivity : Every Single device can be connected to every other device on network making the network mesh. Each device is connected to its left and right person only, creating linear structure.

Administration : From an administrator's point of view, a network can be private network which belongs a single autonomous system and cannot be accessed outside its physical or logical domain. A network can be public which is accessed by all.

Network Architecture : Computer networks can be divided into different types such as Client-Server, peer-to-peer or hybrid. Two systems can be connected Point-to-Point or in back-to-back fashion. They both reside at the same level and called peers.

There can be hybrid network which involves network architecture of both the above types.

Network Applications

Computer systems and peripherals are connected to form a network. They provide numerous advantages:

- ✍ Resource sharing such as printers and storage devices
- ✍ Exchange of information by means of e-Mails and FTP
- ✍ Information sharing by using Web or Internet
- ✍ Interaction with other users using dynamic web pages

The following are the important benefits of a computer network.

Facilitate communications

Using a network, people can communicate efficiently and easily via email, instant messaging, chat rooms, telephone, video telephone calls, and video conferencing. Networks allow users to communicate using e-mail, newsgroups, and video conferencing etc.

Permit sharing of files, data, and other types of information

In a network, authorized users may access data and information stored on other computers on the network. The capability of providing access to data and information on shared storage devices is an important feature of many networks. Networking of computers helps the users to share data files.

Share network and computing resources

In a networked environment, each computer on a network may access and use resources provided by devices on the network, such as printing a document on a shared network printer. Users can share devices such as printers, scanners, CD-ROM drives, hard drives etc.

Cost

Networking can reduce the cost of equipment's of an organization through sharing printers and other devices. One printer is enough for an organization if they properly manage and handle the printer activities. Printer will be installed on a system and share it for other systems to send print from other systems.

Security

It can be managed security of files and data of an organization. Network provides better services to secure our important files and data while in peer to peer network model data and files can be secured at individual level. Some users can be restricted to access the shared files / folders and it can be secured from un-authorization access of files and folders through operating systems security features.

Uses of networks

Sending e-mail, voice mail, doing research on internet, chat rooms, instant messaging.

Telecommuting, video conferencing, making a phone call.

Transferring Funds wiring money to another location.

Sharing a hardware resource such as a printer.

Sharing data , information, software.

Types of Network

Generally, networks are distinguished based on their geographical span. A network can be as small as distance between your mobile phone and its Bluetooth headphone and as large as the internet itself, covering the whole geographical world.

Personal Area Network

A Personal Area Network (PAN) is a computer network used for communication among computer and different information technological devices close to one person.

Some examples of devices that are used in a PAN are personal computers, printers, fax machines, telephones, PDAs, scanners, and even video game consoles. A PAN may include wired and wireless devices. The reach of a PAN typically extends to 10 meters.



Local Area Network

LAN is widely used to allow resources to be shared between personal computers or workstations. The resources to be shared can be hardware like a

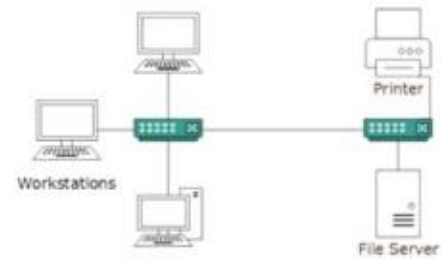
Must Know!



Piconet is Bluetooth-enabled Personal Area Network which may contain up to 8 devices connected together in a master-slave fashion.

printer or softwares or data. One of the computer in a network can become a server serving all the remaining computers called clients. Software can be stored on the server and it can be used by the remaining clients.

The term LAN can also refer just to the hardware and software that allows you to connect all the devices together.



Metropolitan Area Network

A Metropolitan Area Network (MAN) is a network that spans a whole metropolitan area. It is referred to as a high speed-network that covers a city. MANs use similar technology to LANs but cover a much wider geographic region. A consortium of users or a single network provider that sells the service to the other users usually manages a MAN.



MAN can help an organization to connect all of its offices in a city.

Must Know!



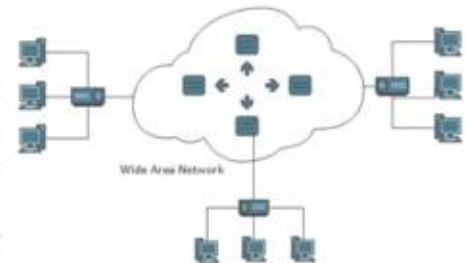
Ethernet is most widely employed LAN technology and uses Star topology.

Wide Area Network

A Wide Area Network (WAN) is a network that covers a large geographic area such as a city, country or the world. Using communications channel that combines many types of media such as telephone lines, cables, and radio waves. A WAN can be one large network or can consist of two or more LANs connected together. The Internet is the world's largest WAN.

Internetwork

Internet is widely deployed on World Wide Web services using HTML linked pages and is accessible by client software known as Web Browsers. When a user requests a page using some web browser located on some Web Server anywhere in the world, the Web Server responds with the proper HTML page. The communication delay is very low.



Home Network

A home network is a residential LAN which is used for communication between digital devices typically deployed in the home, usually a small number of personal computers and accessories, such as printers and mobile computing devices. An important function is the sharing of Internet access, often a broadband service through a cable TV or Digital Subscriber Line (DSL) provider. Many phones, printers, mp3 players, video game consoles, and Digital Video Recorders (DVRs) are equipped with wireless cards and often require very little setup to connect them to your home network.

Network Topologies

A network topology is the physical layout of computers, cables, and other components on a network. There are a number of different network topologies, and a network may be built using multiple topologies. The different types of network layouts are Bus topology, Star topology, Mesh topology, Ring topology, Hybrid topology and Wireless topology.



Must Know!



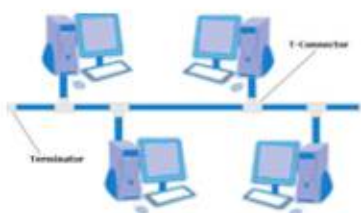
Point-to-point networks contains exactly two hosts such as computer, switches or routers, servers connected back to back using a single piece of cable.

Bus Topology

In this topology at any instant only one computer acts as master and it is allowed to transmit (broadcast). The others are supposed to listen. If two or more machines want to transmit simultaneously then an arbitration mechanism has to be used for resolving the conflict. It is possible to have a centralized or distributed type arbitration mechanism.

Star Topology

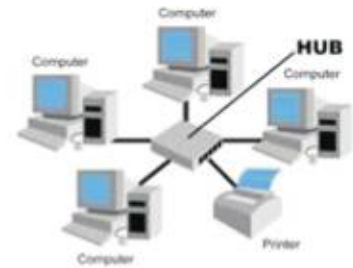
A star network topology, in its simplest form, consists of one central, or hub computer, which acts as a router to transmit messages. All devices are connected to the central computer (hub). All data passes through the hub. If a device fails, there is no effect on the network, only if the hub fails will the network is affected.



Ring Topology

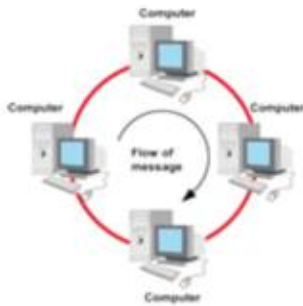
A ring network is a topology where each device is

connected to two others, so as to create a ring or closed loop. Data transmitted on a ring network travels in one direction on the ring from device to device until it reaches its destination. If a device fails, devices before it are not affected, it is the devices after it that are affected.



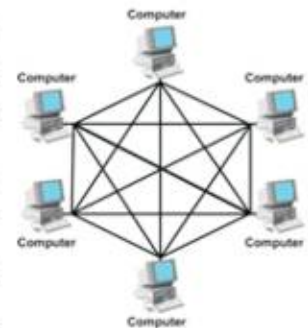
Mesh Topology

Mesh Topology is not commonly used these days. In Mesh topology, every network device is connected to other network devices. Mesh topology is costly because of the extra cables needed and it is very complex and difficult to manage. Devices are connected with many redundant interconnections between network nodes. In a true mesh topology every node has a connection to every other node in the network. Each node is connected to an arbitrary number of neighbors in such a way that there is at least one transversal from any node to any other.



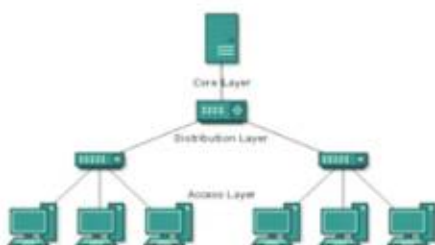
Tree Topology

This topology divides the network in to multiple levels/layers of network. Mainly in LANs, a network is bifurcated into three types of network devices. The lowermost is access-layer where computers are attached. The middle layer is known as distribution layer, which works as mediator between upper layer and lower layer. The highest layer is known as core layer, and is central point of the network, i.e. root of the tree from which all nodes fork.



Networking Hardware

Networking hardware includes all computers, peripherals, interface cards and other equipment needed to perform data-processing and communications within the network. Read the terms below to learn more about those pieces of networking hardware.



Nodes : An active device attached to a computer network or other telecommunications network, such as a computer or a switch, or a point in a network topology at which lines intersect or branch.

Server : A server is a computer program running to

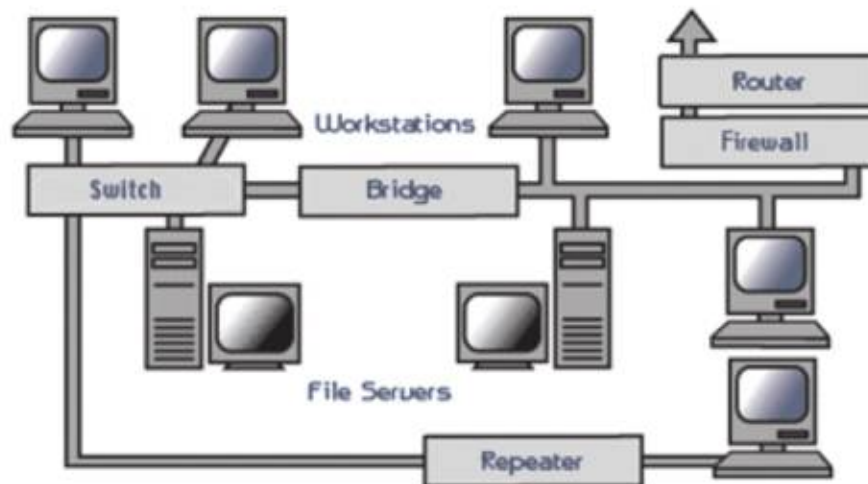
Must Know!



A network structure whose design contains more than one topology is said to be hybrid topology.

serve the requests of other programs, the "clients". Thus, the "server" performs some computational task on behalf of "clients". The clients either run on the same computer or connect through the network.

In some small networks, every machine is equally likely to have a resource that another machine needs to use.



Workstations : Computers that humans use are broadly categorized as workstations. A typical workstation is a computer that is configured with a network interface card, networking software, and the appropriate cables. Workstations do not necessarily need large storage hard drives, because files can be saved on the file server. Almost any computer can serve as a network workstation.

Laptops/Mobile Devices : Laptops and other mobile devices are becoming more and more common. These devices typically have modest internal storage, but enough power to serve as a workstation for users on the go. These machines nearly always have a wireless adapter to allow quick network connections without cumbersome cabling. In a school environment with good wireless coverage, a mobile device user can move about the campus freely, and remain continuously connected to the network.

Network interface controller (NIC)

Expansion card that is inserted into the computer to connect it to a network. Also call LAN adapter. Most are designed for a particular type of network,

protocol and media, although some can serve multiple networks.

A network interface controller (also known as a network interface card, network adapter, LAN adapter and by similar terms) is a computer hardware component that connects a computer to a computer network.

Modem

Short for modulator-demodulator. This device enables a computer to transmit data over telephone or cable lines. It converts digital signals to analog signals and vice versa. (modulation/de-modulation). Computer information is stored digitally whereas information transmitted over telephone lines is transmitted in the form of analog waves. The device converts between the two forms. These devices can be internal or external. The internal ones come as an expansion board that you can insert into a vacant expansion slot.



Must Know!



In addition to telephone modems, radio has now become very popular as a means of connecting to the internet. The device that allows you to do this is called the Wi-Fi modem.



Switch

A device that filters and forwards packets between LAN segments. (A packet is a piece of a message transmitted over a packet switching network. The packet contains the destination address in addition to the data). Most switches are active, that is they electrically amplify the signal as it moves from one device to another. Switches no longer broadcast network packets as hubs did in the past, they memorize addressing of computers and send the information to the correct location directly.

Hub

Hub is a device that provides a central point for cables in a network. It usually has ports for 8 12 devices. It is a common connection point for devices in a network. They are commonly used to connect segments of a LAN. When a packet arrives at a port, it is copied to the other ports so that all segments of the LAN can see all packets.

Router

A device that connects multiple networks, routes traffic to



appropriate network using the fastest available path. It forwards data packets along networks. It is connected to at least 2 networks. They are located at gateways, the places where 2 or more networks connect. A router translates information from one network to another; it is similar to a super intelligent bridge.

Filters

Not all data packets are equal. Some are more equal than others. If your network is to be kept secure it is often essential that some filtering takes place. For example, some staff wish to work from home with their laptops and they need to access files from within the company network.



In this case a filter would be set up that accepts data packets coming from that particular laptop. Other filtering rules would block unwanted packets trying to come in.

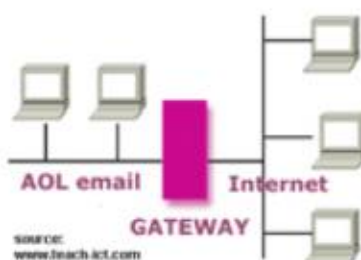
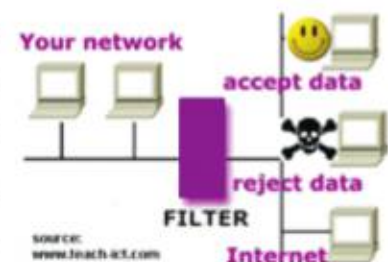
Gateways



A combination of hardware and software that connects networks that use different protocols. A node on a network that serves as an entrance to another network. An earlier term for router. The gateway node often acts as a proxy server and firewall.

Repeaters

A device that accepts a signal from a medium, amplifies it, retransmits it. Solves the problem of attenuation (weakening of a signal due to distance). It regenerates or replicates the signal. It can also relay messages between sub-networks that use different protocols or cable types. Repeaters can be separate devices or they can be incorporated into a hub. They are used when the total length of your network cable exceeds the standards set for the type of cable being used.



Bridge

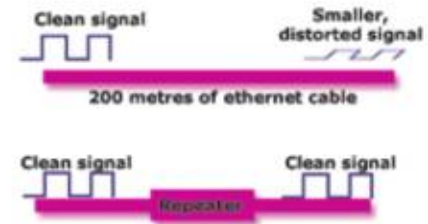
A device that connects 2 LANs of the same protocol. A bridge is a device that allows you to segment a large network into two smaller, more efficient networks. If you are adding to an older wiring scheme and want the new network to be up-to-date, a bridge can connect the two. A

bridge monitors the information traffic on both sides of the network so that it can pass packets of information to the correct location.

Must Know!



Bridges can be used to connect different types of cabling, or physical topologies.



Firewalls

A firewall is a networking device that is installed at the entrance to a LAN when connecting a networks together, particularly when connecting a private network to a public network, such as the internet. The firewall uses rules to filter traffic into and out of the private network, to protect the private network users and data from malicious hackers.

Transmission Media

The media over which the information between two computer systems is sent, called transmission media. Transmission media comes in two forms. Guided Media and Unguided Media.

Guided Media : All communication wires/cables are guided media, such as UTP, coaxial cables, and fiber Optics. In this media, the sender and receiver are directly connected and the information is send (guided) through it.

Twisted Pair Cable

A thin-diameter copper wire commonly used for telephone and network cabling. Generally for analog transmission.

It is easy to tap into and a low frequency transmission medium.

It is small in size and easy to install and covers limited distance, usually less than 100 meters.

It is the most popular medium for LANs and is generally the best option for school networks.

There are two types of twisted pair cables: Shielded Twisted Pair (STP) Cable And Unshielded Twisted Pair (UTP) Cable.

Coaxial Cable

A cable consisting of a conducting outer metal tube enclosing and insulated from a central conducting core, used for high-frequency transmission of telephone, telegraph, and television signals. It is common in cable television

applications. There are two types of coaxial cable: Thin coaxial cable and Thick coaxial cable.



Fiber Optics

A thin glass strand designed for light transmission. Optical fibers transmit data at a faster rate. They are also able to carry more signals. They are more secure, because taps in the line can be detected. Smaller size (thinner, lighter). Fiber optic cable is very expensive. There are two types of fibre optic cable: Single Mode Fibre (SMF) and Multi Mode Fibre (MMF).

Unguided Media : Wireless or open air space is said to be unguided media, because there is no connectivity between the sender and receiver. Information is spread over the air, and anyone including the actual recipient may collect the information.

Wireless media transmit data through high frequency radio signals or infrared light beams. The medium cannot be seen with the naked eye. Wireless media is used when it is inconvenient, impractical or impossible to install physical cables.

Microwave



- ✍ High frequency radio waves that provide high-speed transmission.
- ✍ Signals are sent from one microwave station to another.

- ✍ The microwave must be transmitted in a straight line with no obstructions between antennas.
- ✍ This means that microwave antennae are placed in high places (tall building, hill etc.)



Broadcast radio

- ✍ Distributes radio signals through the air over long distances.
- ✍ It is used with AM, FM radio, television stations, CB (citizens band) radio (i.e. walkie-talkie).
- ✍ Bluetooth, HomeRF and 802.11 technologies use broadcast radio signals.
- ✍ Speeds range from 1 Mbps to 54 Mbps depending on the technology.

Cellular radio

Form of broadcast radio used for mobile communication. (Handheld computers, Phones etc.)

Transmission speeds range from 9.6Kbps to 2Mbps depending on the generation.

Satellite

A satellite is a space station that receives microwave signals from an earth based station (downlink), amplifies the signals then broadcasts the signals back over a wide area (uplink).

VSAT (very small aperture terminal) a small communications satellite.

Transmission speed is up to 1Gbps.

More expensive/harder to fix problems.

Affected by bad weather (e.g. you lose certain channels whenever it is raining).

Infra-Red

Sends signals using infrared light waves.

Also uses line-of-sight transmission.

Signals only travel for short distances.

Used by remote controls, wireless devices such as mouse/printer/digital camera.

Laser : The laser transmission direct line of sight. It is unidirectional like microwave, but has much higher speed than microwaves. The laser transmission requires the use of a laser transmitter and a photo sensitive receiver at each end.

Bluetooth : Bluetooth is a proprietary open wireless technology standard for exchanging data over short distances from fixed and mobile devices, creating Personal Area Networks (PANs) with high levels of security. It can connect several devices, overcoming problems of synchronization.

Wi-Fi : Wi-Fi is a mechanism for wirelessly connecting electronic devices. A device enabled with Wi-Fi, such as a personal computer, video game console, smart phone, or digital audio player, can connect to the Internet via a wireless network access point.

WiMax : WiMAX (Worldwide Interoperability for Microwave Access) is a communication technology for wirelessly delivering high-speed Internet service to large geographical areas. It is a part of a “fourth generation,” or 4G, of wireless-communication technology. WiMax hope to establish a global area network in which signals could reach.



A Tick (3) the correct option.

(NEP) Analytical Skills

1. Identify the odd term amongst the following group.

a. Coaxial cable	☐	b. Optical fibre	☐
c. Twisted pair wire	☐	d. Microwaves	☐
2. Which of the following items is not used in Local Area Networks (LANs)?

a. Computer	☐	b. Modem	☐	c. Printer	☐	d. Cable	☐
-------------	--------------------------	----------	--------------------------	------------	--------------------------	----------	--------------------------
3. Full form of LAN is:

a. Legal All Network	☐	b. Legal Area Network	☐
c. Local Area Network	☐	d. Lost Area Network	☐
4. A Local Area Network is a:

a. network within a building or site.	☐
b. network across a city, country or even the world.	☐
c. network that contains servers across the world.	☐
d. a network with 2000 or more computers.	☐
5. A Wide Area Network is a:

a. network within a building or site.	☐
b. network across a city, country or even the world.	☐
c. network that contains no server.	☐
d. network with fewer than two computers.	☐
6. Networks are useful because:

a. no one can copy your files.	☐
b. you only need one printer per network.	☐
c. the computers use less electricity.	☐
d. each user of the network can access files more easily.	☐

7. This cable is made of two plastic insulated copper wires twisted together to form a single media.

a. Twisted Pair Cable

☐

b. Coaxial Cable

☐

c. Fibre Cable

☐

d. Shielded Cable

☐

8. This cable is capable of carrying high frequency signals than that of twisted pair cable.

a. Twisted Pair Cable

☐

b. Coaxial Cable

☐

c. Fibre Cable

☐

d. Shielded Cable

☐

B Fill in the blanks.

1. Nodes are also called _____.

2. _____ is a unit of measure of the rate of signal modulation.

3. In _____ transmission data is transferred in the form of characters.

4. The twisted pair cable consists of _____ identical wires.

5. In _____ all devices share single communication line or cable.

6. The most common network interface connections are _____ and _____.

7. _____ can contain connections to attach coaxial, twisted pair, or fiber optics cable.

8. There are two types of twisted pair cables _____ and _____.

C Write T for True or F for False.

1. Two computers are said to be inter connected if they are capable of exchanging information.

☐

2. Workstation provides service to the network.

☐

3. Optical fibre cable consists of a solid wire core.

☐

4. Repeater is used to regenerate data and voice signal.

☐

5. Nodes helps in the communication between the server and the different nodes.

☐

6. Coaxial cables are capable of transmitting digital signals at a very high rate of 10 MBps.

☐

7. Electronic waves at microwave frequencies cannot bend round hills.

☐

8. Microwave transmission is possible for long distances without repeaters etc.

☐

D Answer the following questions.

1. What is a computer network? Explain with an example.

2. Differentiate between optical fibre and co-axial cable.
3. Differentiate between Bluetooth and Infrared with example.
4. Define LAN, MAN and WAN.
5. Write the use of Bus, Star and Ring Topology.
6. What are the hardware components required for networking?
7. Differentiate between Switches and Repeaters.
8. Differentiate between Twisted Pair Cable and Coaxial Cable.

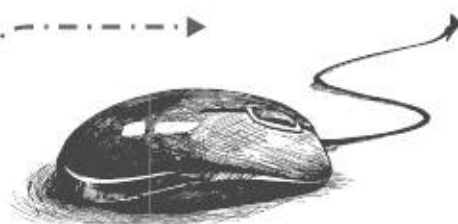
E Define the following.

Routers

Firewalls

Microwave Transmission

Infrared Transmission



W o r k s h e e t



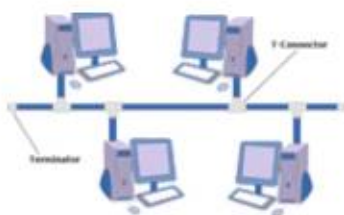
A Complete the table below with information collected about the computers and the devices in the computer laboratory.

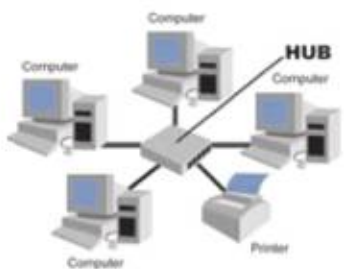
(NEP) Critical Thinking

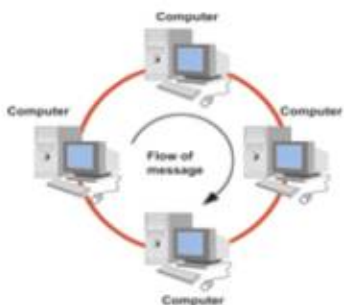
Hardware	Available or not		Numbers	Wired or Wireless	Function (briefly)
	Have	Haven't			
Computers					
Server computers					
Switches					
Routers					
Printers					
Scanners					
Multimedia projectors					
Other					

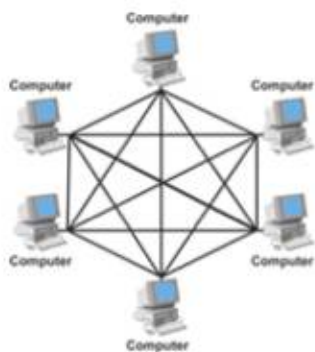
B Observe different network topologies and write about them.

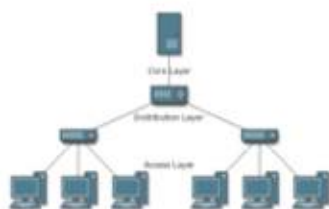
(NEP) Subject Enrichment











Mail Merge in MS-Word 2013

Lesson at a Glance

Mail Merge Process
Mailing List

Mailing Documents
Performing the Merge

Mail merge consists of combining mail and letters. This feature is usually employed in a word processing document which contains fixed text and variables which acts as placeholders that are replaced by text from the data source word to envelopes for mailing labels for mass mailings from a letter. A word processor is a device or computer program that provides for input editing, formatting and output of text often with some additional features.

Mail merge is a tool for writing a personalised email or letter to many people simultaneously. It imports data from other sources like worksheets and uses it to replace placeholders throughout the message with the relevant information for each individual that is being texted.

The mail merge feature is also known as the merge field.

Mail Merge Process

Mail merge is a useful tool that will allow you to easily produce multiple letters labels, envelopes, name tags and more using information stored in a list database or spreadsheet.

- **Main Document :** This is the main body of the letter whose contents are same for all recipients. To create the main document, simply type the text in a word file. The main document should have space for inserting data from the data source.
- **Data Source :** The data source is also known as mailing list and carries information about the recipients. This data is inserted at the desired positions in the main document. We can also edit the data source while doing the mail merge.

- **Merged Document :** This document is a combination of the main document and the mailing list. Information is pulled from your mailing list and inserted in your main document, resulting in the merged document the letter, email, labels, or envelopes personalized for different people on the mailing list.

Mailing Document

- **Letters :** Create and print a batch of personalized letters for everyone on your mailing list. Each letter prints on a separate sheet of paper.
- **Email :** Create and send email messages for each person on your mailing list with custom information inserted from the mailing list.
- **Envelopes :** Create and print a batch of envelopes for mailing.
- **Labels :** Create and print sheets of mailing labels; each label is a unique mailing address.

Mailing List

Word can pull data from a variety of data sources to perform mail merge. If you don't have a mailing list, you can create one during the mail merge process. Here's a list of a few, simple data sources you can use for mail merge.

- **Microsoft Excel spreadsheet** An Excel spreadsheet works well as a data source for mail merge if all data is well-formatted and on one sheet so that it can be read by Word. For more information, see [Prepare your Excel data source for a mail merge in Word](#).
- **Microsoft Outlook Contact List** You can retrieve your Outlook Contact List in Word. For more information, see [Use Outlook contacts as a data source for a mail merge](#).
- **Word data file** Your Word document should contain a single table. The first row of the table must contain headings, and the other rows must contain the records you want to merge. For more information, see [Set up a mail merge list with Word](#).

The Mail Merge file will hold special symbols (Mail Merge Fields), which will substitute the name and address and other relevant information stored in the data file.

To use Mail Merge :

1. Open an existing Word document, or create a new one.

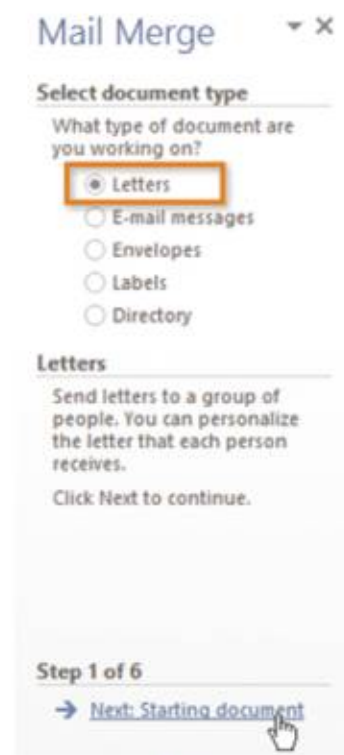
2. From the Mailings tab, click the Start Mail Merge command and select Step by Step Mail Merge Wizard from the drop-down menu.



The Mail Merge pane appears and will guide you through the six main steps to complete a merge. The following example demonstrates how to create a form letter and merge the letter with a recipient list.

Performing the Merge

Step 1: Choose the type of document you want to create. Here, we'll select Letters. Then click Next: Starting document to move to Step 2.



You are given information on the choices available at each step. When you complete a step, click the Next button at the bottom of the Task Pane to move forward to the next step. If you make a mistake, click the Previous button to retrace your steps.

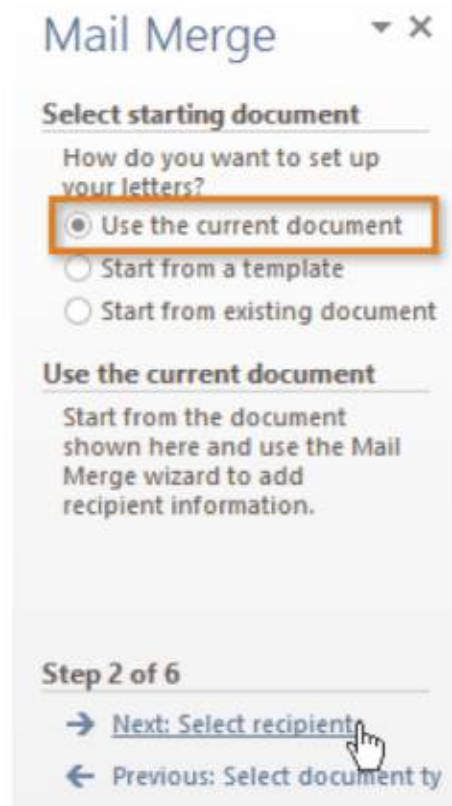
Step 2 : Select Starting Document

You have a choice of using:

Use the current document	The blank document displayed on screen
Start from a template	if you have previously created a template of your choice, would prefer to use one of Word's own Mail Merge templates
Start from existing document	This could be a document that already contains pre-typed text

Choose **Use the current document**

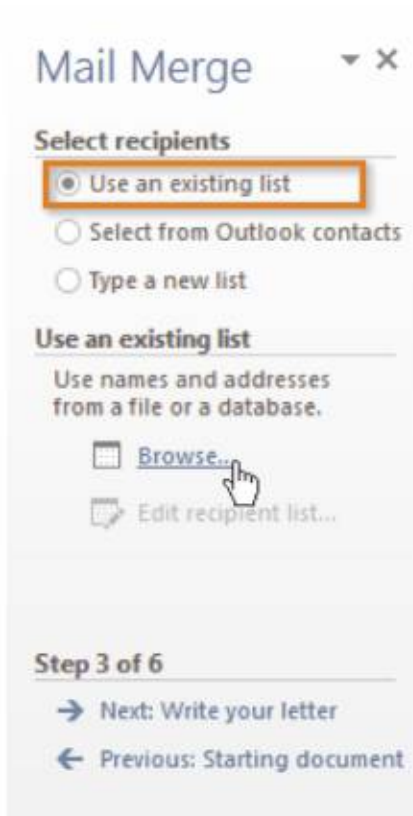
Step 3 : Select recipients.



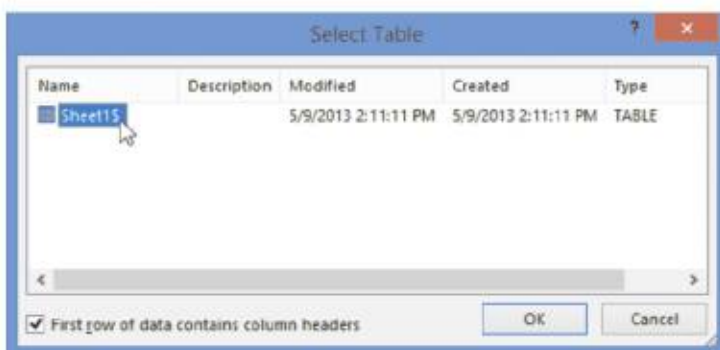
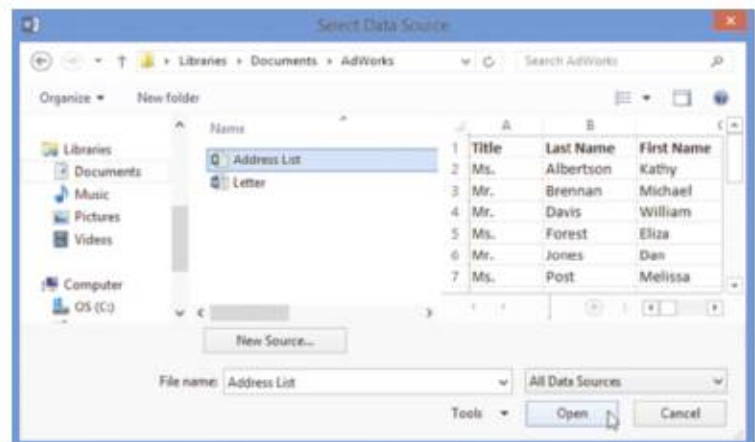
Now you'll need an address list so Word can automatically place each address into the document. The list can be in an existing file, such as an Excel workbook, or you can type a new address list from within the Mail Merge Wizard.

1. From the Mail Merge task pane, select Use an existing list, then click Browse... to select the file.

2. Locate your file and click Open.



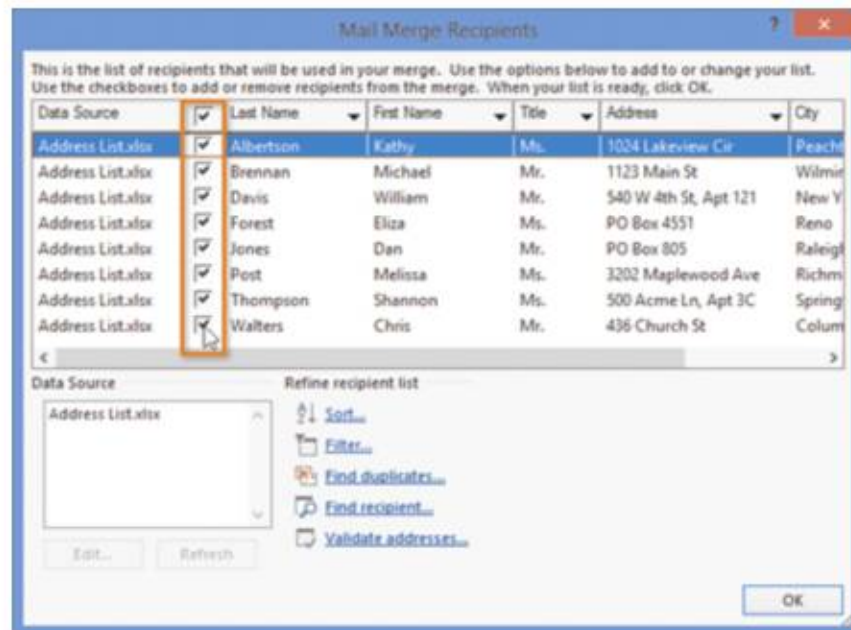
3. If the address list is in an Excel workbook, select the worksheet that contains the list and click OK.



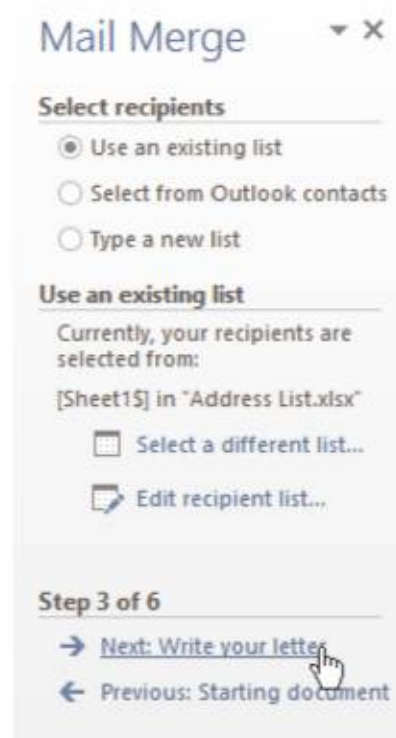
4. In the Mail Merge Recipients dialog box, you can check or uncheck each box to control which recipients are included in the merge. By default, all

recipients should be selected. When you're done, click OK.

5. From the Mail Merge task pane, click Next: Write your letter to move to Step 4.



If you don't have an existing address list, you can click the Type a new list button and click Create. You can then type your address list.



Step 4 : Write Your Letter

In the Mail Merge task pane, you have four options :

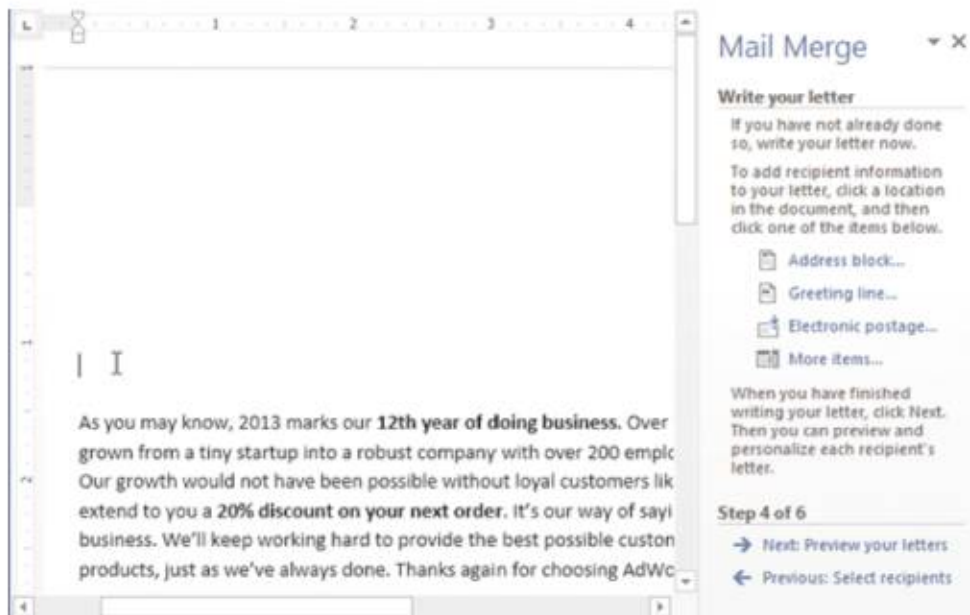
Now you're ready to write your letter. When it's printed, each copy of the letter

Address block	Use this option to insert a formatted address
Greeting Line	Use this option to insert a formatted salutation
Electronic postage	Use this option to insert electronic postage
More items	Use this option to insert individual merge fields

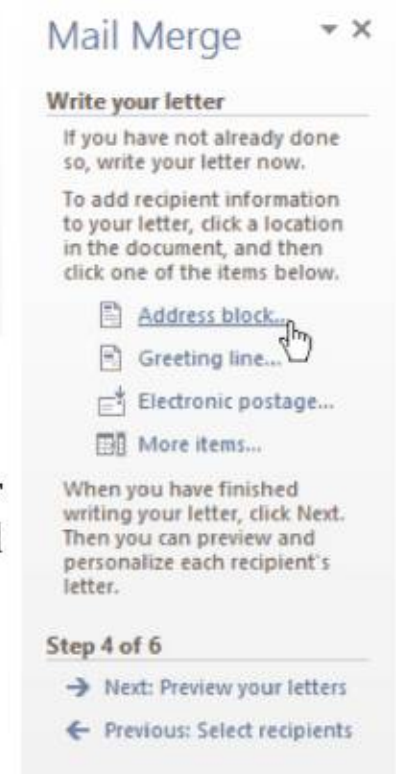
will basically be the same; only the recipient data (such as the name and address) will be different. You'll need to add placeholders for the recipient data so Mail Merge knows exactly where to add the data.

To insert recipient data :

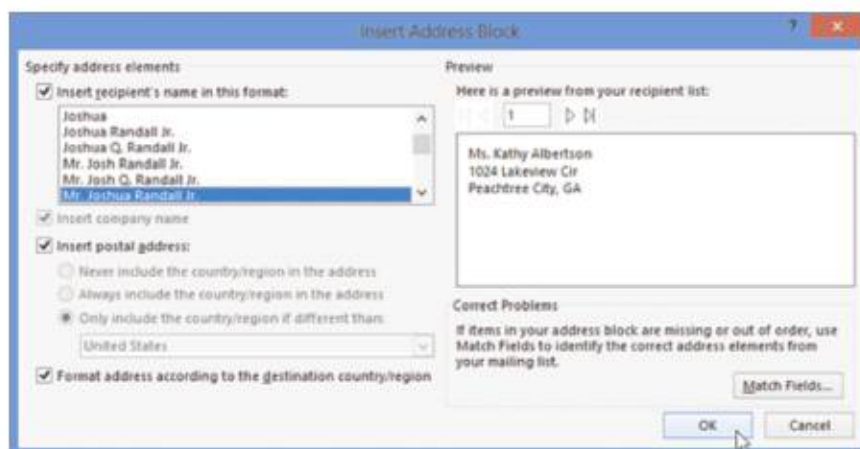
1. Place the insertion point in the document where you want the information to appear.
2. Choose one of the four placeholder options: Address block, Greeting line, Electronic postage, or More items.



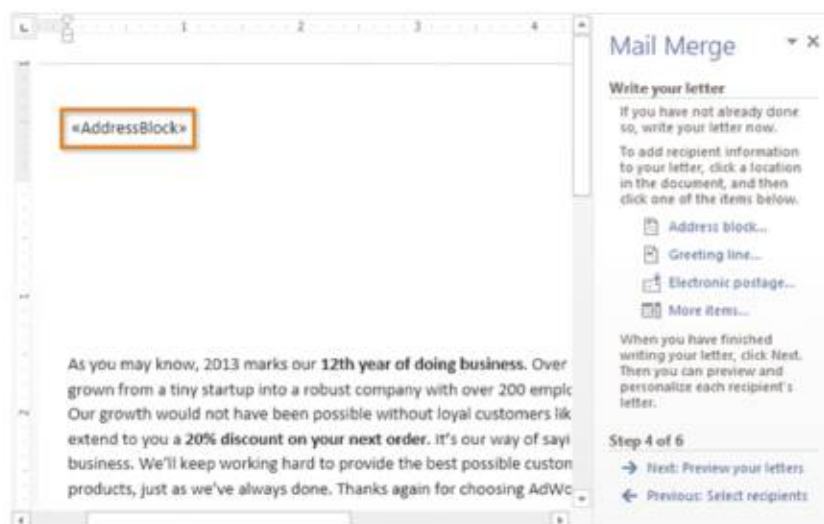
3. Depending on your selection, a dialog box may appear with various options. Select the desired options and click OK.



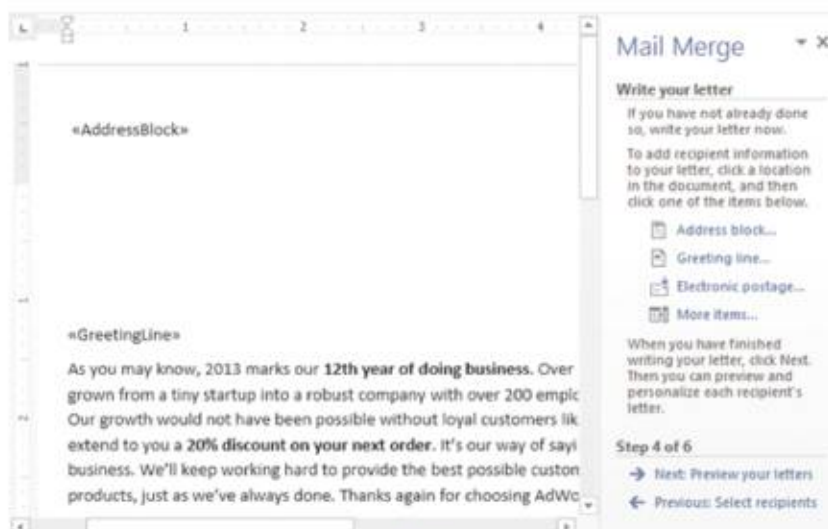
4. A placeholder will appear in your document (for example, «AddressBlock»).



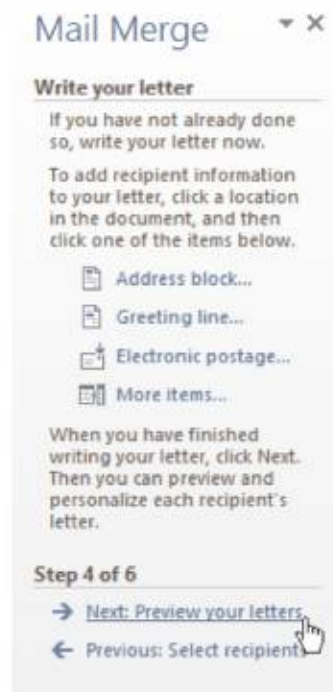
5. Repeat these steps each time you need to enter information from your data record. In our example, we'll add a Greeting line.



6. When you're done, click Next: Preview your letters to move to Step 5.



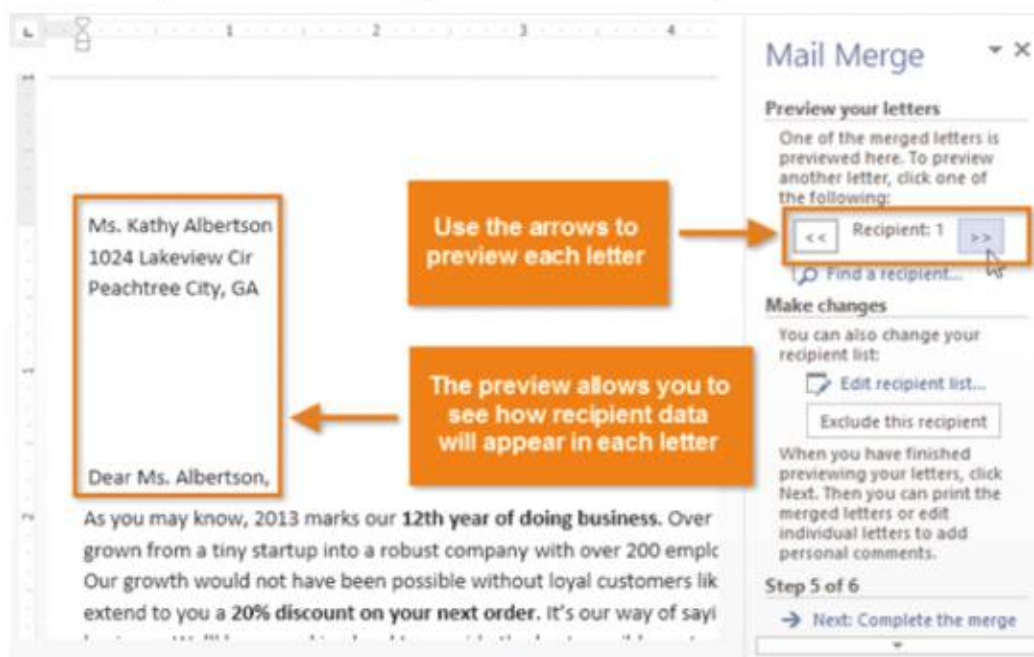
For some letters, you'll only need to add an Address block and Greeting line.



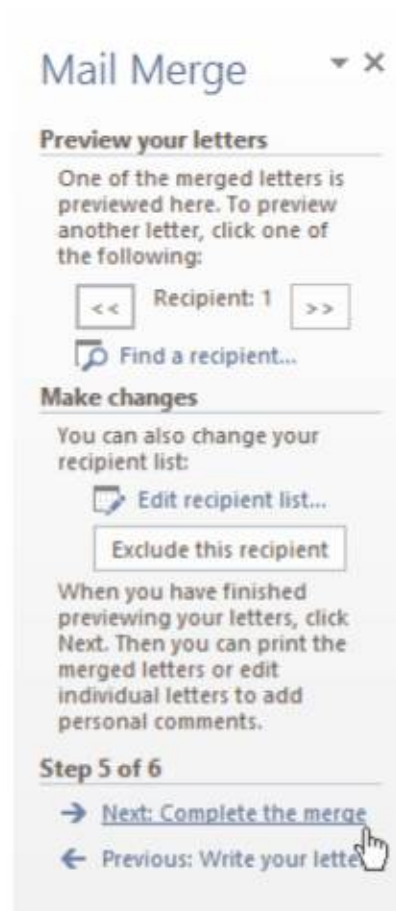
Sometimes, however, you may want to place recipient data within the body of the letter to personalize it even further.

Step 5 Next : Preview your letters.

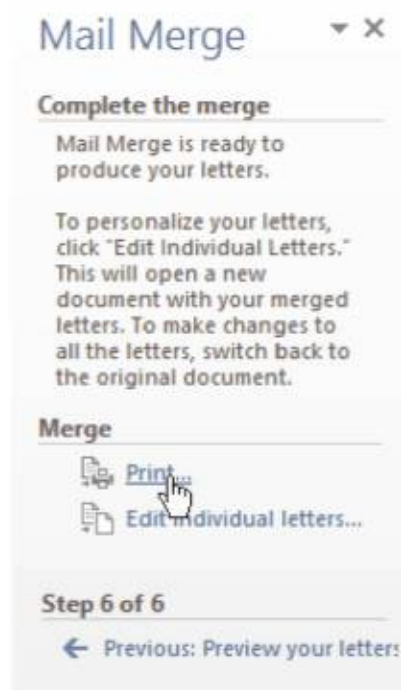
1. Preview the letters to make sure the information from the recipient list appears correctly in the letter. You can use the left and right scroll arrows to view each document.
2. Click Next: Complete the merge to move to Step 6.



Step 6 : Complete the Merge



1. Click Print... to print the letters.
2. A dialog box will appear. Click All, then click OK.



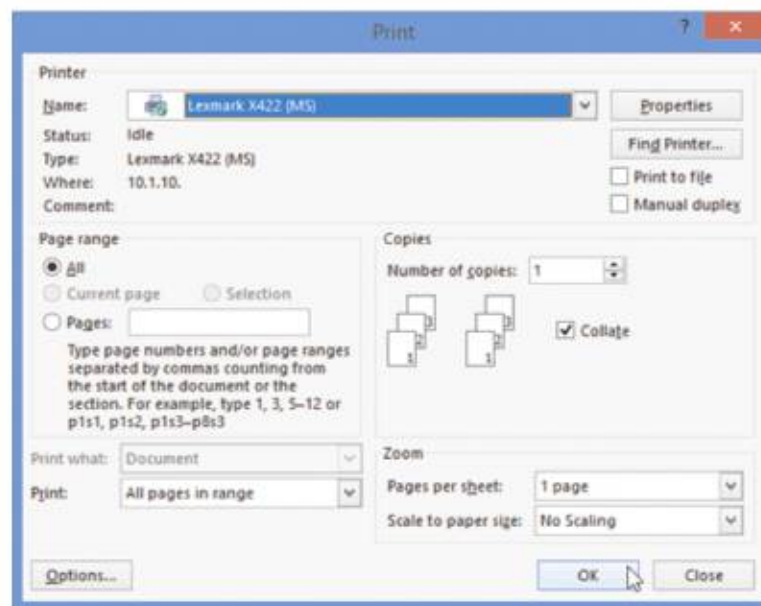
3. The Print dialog box will appear. Adjust the print settings if needed, then click OK. The letters will be printed.



Saving the letters

The merged letters are all contained in one file. Save this file

Click on the File tab on the Word Ribbon. You can copy and paste any of these letters into an individual file.



A Tick(3) the correct option.

(NEP) Analytical Skills

1. You also can create and print mailing labels and envelopes by using:

a. Address block	☐	b. Mail merge	☐
c. Greeting Line	☐	d. Electronic postage	☐

2. This document contains text and graphics that are identical for each version of the merged document.

a. Your main document	☐	b. Your mailing list	☐
c. Your address block	☐	d. Your greeting Line	☐
3. This document contains the data that is used to fill in information in your main document.

a. Your main document	☐	b. Your mailing list	☐
c. Your address block	☐	d. Your greeting Line	☐
4. Use this option to insert a formatted address.

a. Address block	☐	b. Main document	☐
c. Mailing list	☐	d. Greeting Line	☐
5. Use this option to insert a formatted salutation.

a. Electronic postage	☐	b. Greeting Line	☐
c. Mailing list	☐	d. Main document	☐
6. Use this option to insert electronic postage.

a. Electronic postage	☐	b. Greeting Line	☐
c. Mailing list	☐	d. Main document	☐

B Fill in the blanks.

1. _____ consists of combining mail and letters.
2. _____ is a combination of the main document and the mailing list.
3. A _____ appears to the right of the document and is visible throughout the entire Mail Merge procedure.
4. _____ option to insert individual merge fields.
5. The _____ file will hold special symbols, which will substitute the name and address and other relevant information stored in the data file.
6. _____ create and print a batch of personalized letters for everyone on your mailing list.

C Write T for True or F for False.

1. There are three documents involved in creating and printing letters and emails, or labels and envelopes using the mail merge process. ☐
2. An example of identical content is the return address on the envelope or in the body of a letter or an email message. ☐
3. Word can pull data from a variety of data sources to perform mail merge. ☐

4. The label results button turns orange when selected. ☐
5. You can refine the recipient list by selecting any of the options below, mail, order, find duplicates, etc. ☐

D Answer the following questions.

1. What is Mail Merge?
2. What you can create with the use of mail merge?
3. How information is organised in a data file?
4. How main document is created?
5. What is data source?
6. Define merge fields.
7. List the variety of formats of data file.

LAB ACTIVITY

A. Mail Merge - Practice Exercise One Using the Mail Merge Helper

(NEP) Experimental Skills

1. Create the Main Document.
2. Use the following information to determine the fields needed for your data source and enter the records. Save the data source class info.

Field Names	Record 1	Record 2	Record 3
Title	Ms.	Mr.	Mr.
FirstName	Mary	Harry	Jim
LastName	Smith	Jones	Walker
JobTitle	President	Comptroller	Manager
Company	New Company	ABC Company	XYZ Company
Address1	124 Main Street	890 Fifth Avenue	412 Central Avenue
City	Duluth	Smithtown	Oranagetown
State	MN	NY	NJ
PostalCode	67774	87765	7984
Class	Word	Word	Excel

Class_Date

10/28/2022

10/25/2022

10/30/2022

3. Type the following main document and save it as ClassAct.
4. Save the merged document as MergeClassAct.
5. Use the template below to create the main document.

Date (enter the date using the insert command)

«Title»

«FirstName»«LastName»

«JobTitle»

«Company»

«Address1»

«City», «State» «PostalCode»

Dear «FirstName»

Thank you for enrolling in our class. All classes begin at 9 a.m. and end at 4 p.m.

We look forward to seeing you in our «Class» on «ClassDate».

Sincerely

Advance MS- Excel

Lesson at a Glance

Basic Sorting
Filtering Data

Custom Sorting
Advanced Filtering

Sorting Multiple Levels

Microsoft Excel is a spreadsheet software used extensively for calculation and analytical work. In Excel, data is organized in rows and columns. Rows are horizontal sections and columns are vertical sections. These rows and columns intersect to form cells. Data is stored in these cells. A cell is identified by its column letter and row number, e.g. A5, C6.

Basic Sorting

The arrangement of keeping the data either in ascending or descending order is known as sorting of Data. The ascending order arranges the data from A to Z or 0 to 9 while descending order arranges the data from Z to A or 9 to 0.

To sort a sheet :

1. Select a cell in the column you want to sort by.

C2				Chen		
	A	B	C	D	E	F
1	Homeroom #	First Name	Last Name	T-Shirt Size	Payment Method	
2	105	Christiana	Chen	Medium	Cash	
3	105	Melissa	White	Small	Debit Card	
4	105	Esther	Yaron	Small	Check	
5	135	Anisa	Naser	Small	Check	
6	135	Chantal	Weller	Medium	Cash	
7	220-A	Juan	Flores	X-Large	Pending	
8	220-B	Malik	Reynolds	Small	Cash	
9	220-B	Avery	Kelly	Medium	Debit Card	
10	105	Derek	MacDonald	Large	Cash	

2. Select the Data tab on the Ribbon, then click the Ascending command to Sort A to Z, or the Descending command to Sort Z to A.



3. The worksheet will be sorted by the selected column. Here, the worksheet is now sorted by last name.

C2					Ackerman	
	A	B	C	D	E	F
1	Homeroom #	First Name	Last Name	T-Shirt Size	Payment Method	
2	110	Kris	Ackerman	Large	Money Order	
3	105	Nathan	Albee	Medium	Check	
4	220-B	Samantha	Bell	Medium	Check	
5	110	Matt	Benson	Medium	Money Order	
6	105	Christiana	Chen	Medium	Cash	
7	110	Gabriel	Del Toro	Medium	Cash	
8	220-A	Brigid	Ellison	Small	Cash	
9	220-A	Juan	Flores	X-Large	Pending	
10	220-B	Tyrese	Hanlon	X-Large	Debit Card	

Custom Sorting

You can use a Custom List to identify your own sorting order, such as days of the week or in this example, T-shirt sizes from smallest to largest.

To create a custom sort :

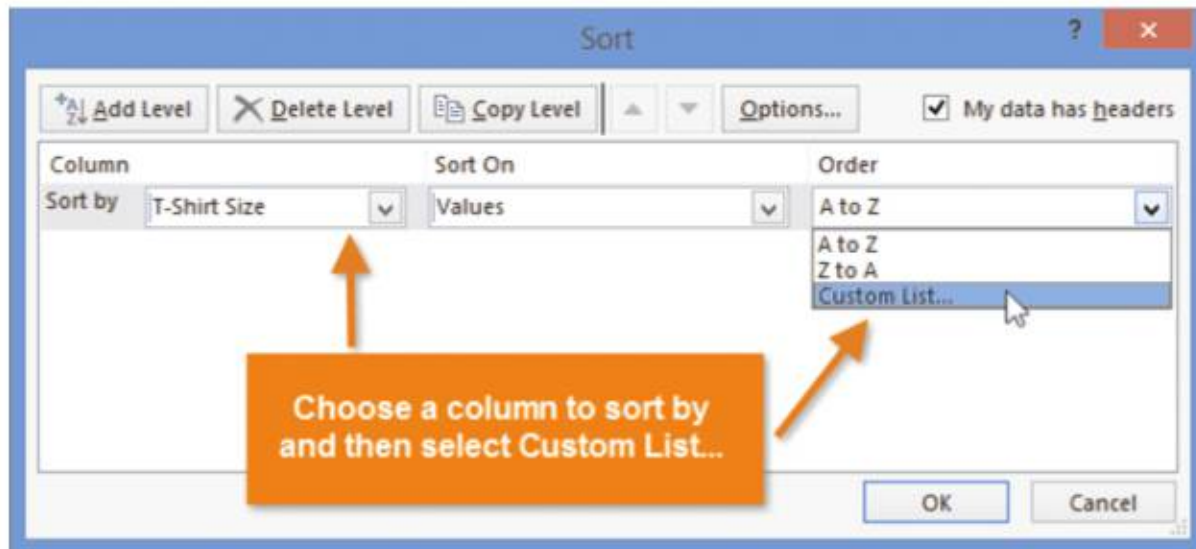
1. Select a cell in the column you want to sort by.

D2					Large	
	A	B	C	D	E	F
1	Homeroom #	First Name	Last Name	T-Shirt Size	Payment Method	
2	110	Kris	Ackerman	Large	Money Order	
3	105	Nathan	Albee	Medium	Check	
4	220-B	Samantha	Bell	Medium	Check	
5	110	Matt	Benson	Medium	Money Order	
6	105	Christiana	Chen	Medium	Cash	
7	110	Gabriel	Del Toro	Medium	Cash	
8	220-A	Brigid	Ellison	Small	Cash	
9	220-A	Juan	Flores	X-Large	Pending	
10	220-B	Tyrese	Hanlon	X-Large	Debit Card	

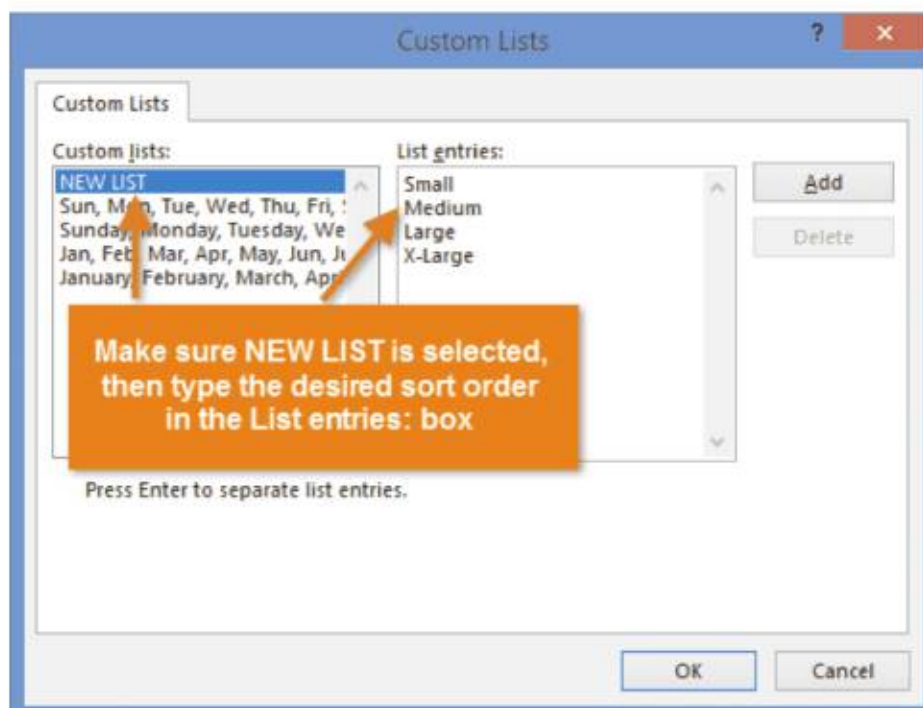
2. Select the Data tab, then click the Sort command.



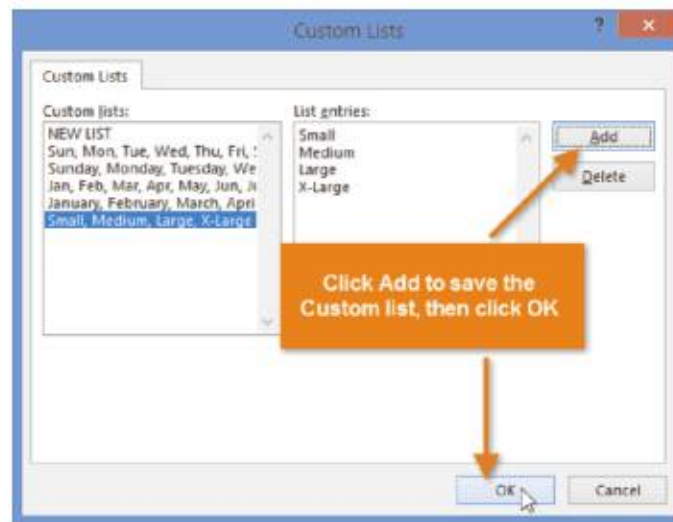
3. The Sort dialog box will appear. Select the column you want to sort by, then choose Custom List... from the Order field.



4. The Custom Lists dialog box will appear. Select NEW LIST from the Custom Lists: box.
5. Type the items in the desired custom order in the List entries: box.



6. Click Add to save the new sort order. The new list will be added to the Custom lists: box. Make sure the new list is selected, then click OK.



7. The Custom Lists dialog box will close. Click OK in the Sort dialog box to perform the custom sort.



8. The worksheet will be sorted by the custom order.
To sort by cell formatting :

	A	B	C	D	E	F
1	Homeroom #	First Name	Last Name	T-Shirt Size	Payment Method	
2	220-A	Brigid	Ellison	Small	Cash	
3	220-B	Michael	Lazar	Small	Cash	
4	135	Anisa	Naser	Small	Pending	
5	220-A	Christopher	Peyton-Gomez	Small	Check Bounced	
6	220-B	Malik	Reynolds	Small	Cash	
7	220-B	Windy	Shaw	Small	Cash	
8	105	Melissa	White	Small	Debit Card	
9	105	Esther	Yaron	Small	Check	
10	105	Nathan	Albee	Medium	Check	
11	220-B	Samantha	Bell	Medium	Check	
12	220-B	Avery	Kelly	Medium	Debit Card	
13	220-A	Chevonne	Means	Medium	Money Order	
14	135	James	Panarella	Medium	Check	
15	135	Chantal	Weller	Medium	Cash	
16	110	Kris	Ackerman	Large	Money Order	
17	105	Derek	MacDonald	Large	Cash	

You can also choose to sort your worksheet by formatting rather than cell content. This can be especially helpful if you add color coding to certain cells.

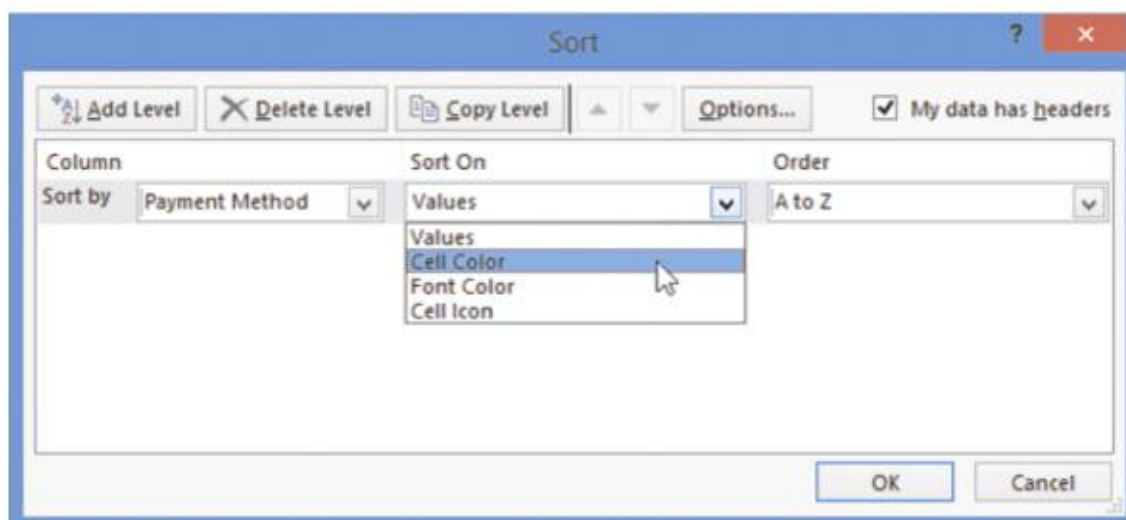
1. Select a cell in the column you want to sort by. Here, we'll select cell E2.
2. Select the Data tab, then click the Sort command.

E2					Check Bounced	
	A	B	C	D	E	F
1	Homeroom #	First Name	Last Name	T-Shirt Size	Payment Method	
2	220-A	Christopher	Peyton-Gomez	Small	Check Bounced	
3	220-B	Malik	Reynolds	Small	Cash	
4	220-B	Windy	Shaw	Small	Cash	
5	220-B	Michael	Lazar	Small	Cash	
6	135	Anisa	Naser	Small	Pending	
7	220-A	Brigid	Ellison	Small	Cash	
8	105	Melissa	White	Small	Debit Card	
9	105	Esther	Yaron	Small	Check	
10	135	Chantal	Weller	Medium	Cash	

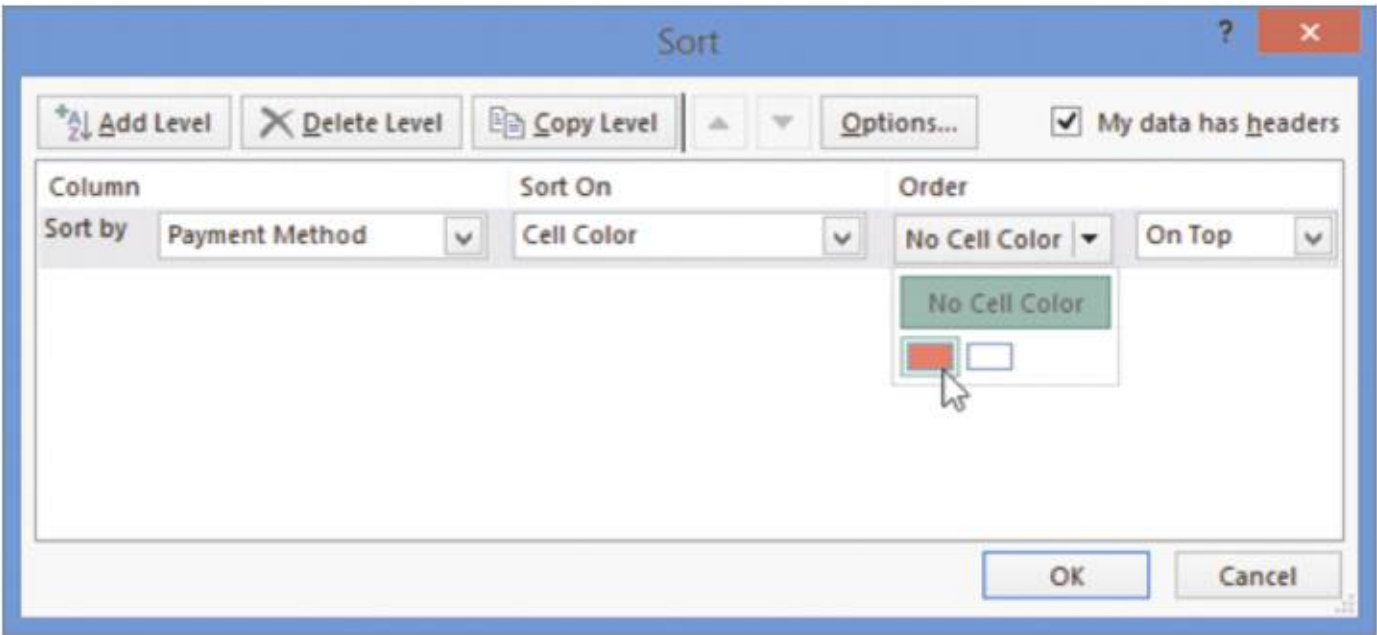
3. The Sort dialog box will appear. Select the column you want to sort by, then decide whether you'll sort by Cell Color, Font Color, or Cell Icon from the Sort On field.



4. Choose a color to sort by from the Order field.
5. Click OK. Here, the worksheet is now sorted by cell color, with the light red



cells on top. This allows us to see which orders still have outstanding payments.



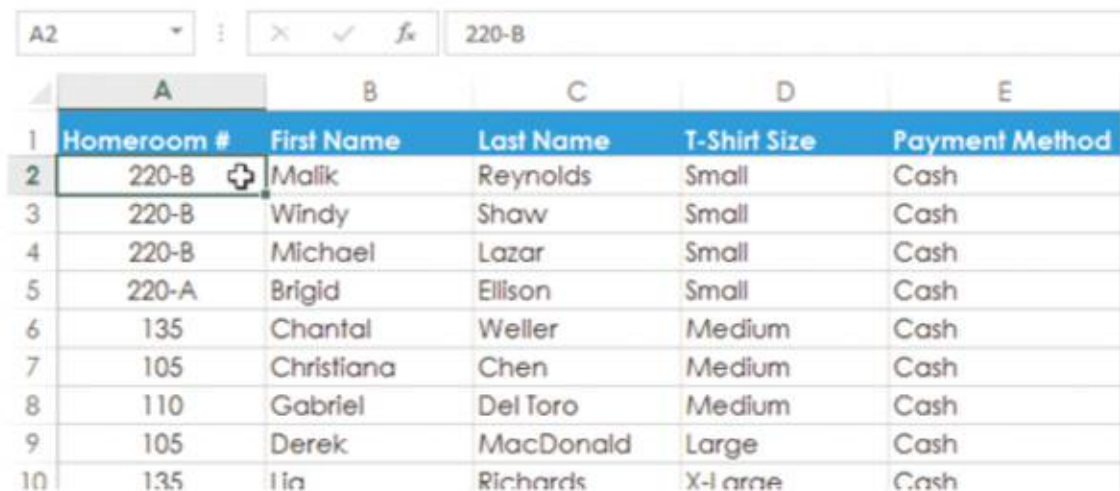
Sorting Multiple Levels

Another feature of custom sorting sorting multiple level sallows you to identify which columns to sort by and when, giving you more control over the organization of your data. For example, you could sort by more than one cell color such as red, then yellow, then green, to indicate different levels of priority or, as seen below, you could sort students by home room number, then by last name.

E2					Check Bounced	
	A	B	C	D	E	F
1	Homeroom #	First Name	Last Name	T-Shirt Size	Payment Method	
2	220-A	Christopher	Peyton-Gomez	Small	Check Bounced	
3	105	Sidney	Kelly	Medium	Check Bounced	
4	135	Anisa	Naser	Small	Pending	
5	110	Regina	Olivera	Large	Pending	
6	220-A	Juan	Flores	X-Large	Pending	
7	135	Alex	Yuen	Large	Wrong Amount	
8	220-B	Malik	Reynolds	Small	Cash	
9	220-B	Windy	Shaw	Small	Cash	
10	220-B	Michael	Lazar	Small	Cash	

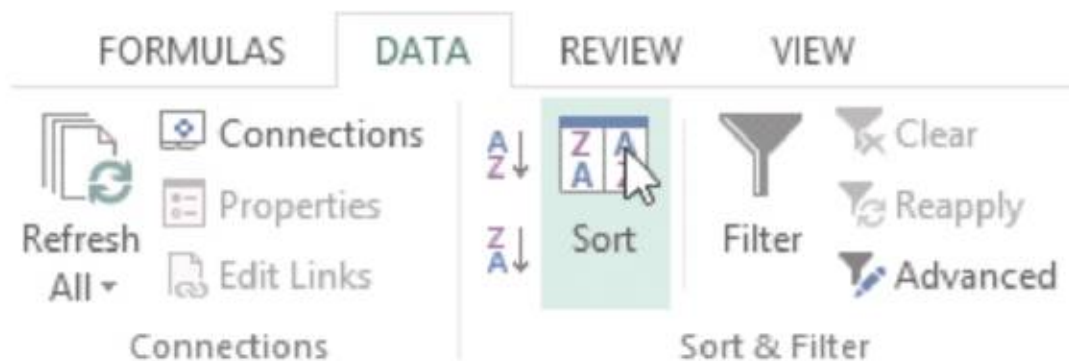
To add a level :

1. Select a cell in the column you want to sort by.



	A	B	C	D	E
1	Homeroom #	First Name	Last Name	T-Shirt Size	Payment Method
2	220-B	Malik	Reynolds	Small	Cash
3	220-B	Windy	Shaw	Small	Cash
4	220-B	Michael	Lazar	Small	Cash
5	220-A	Brigid	Ellison	Small	Cash
6	135	Chantal	Weller	Medium	Cash
7	105	Christiana	Chen	Medium	Cash
8	110	Gabriel	Del Toro	Medium	Cash
9	105	Derek	MacDonald	Large	Cash
10	135	Lia	Richards	X-Large	Cash

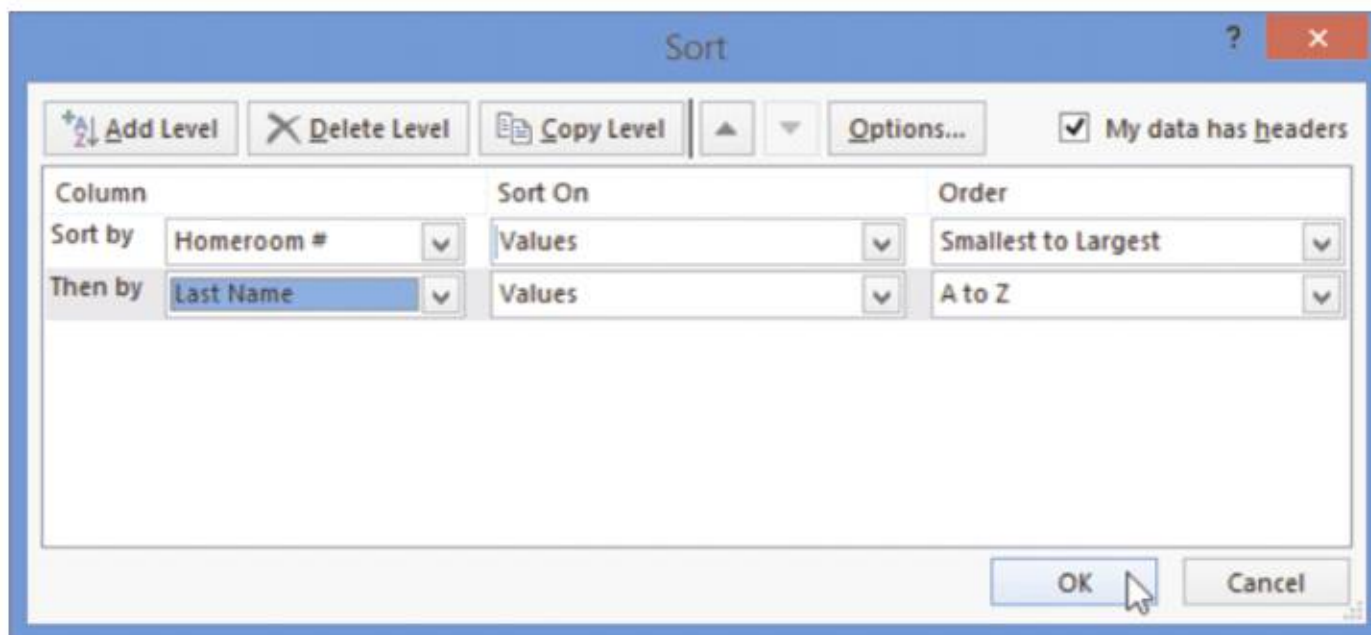
2. Click the Data tab, then select the Sort command.



3. The Sort dialog box will appear. Select the first column you want to sort by.
4. Click Add Level to add another column to sort by.



5. Select the next column you want to sort by, then click OK.



6. The worksheet will be sorted according to the selected order.

A2

:

✕

✓

fx

105

	A	B	C	D	E	F
1	Homeroom #	First Name	Last Name	T-Shirt Size	Payment Method	
2	105	Nathan	Albee	Medium	Check	
3	105	Christiana	Chen	Medium	Cash	
4	105	Sidney	Kelly	Medium	Check Bounced	
5	105	Derek	MacDonald	Large	Cash	
6	105	Melissa	White	Small	Debit Card	
7	105	Esther	Yaron	Small	Check	
8	110	Kris	Ackerman	Large	Money Order	
9	110	Matt	Benson	Medium	Money Order	
10	110	Gabriel	Del Toro	Medium	Cash	

Copy Level will add a level by duplicating the one you have selected and allowing you to modify the sorting criteria. This is useful if you need to sort multiple levels that share some criteria, such as the same Column, Sort On, or Order.

Filtering Data

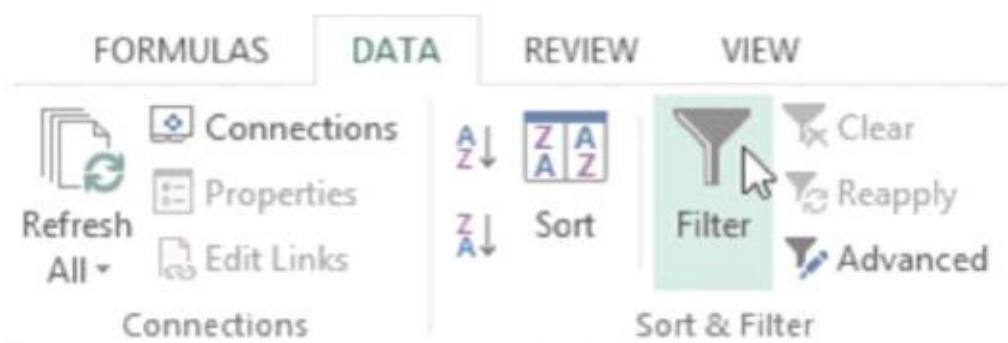
Filters can be applied in different ways to improve the performance of your worksheet. You can filter text, dates, and numbers. You can even use more than one filter to further narrow your results.

To filter data :

1. In order for filtering to work correctly, your worksheet should include a header row, which is used to identify the name of each column.

	A	B	C	D	E	F
1	ID #	Type	Equipment Detail	Checked Out	Checked In	Checked Out By
2	3000	Camera	Saris Lumina Digital Camera	12-May-13	15-May-13	Shannon Nguyen
3	3005	Camera	Saris Zoom Z-60 Digital Camera	27-Jul-13	06-Aug-13	Sela Shepard
4	3070	Camera	Omega PixL Digital Camcorder	06-Oct-13		Min Seung
5	1021	Laptop	15" EDI SmartPad L200-3	15-Sep-13	01-Oct-13	Sofie Ragnar
6	1022	Laptop	15" EDI SmartPad L200-3	14-Aug-13	16-Aug-13	Hank Sorenson
7	1023	Laptop	15" EDI SmartPad L200-3	08-Aug-13	15-Aug-13	Jennifer Weiss
8	1025	Laptop	15" EDI SmartPad L200-4X	26-Sep-13	04-Oct-13	Min Seung
9	1031	Laptop	17" Saris X-10 Laptop	04-Oct-13		Nick Ortiz
10	1032	Laptop	17" Saris X-10 Laptop	19-Sep-13		Stanley Geyer
11	1033	Laptop	17" Saris X-10 Laptop	24-Sep-13	26-Sep-13	George D'Agosta

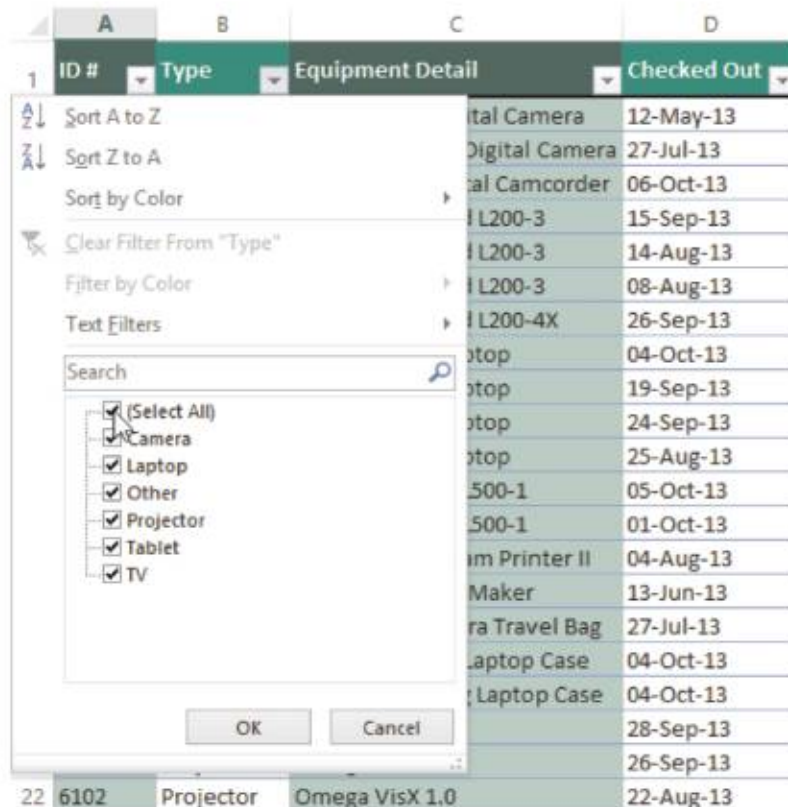
2. Select the Data tab, then click the Filter command.



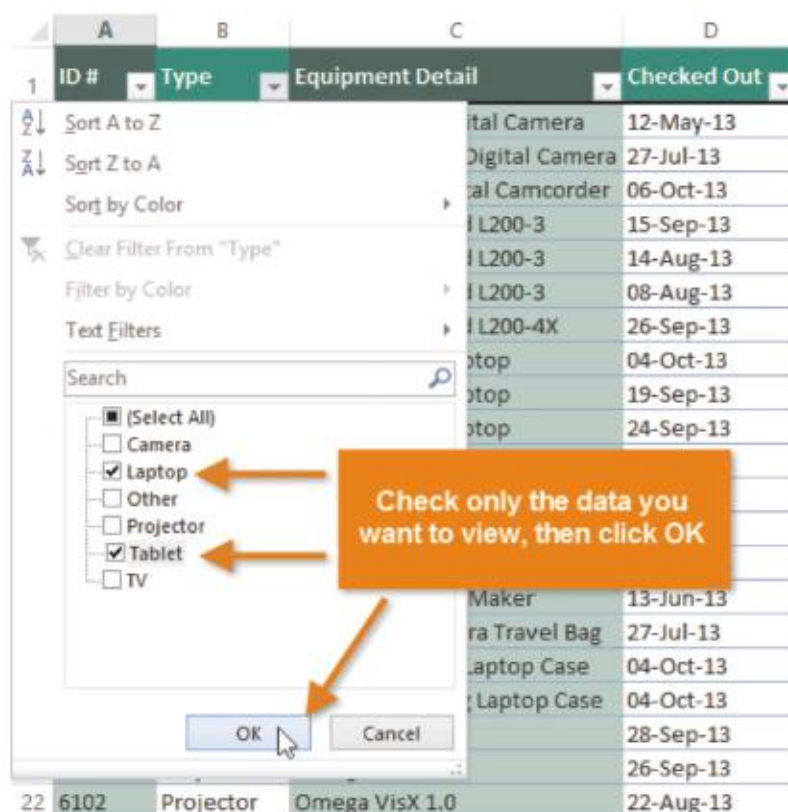
3. A drop-down arrow will appear in the header cell for each column.
4. Click the drop-down arrow for the column you want to filter.

	A	B	C	D	E	F
1	ID #	Type	Equipment Detail	Checked Out	Checked In	Checked Out By
2	3000	Camera	Saris Lumina Digital Camera	12-May-13	15-May-13	Shannon Nguyen
3	3005	Camera	Saris Zoom Z-60 Digital Camera	27-Jul-13	06-Aug-13	Sela Shepard
4	3070	Camera	Omega PixL Digital Camcorder	06-Oct-13		Min Seung
5	1021	Laptop	15" EDI SmartPad L200-3	15-Sep-13	01-Oct-13	Sofie Ragnar
6	1022	Laptop	15" EDI SmartPad L200-3	14-Aug-13	16-Aug-13	Hank Sorenson
7	1023	Laptop	15" EDI SmartPad L200-3	08-Aug-13	15-Aug-13	Jennifer Weiss
8	1025	Laptop	15" EDI SmartPad L200-4X	26-Sep-13	04-Oct-13	Min Seung
9	1031	Laptop	17" Saris X-10 Laptop	04-Oct-13		Nick Ortiz
10	1032	Laptop	17" Saris X-10 Laptop	19-Sep-13		Stanley Geyer
11	1033	Laptop	17" Saris X-10 Laptop	24-Sep-13	26-Sep-13	George D'Agosta

5. The Filter menu will appear.
6. Uncheck the box next to Select All to quickly deselect all data.



7. Check the boxes next to the data you want to filter, then click OK.



- The data will be filtered, temporarily hiding any content that doesn't match the criteria.

	A	B	C	D	E	F
1	ID #	Type	Equipment Detail	Checked Out	Checked In	Checked Out By
5	1021	Laptop	15" EDI SmartPad L200-3	15-Sep-13	01-Oct-13	Sofie Ragnar
6	1022	Laptop	15" EDI SmartPad L200-3	14-Aug-13	16-Aug-13	Hank Sorenson
7	1023	Laptop	15" EDI SmartPad L200-3	08-Aug-13	15-Aug-13	Jennifer Weiss
8	1025	Laptop	15" EDI SmartPad L200-4X	26-Sep-13	04-Oct-13	Min Seung
9	1031	Laptop	17" Saris X-10 Laptop	04-Oct-13		Nick Ortiz
10	1032	Laptop	17" Saris X-10 Laptop	19-Sep-13		Stanley Geyer
11	1033	Laptop	17" Saris X-10 Laptop	24-Sep-13	26-Sep-13	George D'Agosta
12	1034	Laptop	17" Saris X-10 Laptop	25-Aug-13	27-Aug-13	Jay Peralta
26	1011	Tablet	Saris SlimTab Pro	04-Aug-13		Jay Peralta
27	1012	Tablet	Saris SlimTab Pro	29-Sep-13		August Zorn

To add another filter :

Filters are additive, meaning you can use as many as you need to narrow your results.

- Click the drop-down arrow for the column you want to filter.

	A	B	C	D	E	F
1	ID #	Type	Equipment Detail	Checked Out	Checked In	Checked Out By
5	1021	Laptop	15" EDI SmartPad L200-3	15-Sep-13	01-Oct-13	Sofie Ragnar
6	1022	Laptop	15" EDI SmartPad L200-3	14-Aug-13	16-Aug-13	Hank Sorenson
7	1023	Laptop	15" EDI SmartPad L200-3	08-Aug-13	15-Aug-13	Jennifer Weiss
8	1025	Laptop	15" EDI SmartPad L200-4X	26-Sep-13	04-Oct-13	Min Seung
9	1031	Laptop	17" Saris X-10 Laptop	04-Oct-13		Nick Ortiz
10	1032	Laptop	17" Saris X-10 Laptop	19-Sep-13		Stanley Geyer
11	1033	Laptop	17" Saris X-10 Laptop	24-Sep-13	26-Sep-13	George D'Agosta
12	1034	Laptop	17" Saris X-10 Laptop	25-Aug-13	27-Aug-13	Jay Peralta

- The Filter menu will appear.
- Check or uncheck the boxes depending on the data you want to filter, then click OK.

	A	B	C	D	E	F
1	ID #	Type	Equipment Detail	Checked Out	Checked In	Checked Out By
5	1021	Laptop	15" EDI SmartPad L200-3	15-Sep-13	01-Oct-13	Sofie Ragnar
6	1022	Laptop	15" EDI SmartPad L200-3	14-Aug-13	16-Aug-13	Hank Sorenson
7	1023	Laptop	15" EDI SmartPad L200-3	08-Aug-13	15-Aug-13	Jennifer Weiss
8	1025	Laptop	15" EDI SmartPad L200-4X	26-Sep-13	04-Oct-13	Min Seung
9	1031	Laptop	17" Saris X-10 Laptop	04-Oct-13		Nick Ortiz
10	1032	Laptop	17" Saris X-10 Laptop	19-Sep-13		Stanley Geyer
11	1033	Laptop	17" Saris X-10 Laptop	24-Sep-13	26-Sep-13	George D'Agosta
12	1034	Laptop	17" Saris X-10 Laptop	25-Aug-13	27-Aug-13	Jay Peralta
26	1011	Tablet	Saris SlimTab Pro	04-Aug-13		Jay Peralta
27	1012	Tablet	Saris SlimTab Pro	29-Sep-13		August Zorn

4. The new filter will be applied.

	A	B	C	D	E	F
1	ID #	Type	Equipment Detail	Checked Out	Checked In	Checked Out By
6	1022	Laptop	15" EDI SmartPad L200-3	14-Aug-13	16-Aug-13	Hank Sorenson
7	1023	Laptop	15" EDI SmartPad L200-3	08-Aug-13	15-Aug-13	Jennifer Weiss
12	1034	Laptop	17" Saris X-10 Laptop	25-Aug-13	27-Aug-13	Jay Peralta
26	1011	Tablet	Saris SlimTab Pro	04-Aug-13		Jay Peralta

To clear a filter :

After applying a filter, you may want to remove or clear it from your worksheet so you'll be able to filter content in different ways.

1. Click the drop-down arrow for the filter you want to clear.
2. The Filter menu will appear.
3. Choose Clear Filter From [COLUMN NAME] from the Filter menu.

	A	B	C	D	E	F
1	ID #	Type	Equipment Detail	Checked Out	Checked In	Checked Out By
6	1022	Laptop	15" EDI SmartPad L200-3	14-Aug-13	16-Aug-13	Hank Sorenson
7	1023	Laptop	15" EDI SmartPad L200-3	08-Aug-13	15-Aug-13	Jennifer Weiss
12	1034	Laptop	17" Saris X-10 Laptop	25-Aug-13	27-Aug-13	Jay Peralta
26	1011	Tablet	10" Saris SlimTab Pro	04-Aug-13		Jay Peralta

4. The filter will be cleared from the column. The previously hidden data will be displayed.

	A	B	C	D	E	F
1	ID #	Type	Equipment Detail	Checked Out	Checked In	Checked Out By
5	1021	Laptop	15" EDI SmartPad L200-3	15-Sep-13	01-Oct-13	Sofie Ragnar
6	1022	Laptop	15" EDI SmartPad L200-3	14-Aug-13	15-Aug-13	Hank Sorenson
7	1023	Laptop	15" EDI SmartPad L200-3	08-Aug-13	15-Aug-13	Jennifer Weiss
8	1025	Laptop	15" EDI SmartPad L200-4X	26-Sep-13	04-Oct-13	Min Seung

Advanced Filtering

Searching for data is a convenient alternative to checking or unchecking data from the list. You can search for data that contains an exact phrase, number, date, or simple fragment.

Using advanced text filters

Advanced text filters can be used to display more specific information, such as cells that contain a certain number of characters or data that does not contain a word you specify.

1. Select the Data tab, then click the Filter command. A drop-down arrow will appear in the header cell for each column.
2. Click the drop-down arrow for the column you want to filter.

	A	B	C	D	E	F
1	ID #	Type	Equipment Detail	Checked Out	Checked In	Checked Out By
13	2050	Other	EDI SmartBoard L500-1	05-Oct-13	06-Oct-13	Anthony Liddell
14	2051	Other	EDI SmartBoard L500-1	05-Oct-13	05-Oct-13	Sofie Ragnar
15	3800	Other	U-Go Saris DigiCam Printer II	04-Aug-13	05-Aug-13	Hank Sorenson
16	3900	Other	U-Go Saris Label Maker	13-Jun-13	20-Jun-13	Clint Gosse
17	4800	Other	7N Deluxe Camera Travel Case	27-Jul-13	06-Aug-13	Sela Shepard
18	4900	Other	7N Light Rolling Laptop Case	04-Oct-13		Jay Peralta
19	4905	Other	7N Heavy Rolling Laptop Case	04-Oct-13		Nick Ortiz
31						

3. The Filter menu will appear. Hover the mouse over Text Filters, then select the desired text filter from the drop-down menu.
4. The Custom AutoFilter dialog box will appear. Enter the desired text to the

	A	B	C	D	E	F
1	ID #	Type	Equipment Detail	Checked Out	Checked In	Checked Out By
13	2050			05-Oct-13	06-Oct-13	Anthony Liddell
14	2051			01-Oct-13	05-Oct-13	Sofie Ragnar
15	3800			04-Aug-13	05-Aug-13	Hank Sorenson
16	3900			13-Jun-13	20-Jun-13	Clint Gosse
17	4800			27-Jul-13	06-Aug-13	Sela Shepard
18	4900			04-Oct-13		Jay Peralta
19	4905					Nick Ortiz
31						
32						
33						
34						
35						
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43						
44						

right of the filter, then click OK.

- The data will be filtered by the selected text filter.

Using advanced date filters

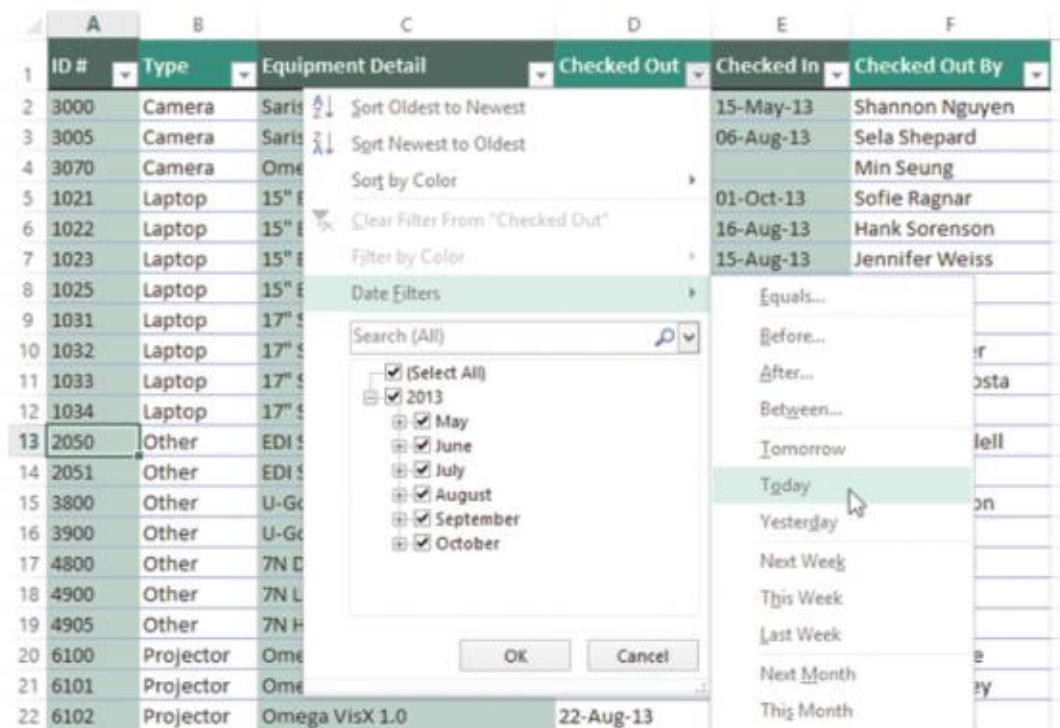
	A	B	C	D	E	F
1	ID #	Type	Equipment Detail	Checked Out	Checked In	Checked Out By
13	2050	Other	EDI SmartBoard L500-1	05-Oct-13	06-Oct-13	Anthony Liddell
14	2051	Other	EDI SmartBoard L500-1	01-Oct-13	05-Oct-13	Sofie Ragnar
15	3800	Other	U-Go Saris DigiCam Printer II	04-Aug-13	05-Aug-13	Hank Sorenson
16	3900	Other	U-Go Saris Label Maker	13-Jun-13	20-Jun-13	Clint Gosse
31						

Advanced date filters can be used to view information from a certain time period, such as last year, next quarter, or between two dates.

- Select the Data tab, then click the Filter command. A drop-down arrow will appear in the header cell for each column.
- Click the drop-down arrow for the column you want to filter.
- The Filter menu will appear. Hover the mouse over Date Filters, then select the desired date filter from the drop-down menu.

	A	B	C	D	E	F
1	ID #	Type	Equipment Detail	Checked Out	Checked In	Checked Out By
2	3000	Camera	Saris Lumina Digital Camera	12-May-13	12-May-13	Shannon Nguyen
3	3005	Camera	Saris Zoom Z-60 Digital Camera	27-Jul-13	27-Jul-13	Sela Shepard
4	1021	Laptop	15" EDI SmartPad L200-3	15-Sep-13	01-Oct-13	Sofie Ragnar
5	1022	Laptop	15" EDI SmartPad L200-3	14-Aug-13	16-Aug-13	Hank Sorenson
6	1023	Laptop	15" EDI SmartPad L200-3	08-Aug-13	15-Aug-13	Jennifer Weiss

4. The worksheet will be filtered by the selected date filter.



Using advanced number filters

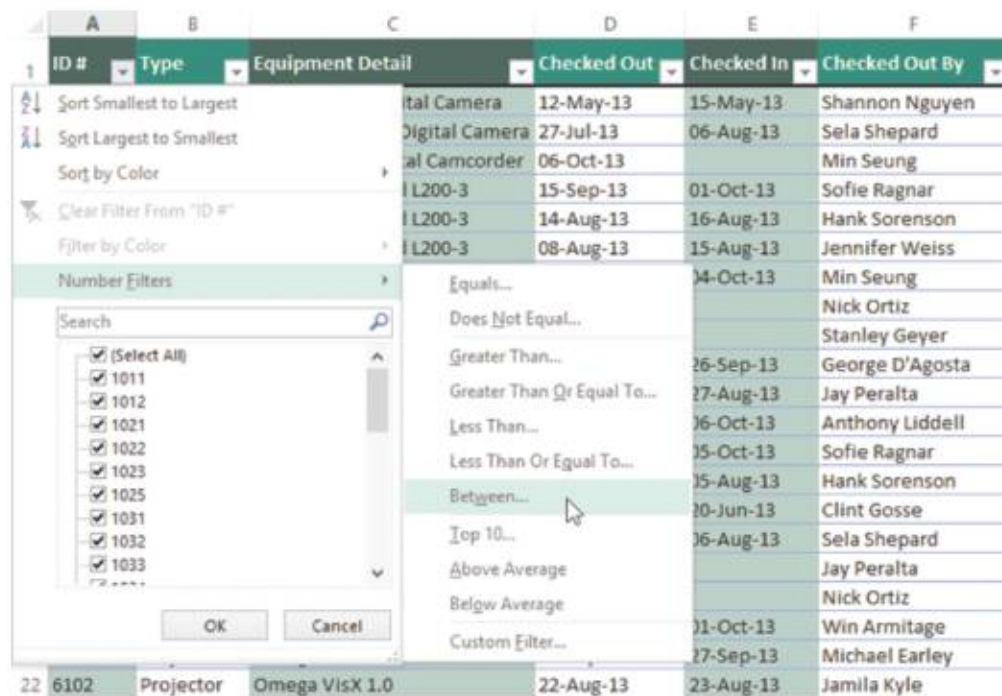
	A	B	C	D	E	F
1	ID #	Type	Equipment Detail	Checked Out	Checked In	Checked Out By
7	3070	Camera	Omega PixL Digital Camcorder	06-Oct-13		Min Seung
31						
32						

Advanced number filters allow you to manipulate numbered data in different ways. For example, in a worksheet of exam grades you could display the top and bottom numbers to view the highest and lowest scores.

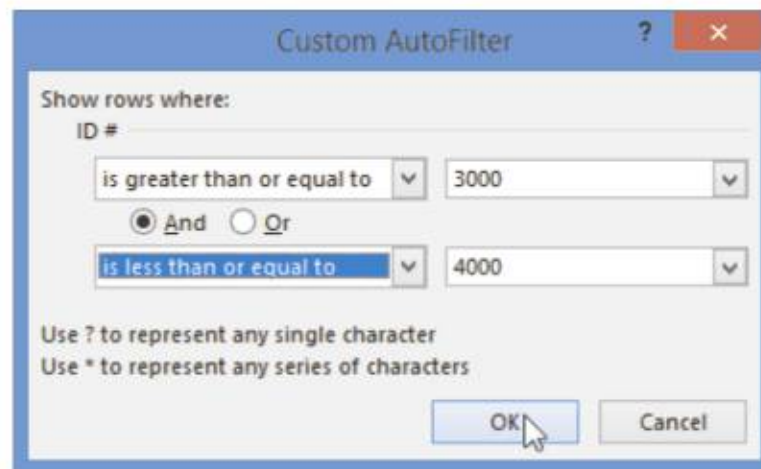
1. Select the Data tab on the Ribbon, then click the Filter command. A drop-down arrow will appear in the header cell for each column.
2. Click the drop-down arrow for the column you want to filter.
3. The Filter menu will appear. Hover the mouse over Number Filters, then

	A	B	C	D	E	F
1	ID #	Type	Equipment Detail	Checked Out	Checked In	Checked Out By
2	3000	Camera	Saris Lumina Digital Camera	12-May-13	15-May-13	Shannon Nguyen
3	3005	Camera	Saris Zoom Z-60 Digital Camera	27-Jul-13	06-Aug-13	Sela Shepard
4	1021	Laptop	15" EDI SmartPad L200-3	15-Sep-13	01-Oct-13	Sofie Ragnar
5	1022	Laptop	15" EDI SmartPad L200-3	14-Aug-13	16-Aug-13	Hank Sorenson
6	1023	Laptop	15" EDI SmartPad L200-3	08-Aug-13	15-Aug-13	Jennifer Weiss

select the desired number filter from the drop-down menu.



4. The Custom AutoFilter dialog box will appear. Enter the desired number(s) to the right of each filter, then click OK.



5. The data will be filtered by the selected number filter.

	A	B	C	D	E	F
1	ID #	Type	Equipment Detail	Checked Out	Checked In	Checked Out By
2	3000	Camera	Saris Lumina Digital Camera	12-May-13	15-May-13	Shannon Nguyen
3	3005	Camera	Saris Zoom Z-60 Digital Camera	27-Jul-13	06-Aug-13	Sela Shepard
4	3070	Camera	Omega PixL Digital Camcorder	06-Oct-13		Min Seung
15	3800	Other	U-Go Saris DigiCam Printer II	04-Aug-13	05-Aug-13	Hank Sorenson
16	3900	Other	U-Go Saris Label Maker	13-Jun-13	20-Jun-13	Clint Gosse
31						



Work Station



A Tick (3) the correct option.

(NEP) Analytical Skills

- In Excel it is organized in rows and columns.
a. File ☐ c. List ☐ b. Data ☐ d. Level ☐
- The arrangement of keeping the data either in ascending or descending.
a. Filtering ☐ c. Copy Level ☐ b. Sorting ☐ d. CustomList ☐
- Which statement is true regarding sorting?
a. You can even use more than one filter to further narrow your results. ☐
b. The data in the spreadsheet will be organized alphabetically. ☐
c. It gives you the ability to work with an enormous amount of data. ☐
d. Manipulate numbered data in different ways. ☐
- It can be applied in different ways to improve the performance of your worksheet.
a. Sorting ☐ c. Filters ☐ b. Copy Level ☐ d. CustomList ☐
- It will add a level by duplicating the one you have selected and allowing you to modify the sorting criteria.
a. Sorting ☐ c. Filters ☐ b. Copy Level ☐ d. CustomList ☐

B Fill in the blanks.

- _____ giving you the ability to sort multiple levels such as department first, then birthdate to group birthdates by department.
- _____ allows you to identify which columns to sort by and when, giving you more control over the organization of your data.
- To instantly clear all filters from your worksheet, click the _____ on the Data tab.
- _____ for data is a convenient alternative to checking or unchecking data from the list.
- Data can be filter either in two ways _____ or _____.

C Write T for True or F for False.

- Filtering options can also be found on the Home tab, condensed into the Sort & Filter command. ☐
- You can use a Custom List to identify your own sorting order, such as days of the week. ☐
- In Excel you can filter text, dates, and numbers. ☐
- You can even use more than one filter to further narrow your results. ☐

5. Sorting options can also be found on the Home tab, condensed into the Sort & Filter command.



D Match the followings.

- | | |
|----------------------------|--|
| 1. Sorting multiple levels | a. Customize the order of your data. |
| 2. Number filters | b. Improve the performance of your worksheet. |
| 3. Date filters | c. To identify your own sorting order |
| 4. Sorting | d. Display more specific information |
| 5. Filters | e. View information from a certain time period |
| 6. Custom List | f. Manipulate numbered data in different ways. |
| 7. Text filters | g. Identify which columns to sort by and when. |

E Answer the following questions.

1. What do you mean by sorting?
2. What is the use of Filtering data?
3. What do you mean by Sorting multiple levels?
4. How sorting is useful in arranging the data.
5. Differentiate between Auto filter and Custom filter.

LAB ACTIVITY

A Create a worksheet given below and perform the following in Excel:

(NEP) Experimental Learning

- (a) Display the name of Students who participated in the Game 'Cricket'.
- (b) Display only the list of Girls who participated in volley ball.
- (c) Display the name of class XII students.

S.No	Student Name	Class	Game	Gender
1.		XII	Cricket	Boy
2.		XI	Football	Boy
3.		XI	Basketball	Girl
4.		XII	Cricket	Boy

5.	XII	Cricket	Boy
6.	XI	TT	Girl
7.	XI	Football	Boy
8.	XI	TT	Girl
9.	XI	Volley ball	Girl
10.	XII	Cricket	Boy
11.	XI	TT	Girl
12.	XI	Basketball	Girl
13.	XII	Cricket	Boy
14.	XI	Football	Boy
15.	XII	Cricket	Boy
16.	XI	Volleyball	Girl
17.	XII	Volleyball	Girl
18.	XII	Cricket	Boy

Animation in PowerPoint

Lesson at a Glance

Animations

Types of Animations

Working with Animations

Animations

An animation effect is a special visual or sound effect added to a text or an object on a slide. It is possible to animate the text and the other objects using the buttons on the Animation tab button or you can have the bullet points appearing one at a time. It makes our presentation more attractive and appealing to catch the audience attention on specific topic of presentation.

PowerPoint provides four types of animations :

Types of Animations

There are several animation effects you can choose from, and they are organized into four types :

- **Entrance** : These effect control how the object enters into a slide during a slide show. For example, with the Bounce animation the object will "fall" onto the slide and then bounce several times.



- **Emphasis** : These effects determine how the objects draw attention to themselves. For example, you can set an object to spin when you click the

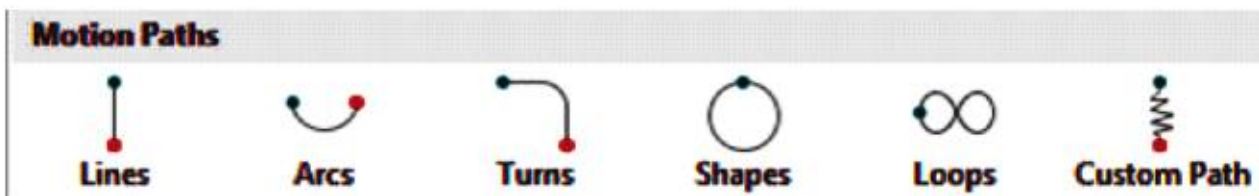


mouse, while already on the slide during the slide show. They are often triggered by a mouse click.

- **Exit** : These control how the object exits the slide. For example, with the Fade animation the object will simply fade away.



- **Motion Paths** : These are similar to Emphasis effects, except the object moves within the slide along a predetermined path, like a circle.



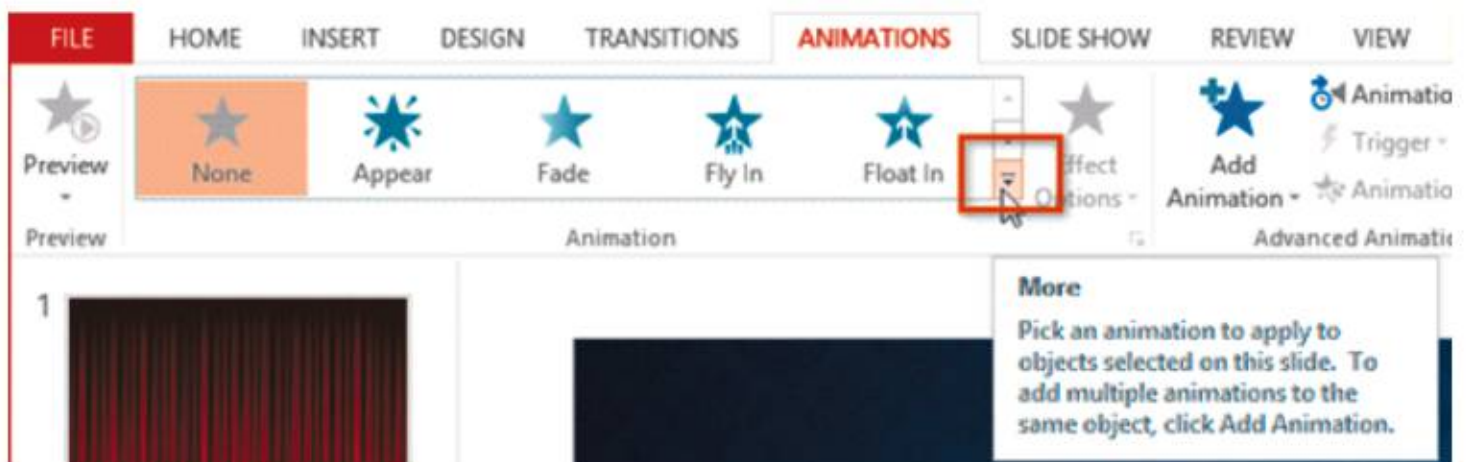
Must Know!

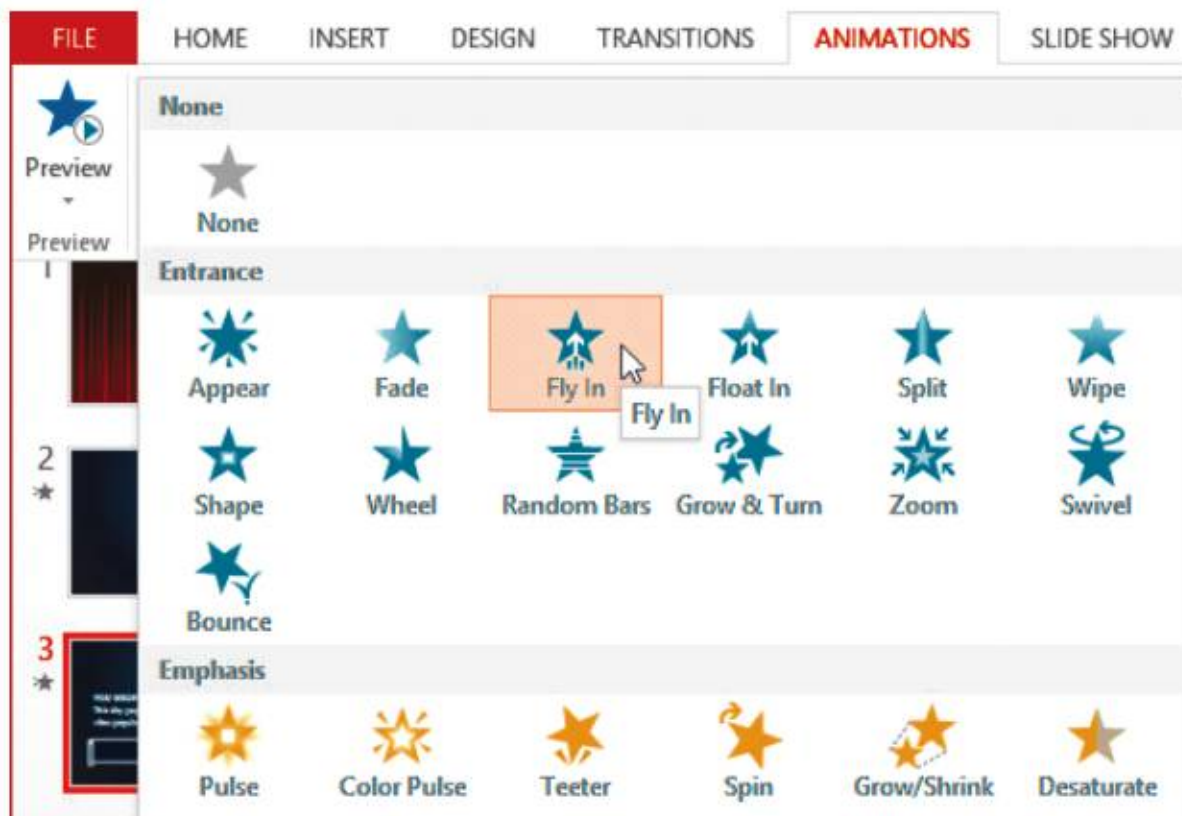


In PowerPoint, you can animate text and objects such as clip art, shapes, and pictures.

To apply an animation to an object :

1. Select the object you want to animate.
2. On the Animations tab, click the More drop-down arrow in the Animation group.
3. A drop-down menu of animation effects will appear. Select the desired effect.





4. The effect will apply to the object. The object will have a small number next to it to show that it has an animation. In the Slide pane, a star symbol also will appear next to the slide.



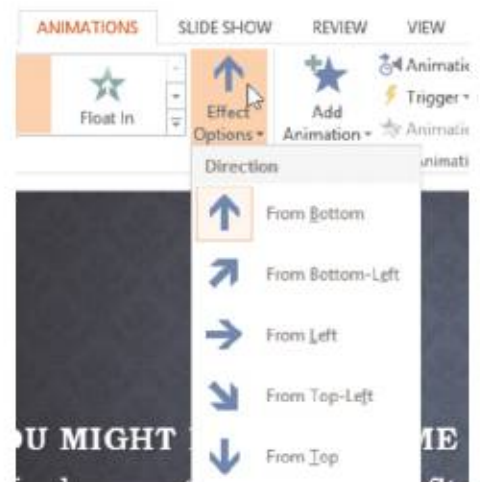
Must Know!



Animation on the slide can be used to draw the audience's attention to specific content or to make the slide easier to read.

Effect options

Some effects will have options you can change. For example, with the Fly In effect you can control which direction the object comes from. These options can be accessed from the Effect Options command in the Animation group.

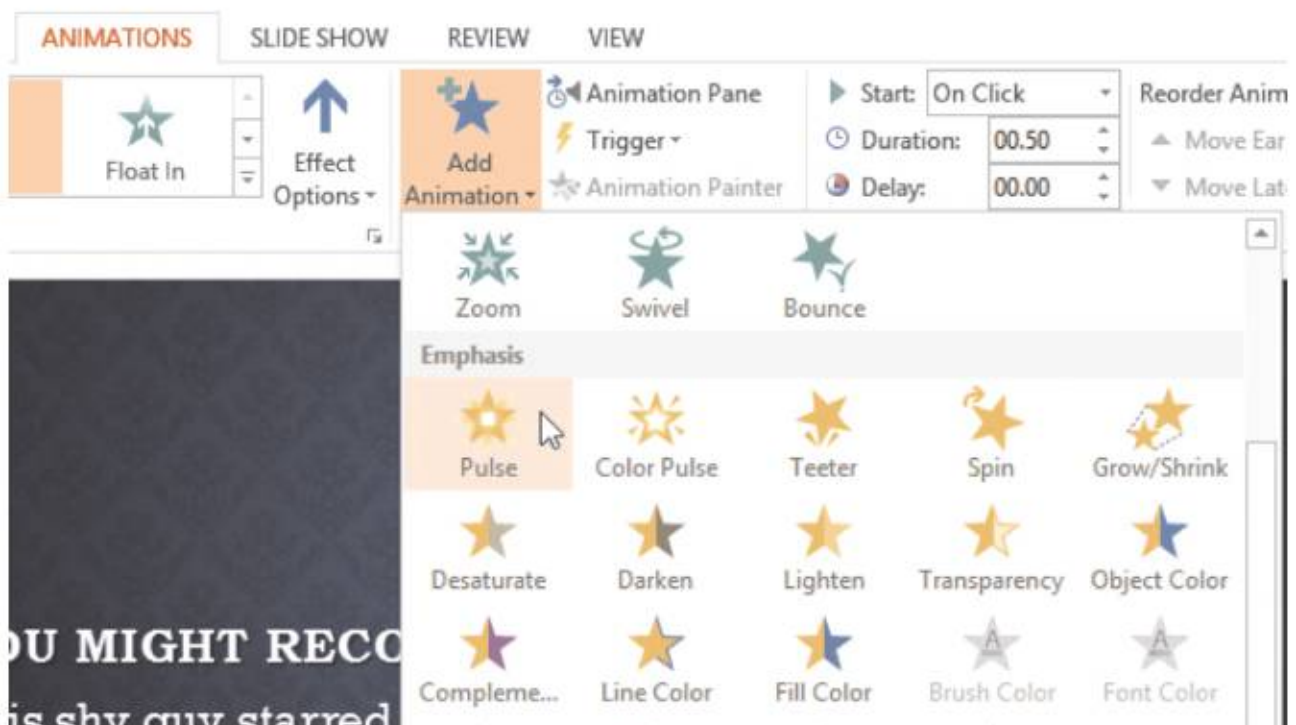


Working with Animations

To add multiple animations to an object:

If you select a new animation from the menu in the Animation group, it will replace the object's current animation. However, you'll sometimes want to place more than one animation on an object, like an Entrance and an Exit effect. To do this, you'll need to use the Add Animation command, which will allow you to keep your current animations while adding new ones.

1. Select an object.
2. Click the Animations tab.
3. In the Advanced Animation group, click the Add Animation command to view the available animations.
4. Select the desired animation effect.

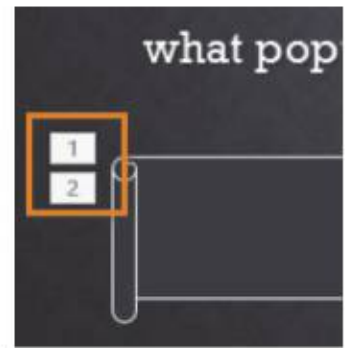


5. If the object has more than one effect, it will have a different number for each effect. The numbers indicate the order in which the effects will occur.

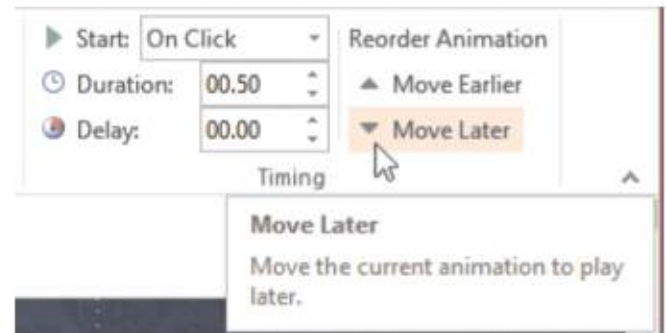


To reorder the animations :

1. Select the number of the effect you want to change.



2. From the Animations tab, click the Move Earlier or Move Later commands to change the ordering.



To preview animations :

Any animation effects you have applied

will show up when you play the slide show. However, you can also quickly preview the animations for the current slide without viewing the slide show.



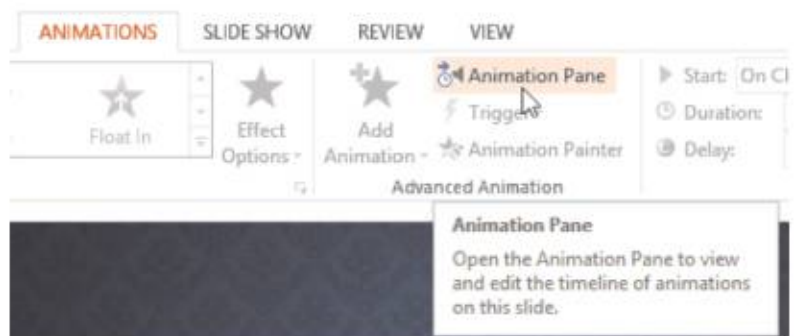
1. Navigate to the slide you want to preview.
2. From the Animations tab, click the Preview command. The animations for the current slide will play.

The Animation Pane

The Animation Pane allows you to view and manage all of the effects that are on the current slide. You can modify and reorder effects directly from the Animation Pane, which is especially useful when you have several effects.

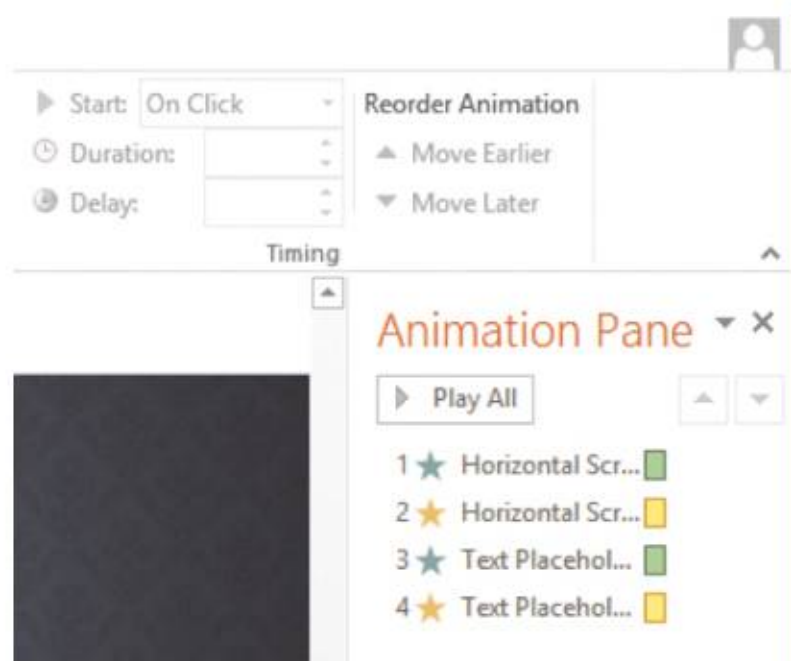
To open the Animation Pane :

1. From the Animations tab, click the Animation Pane command.
2. The Animation Pane will open on the right side of the window. It will show

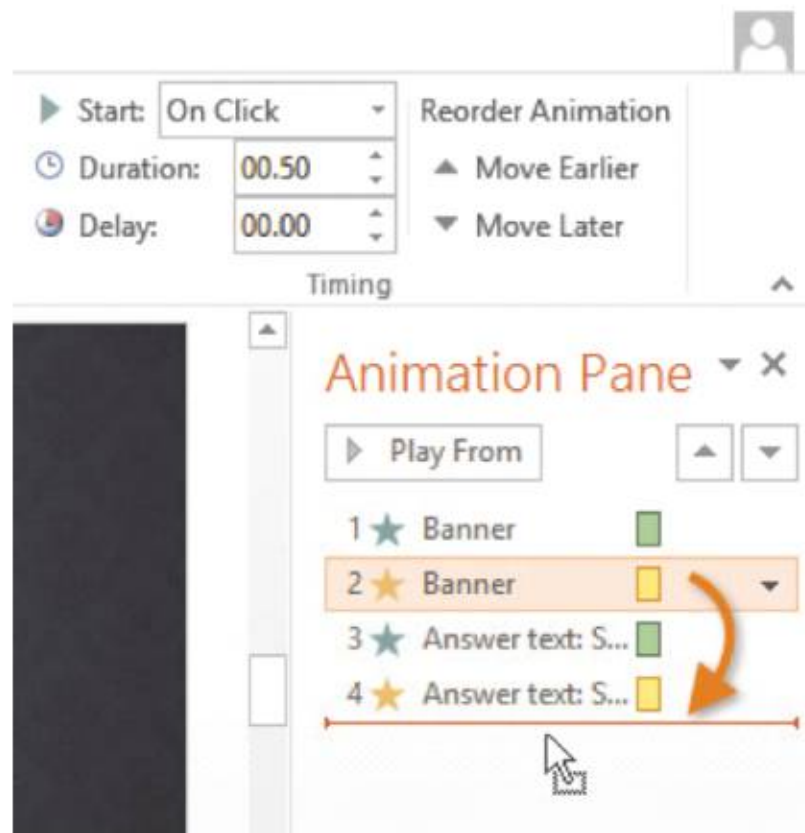


all of the effects for the current slide in the order in which they will appear.

To reorder effects from the Animation Pane :

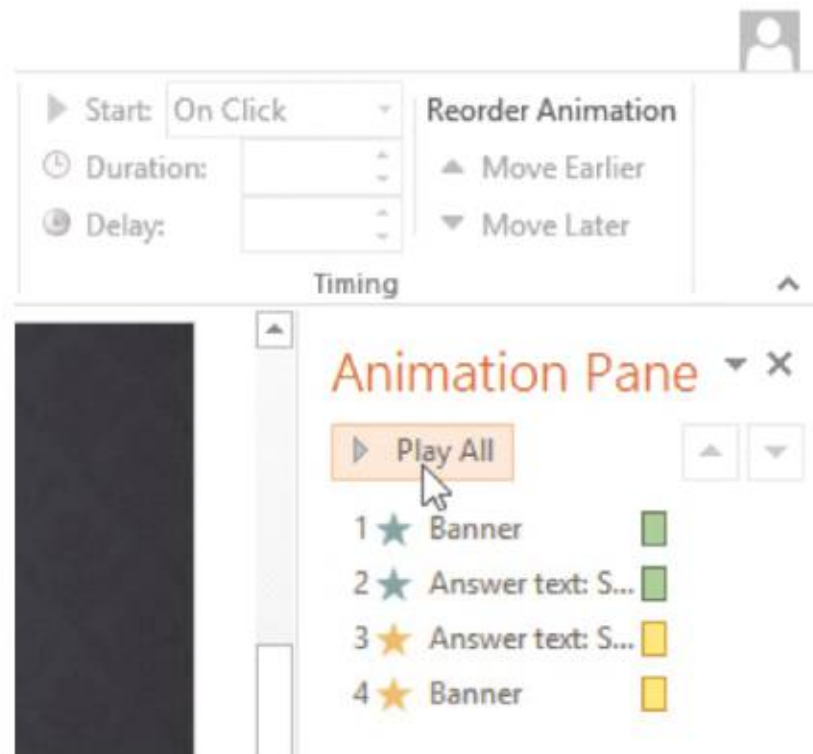


1. On the Animation Pane, click and drag an effect up or down.
2. The effects will reorder themselves.



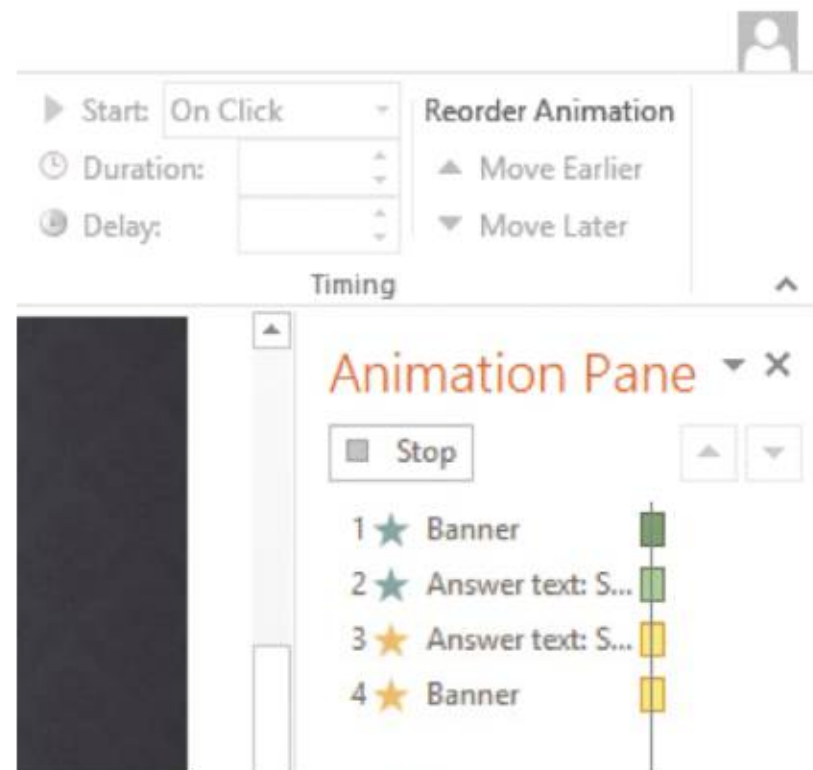
To preview effects from the Animation Pane :

1. From the Animation Pane, click the Play button.
2. The effects for the current slide will play. On the right side of the Animation

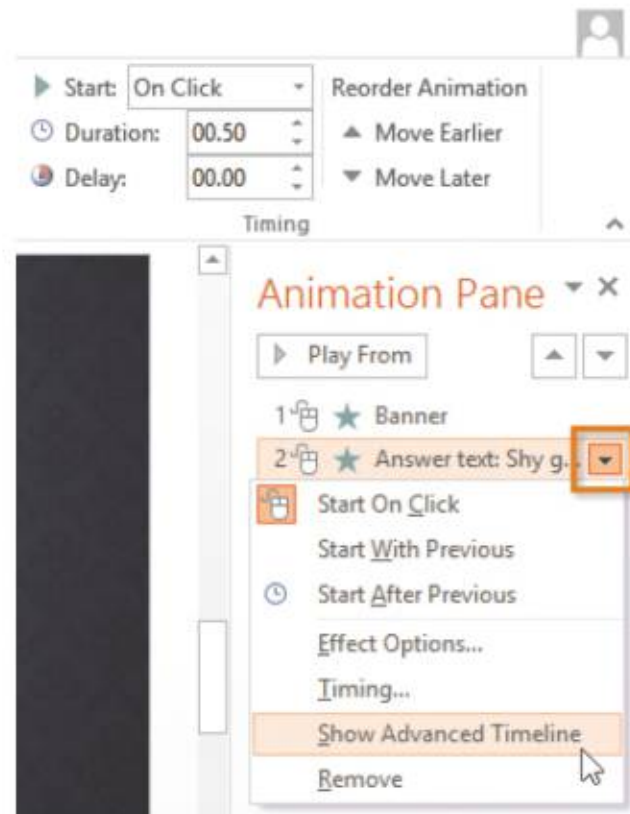


Pane, you will be able to see a timeline that shows the progress through each effect.

If the timeline is not visible, click the drop-down arrow for an effect, then select



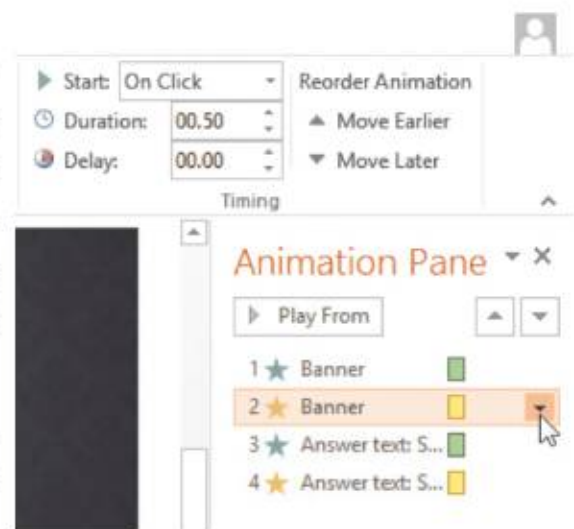
Show Advanced Timeline.



To change an effect's start option :

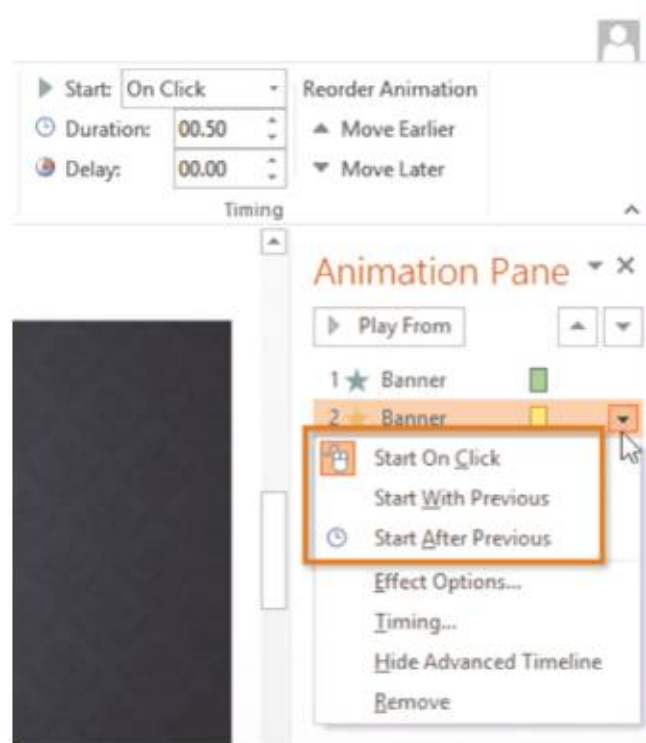
By default, an effect starts playing when you click the mouse during a slide show. If you have multiple effects, you will need to click multiple times to start each effect individually. However, by changing the start option for each effect, you can have effects that automatically play at the same time or one after the other.

1. From the Animation Pane, select an effect. A drop-down arrow will appear next to the effect.
2. Click the drop-down arrow. You will see three start options :
 - Start On Click will cause the animation to start when you click the mouse.
 - Start With Previous will cause the animation to begin with the previous animation; if this is the first object it will begin as soon as you reach the



slide during the slide show.

- Start After Previous will cause the animation to begin after the previous animation ends.



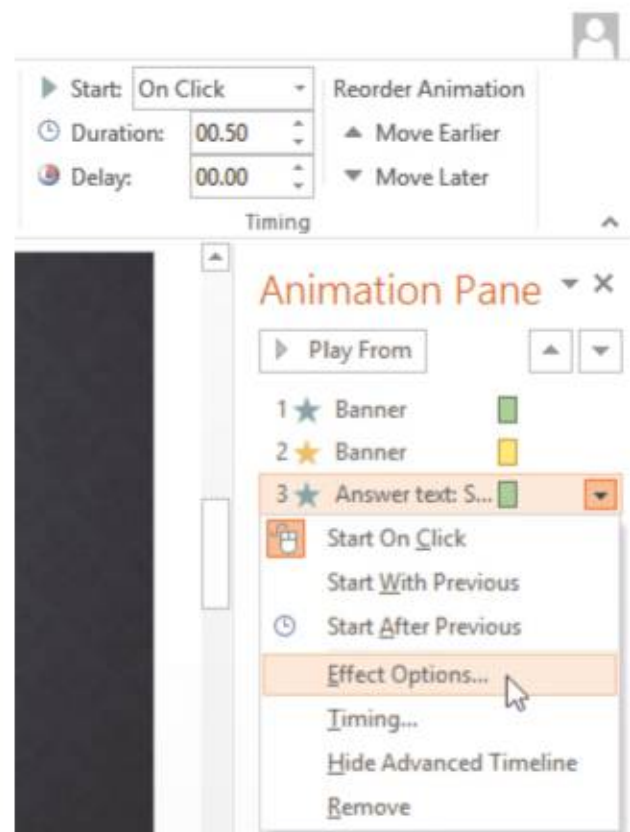
3. Select the desired start option.

The Effect Options Dialog Box

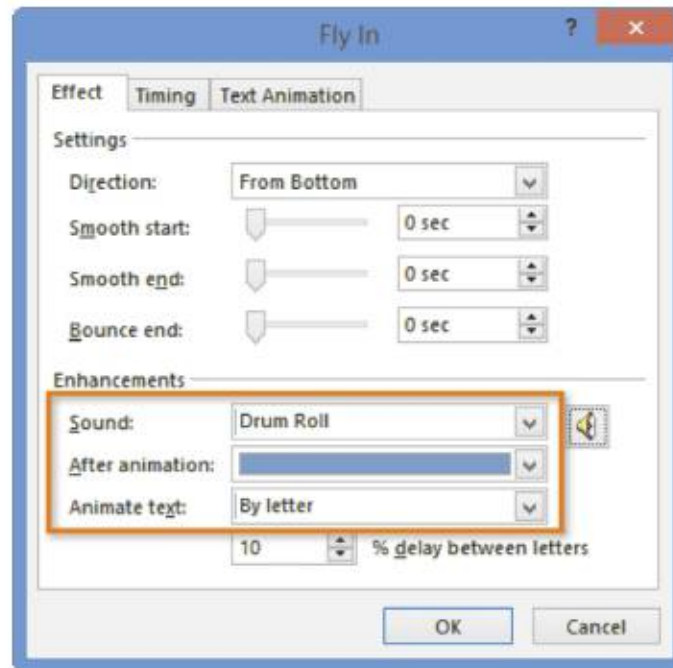
From the Animation Pane, you can access the Effect Options dialog box, which contains more advanced options you can use to fine tune your animations.

To open the Effect Options dialog box :

1. From the Animation Pane, select an effect. A drop-down arrow will appear next to the effect.
2. Click the drop-down arrow, then select Effect Options.
3. The Effect Options dialog box will appear. From here, you can add various enhancements to the effect :
 - **Sound** : Adds a sound effect to the animation.



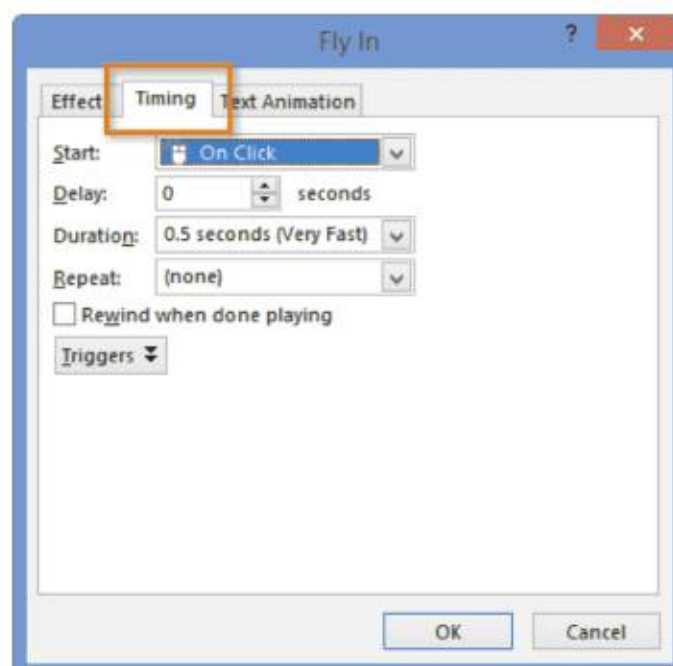
- **After animation** : Changes the color or hides the object after the animation is over.
- **Animate text** : Allows you to animate text all at once, one word at a time, or one letter at a time.



Some effects have additional options you can change. These will vary depending on which effect you have selected.

To change the effect timing :

1. From the Effect Options dialog box, select the Timing tab.



2. From here, you can add a delay before the effect starts, change the duration of the effect, and control whether or not the effect repeats.



A Tick(3) the correct option.

(NEP) Analytical Skills

1. These animation control how the object enters the slide.
 - a. Bounce ☐
 - b. Entrance ☐
 - c. Fly Out ☐
 - d. Fly In ☐
2. These animations occur while the object is on the slide, often triggered by a mouse click.
 - a. Reorder ☐
 - b. Emphasis ☐
 - c. Timeline ☐
 - d. MotionPaths ☐
3. These animation control how the object exits the slide.
 - a. Enter ☐
 - b. Exit ☐
 - c. Spin ☐
 - d. Fade ☐
4. These are similar to Emphasis effects, except the object moves within the slide along a predetermined path, like a circle.
 - a. Motion File ☐
 - b. Motion Paths ☐
 - c. Motion Print ☐
 - d. Motion Slide ☐
5. Allows you to animate text all at once, one word at a time, or one letter at a time.
 - a. Animate word ☐
 - b. Animate letter ☐
 - c. Animate text ☐
 - d. Animate file ☐

B Fill in the blanks.

1. _____ can be applied to any object on the slide and the motions can the automated, timed or trigger.
2. _____ will cause the animation to start when you click the mouse.
3. _____ will cause the animation to begin with the previous animation.

4. _____ will cause the animation to begin after the previous animation ends.
5. _____ changes the color or hides the object after the animation is over.

C Write T for True or F for False.

1. In power-point you can set an object to spin when you click the mouse. ☐
2. Any animation effects you have applied will show up when you play the slide show. ☐
3. The Slide Pane allows you to view and manage all of the effects that are on the current slide. ☐
4. By default, an effect starts playing when you click the mouse during a printing. ☐
5. If you have multiple effects, you will need to click multiple times to start each effect individually. ☐

D Answer the following questions.

1. Explain the four types of animations.
2. How to apply an animation to an object?
3. How to reorder effects from the Animation Pane?
4. How to show Advanced Timeline?
5. How to change an effect's start option?
6. How to open the Effect Options dialog box?

LAB ACTIVITY

A. In these exercises we will get creative with some of the more advanced animation features.

(NEP) Experimental Skills

- (1) Open the file called Custom Animation.pptx
- (2) Click on the man to select him.



- (3) Click the Animationstab on the Ribbon Bar.
- (4) Click  Animation Pane: to display the AnimationPane on the right side of your window.
- (5) Click Add Animation.
- (6) Scroll down the list of animations until you reach the MotionPaths category.
- (7) Click Custom Path. We will create a path that will make the man move to the top of the hill.
- (8) Draw a custom path going from the man to the top of the hill like the example below.
Double click when you reach the end point of the line.



- (9) Click the Play button at the top of the animation pane to see your animation again.
- (10) Click Add Animation.
- (11) Choose Grow/Shrink from the Emphasis category.
- (12) Click the small arrow next to the animation in the Animation Pane.
- (13) From the list which appears select Effect Options. The Grow/Shrink options will appear.
- (14) Change the Size option to Smaller. If you decide later that it's not quite the right size, you can open these options again and enter a Custom % for the size.
- (15) Click OK and you will see a preview animation showing the man growing smaller.
- (16) Click the Play button at the top of the Animation Pane to preview your animation.
- (17) Click the first animation in the list.
- (18) Change the Start option from On Click to With Previous.
- (19) Make the same change with the second animation (the Grow/Shrink animation) so that it plays at the same time as the first animation.
- (20) Press [F5] to preview the slideshow. The man will begin moving and shrinking as soon as the slideshow opens.
- (21) Press [Esc] to exit the slideshow.

More About the Flash

Lesson at a Glance

Creating Movie Clips

Using Masks

Animating Still Photos

Creating Movie Clips

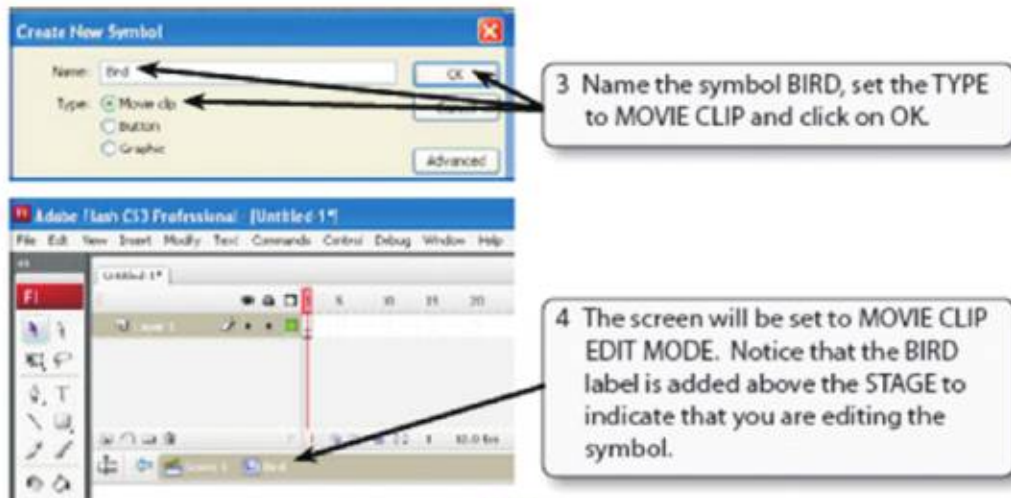
A movie clip symbol has its own independent timeline that loops over and over. A movie clip can be used as a container for an animation. It only play when you test your movie. Symbols can be nested within other symbols. Symbols can have multiple layer within their timeline. Movie clips can be created with motion tweens or with frame-by-frame animations.

Creating a Movie Clip

Movie clips are created as a symbol and then used as required. A simple movie clip of a bird flapping its wings will be created.

A. Setting a Movie Clip Symbol

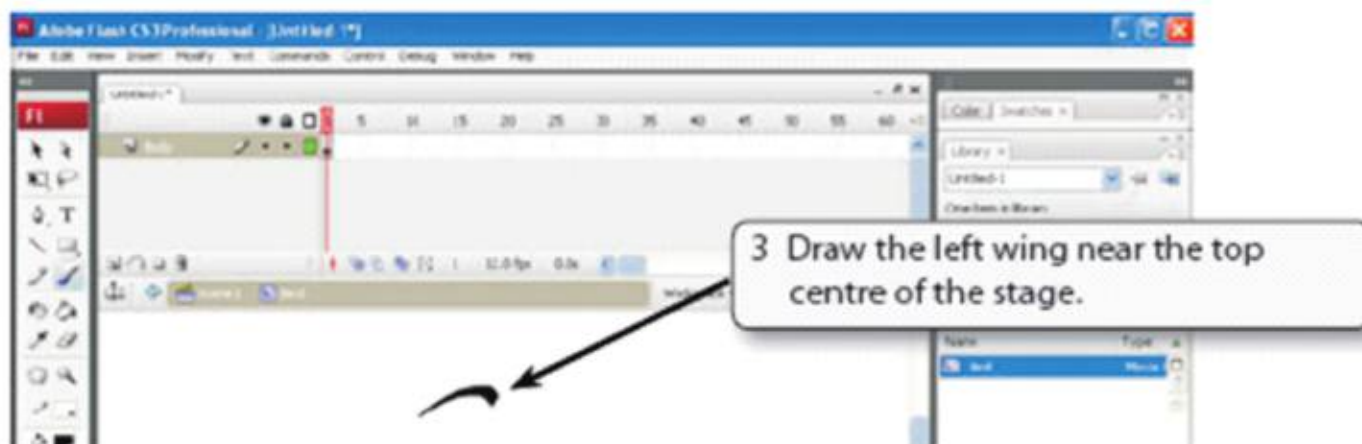
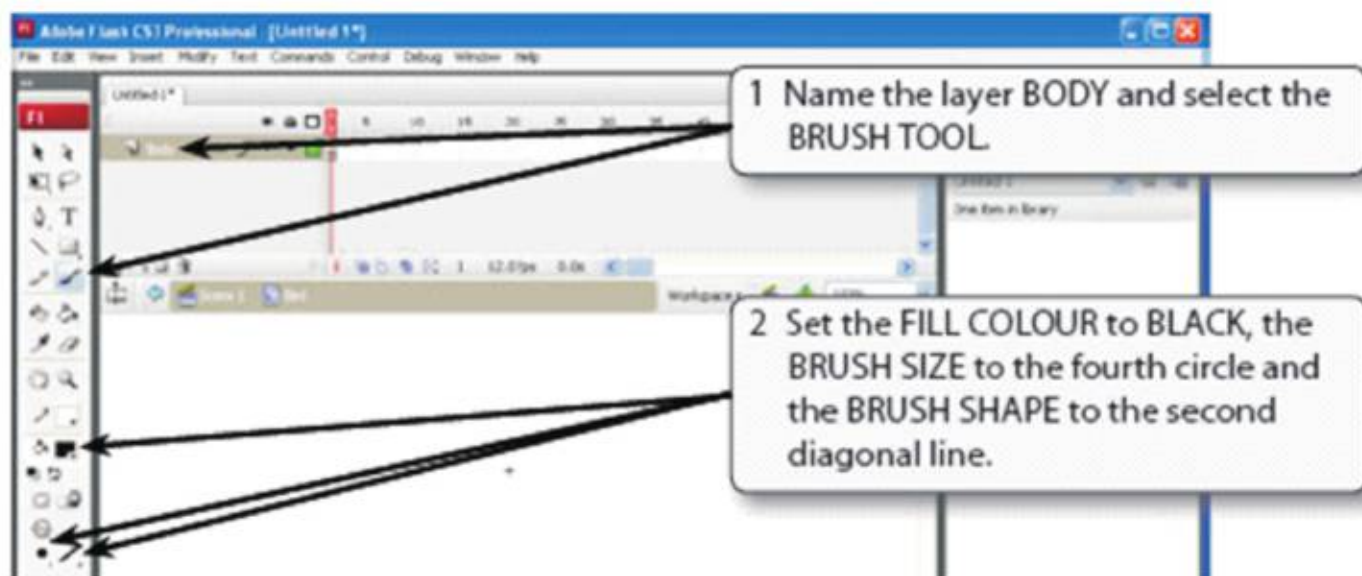
1. Load Flash and create a new Flash File, or close the current file and create a new Flash File.
2. Display the INSERT menu and select NEW SYMBOL.



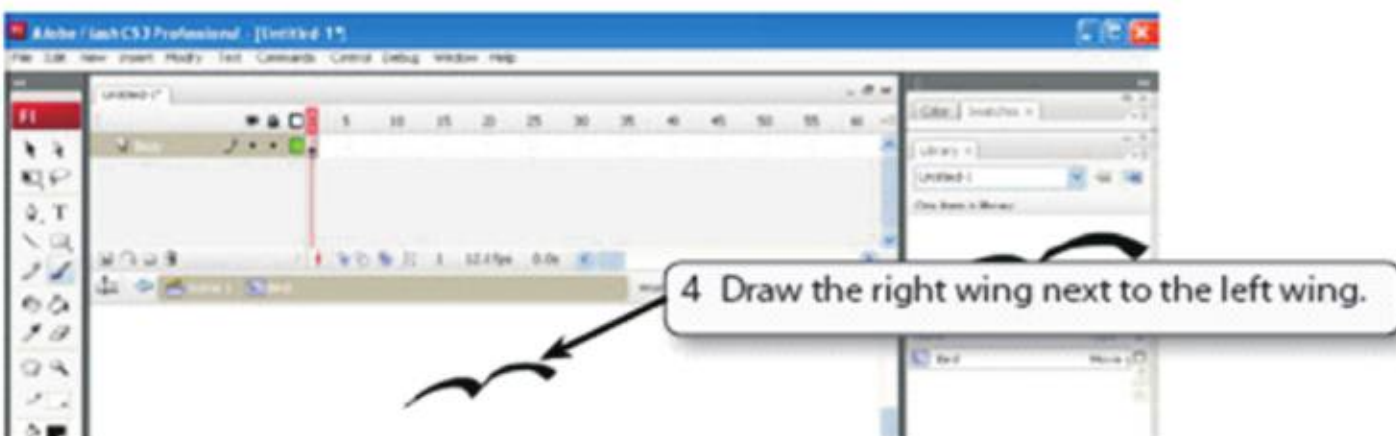
3 Name the symbol BIRD, set the TYPE to MOVIE CLIP and click on OK.

4 The screen will be set to MOVIE CLIP EDIT MODE. Notice that the BIRD label is added above the STAGE to indicate that you are editing the symbol.

B. Drawing the First Position of the Bird

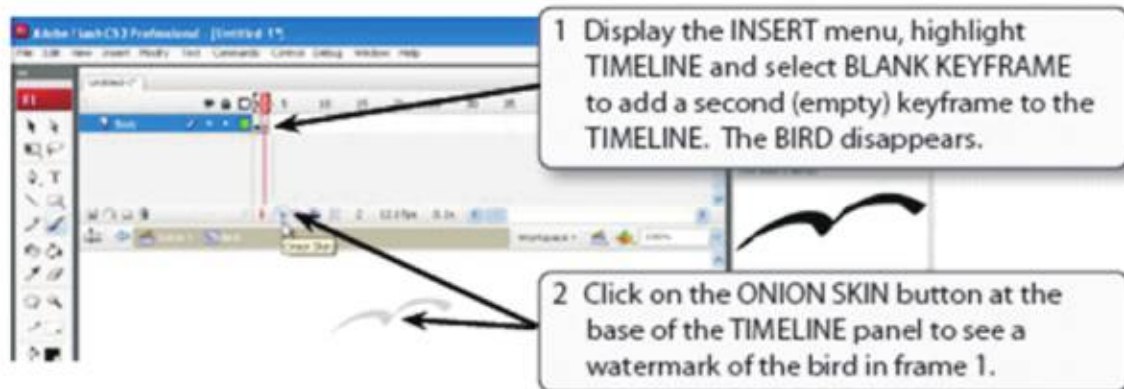


NOTE : It might take a few goes to get the wing looking right. Use CTRL+Z to undo the wing until you produce the shape you want.



NOTE : The LIBRARY panel can be used as a zoom of the symbol as you are creating it. Click on the BIRD symbol to see it. If the LIBRARY panel is not open press CTRL+L to open it.

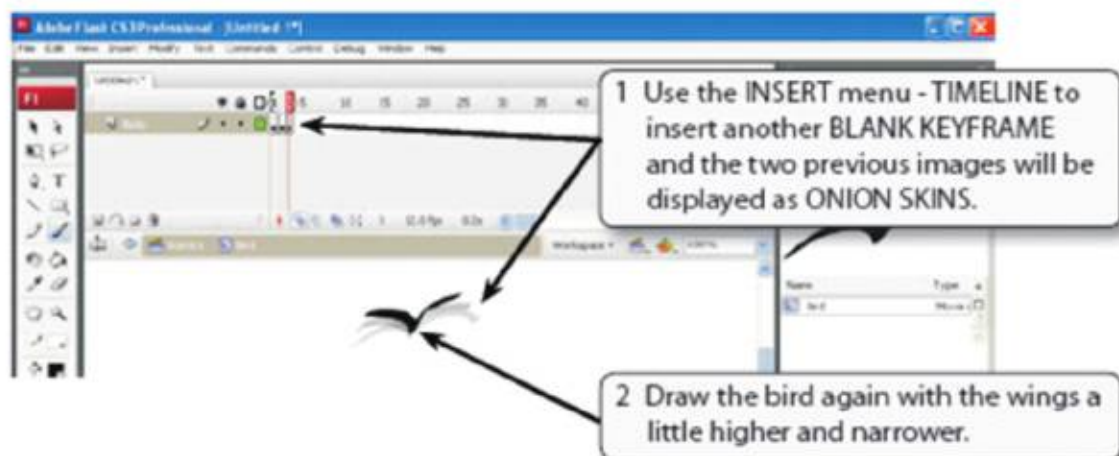
C. Drawing the Second Position of the Bird



- NOTE :**
- (i) The BLANK KEYFRAME will allow you to alter the shape of the bird.
 - (ii) The ONION SKIN image is just a watermark on the screen. It cannot be altered.



D. Setting the Third Position of the Bird



E. Setting the Fourth Position of the Bird

1 Use the INSERT menu - TIMELINE to insert another BLANK KEYFRAME and the three previous images will be displayed as ONION SKINS.

2 Draw the bird again with the wings below and wider than the first image.

3 Run the PLAYHEAD over the frames to see the animation.

4 Click on the SCENE 1 heading at the top of the stage to return the screen to normal.

5. The bird will disappear. It is in the LIBRARY panel.

6. Save the page in your STORAGE folder as: Birds

NOTE : We could have used a MOTION TWEEN or a SHAPE TWEEN as the animation. Movie clips do not have to be KEYFRAME animations.

Using Masks

Some interesting and effective animations can be created using a layer to cover (or mask) part of another layer. The portion of layer being covered can only be seen through whatever shape is drawn in the mask layer. The process is called MASKING.

Creating a Mask over a Background

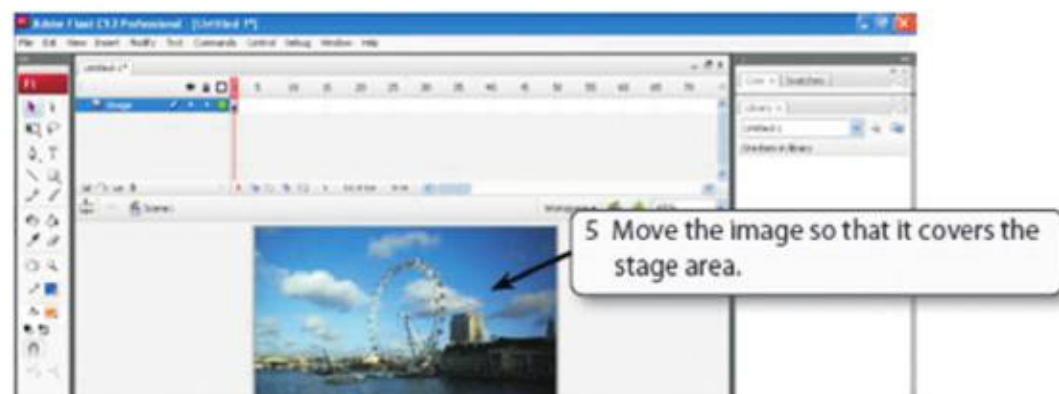
For this first example a mask will be used to display only part of a picture of London.

A. Creating the Background Layer

1. Load Flash and create a new Flash File, or close the current file and create a new Flash File.

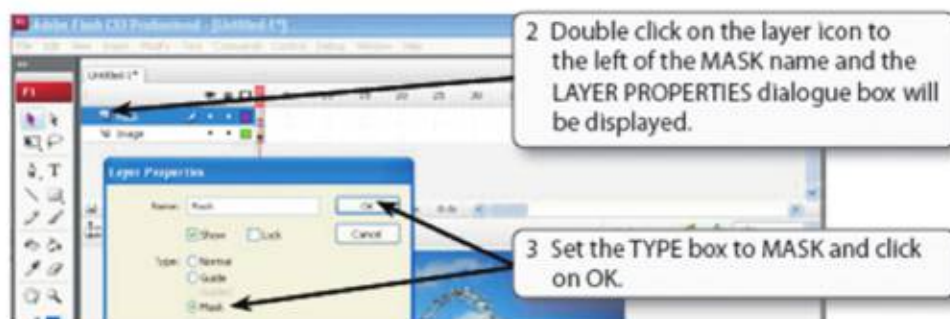


3. Display the FILE menu, highlight IMPORT and select IMPORT TO STAGE.
4. Access the FLASHCS3 SUPPORT FILES, open the CHAPTER 14 folder and import the LONDON image.

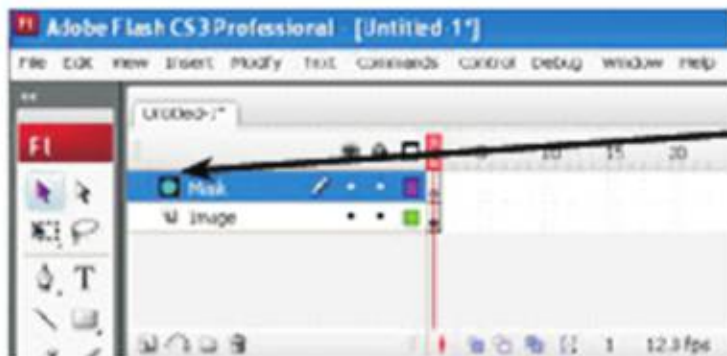


B. Inserting the Mask Layer

In order for a mask to operate a layer must be converted to a MASK layer.



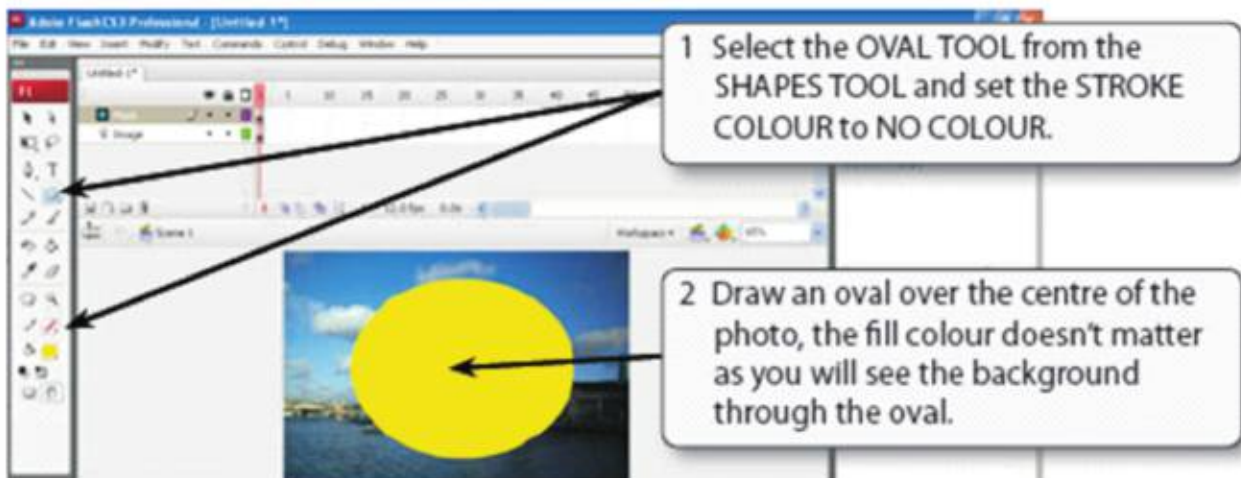
NOTE : You can also RIGHT+CLICK or CONTROL+CLICK on the layer name and select MASK from the SHORTCUTS menu.



4 The MASK layer has a different icon.

C. Creating the Mask

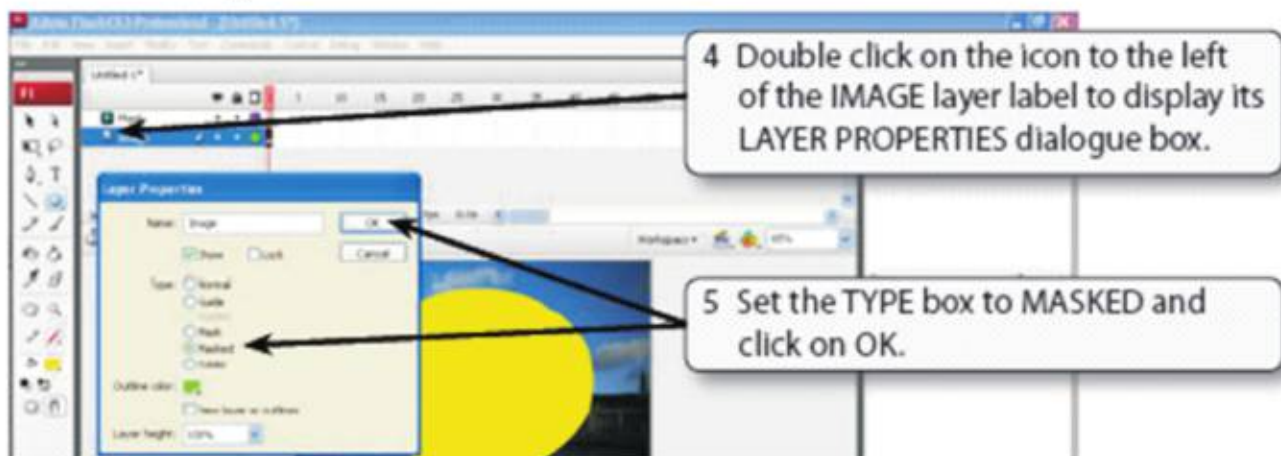
A shape now needs to be drawn in the MASK layer. The background will only be seen through this shape. You can use the shape tools such as the OVAL and RECTANGLE TOOLS to be the mask shape, or use the BRUSH TOOL to draw a more complicated shape.



1 Select the OVAL TOOL from the SHAPES TOOL and set the STROKE COLOUR to NO COLOUR.

2 Draw an oval over the centre of the photo, the fill colour doesn't matter as you will see the background through the oval.

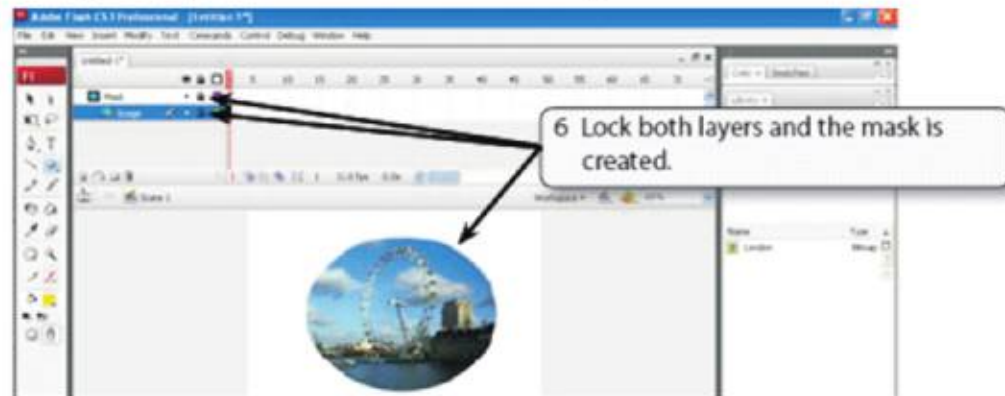
3. The program needs to be told which layer is to be masked. In this case it is the IMAGE layer.



4 Double click on the icon to the left of the IMAGE layer label to display its LAYER PROPERTIES dialogue box.

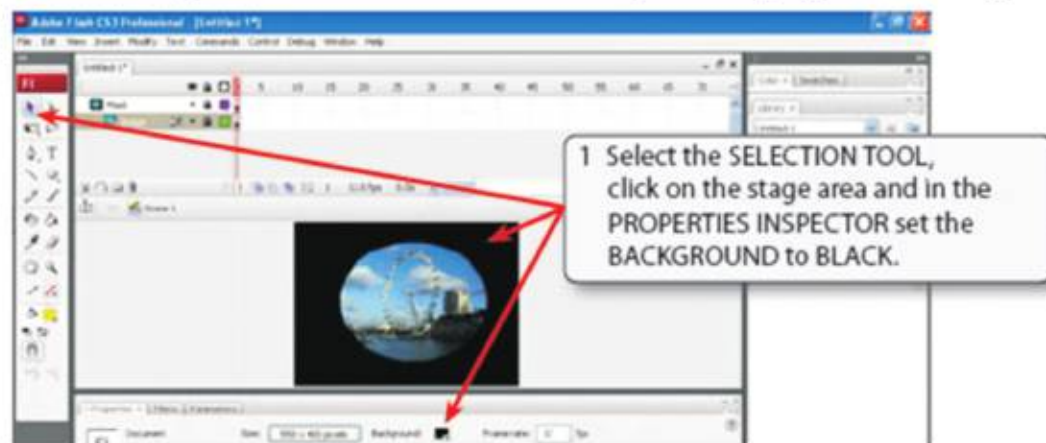
5 Set the TYPE box to MASKED and click on OK.

- NOTE :** (i) The IMAGE layer is indented under the MASK layer to indicate that it is masked by the MASK layer.
- (ii) You can drag a layer to the lower edge of a MASK layer to quickly set it to a MASKED layer.



D. Background Colours

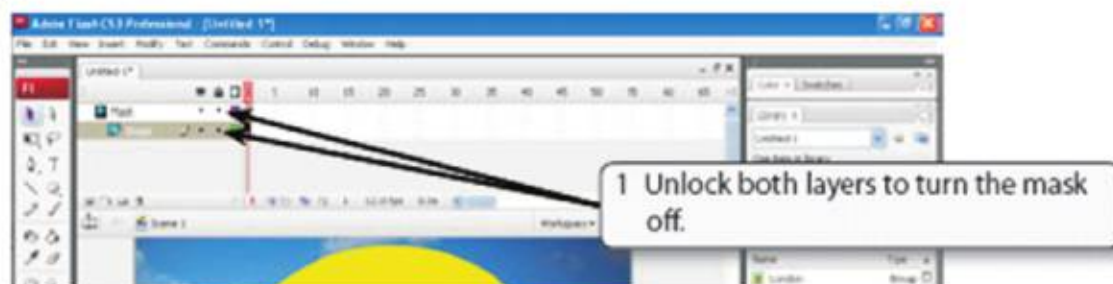
Some interesting effects can be created by changing the background colour.

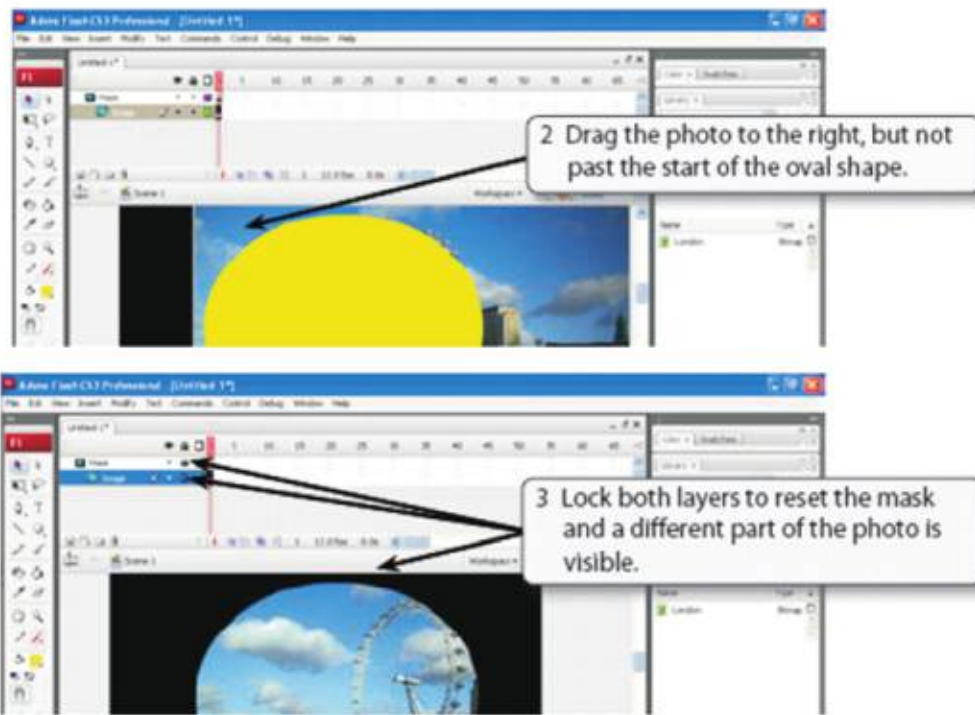


2. Set the ZOOM to 100% to see the effect more clearly.
3. Try some other background colours.

E. Editing the Mask

If you need to edit a mask you simply unlock the layers, make the change then re-lock the layers. Let's say we want to change the part of the background that is seen. You can either move the background or the oval.





Animating Still Photos

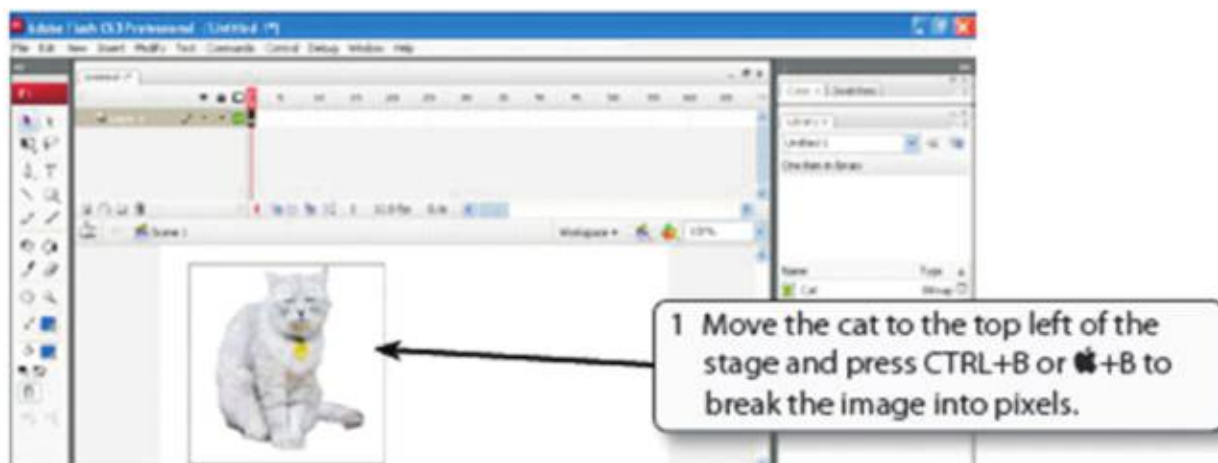
Life can be added to still photos by using Flash to animate parts of them. To illustrate this we will use a photo of a cat and make its head shake and tail wag.

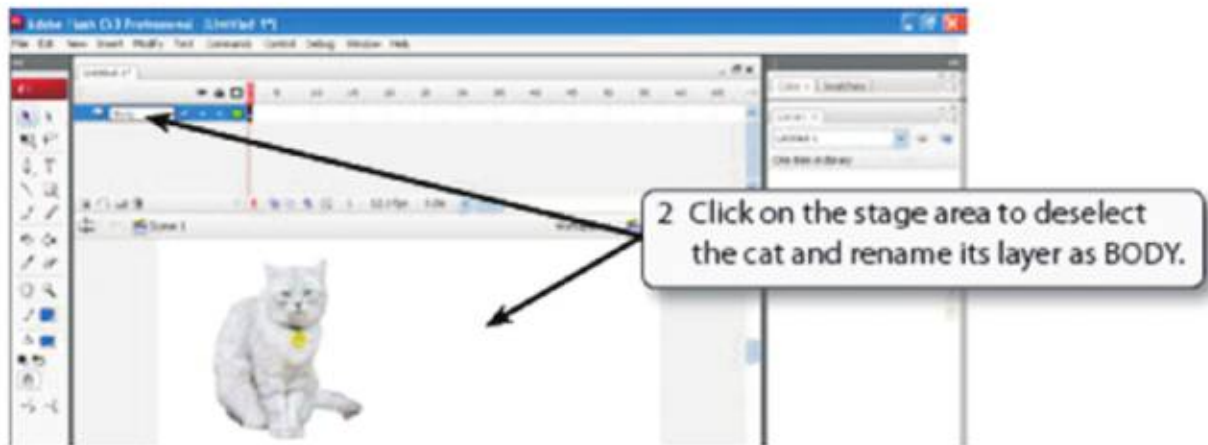
Loading the Prepared File

1. Load Flash or close the current file, and create a new Flash File.
2. Display the FILE menu, highlight IMPORT and select IMPORT TO STAGE.
3. Access the FLASHCS3 SUPPORT FILES, open the CHAPTER 19 folder and load the CAT image.

Breaking the Photo Apart

In order to edit the photo it must be broken apart into individual pixels.

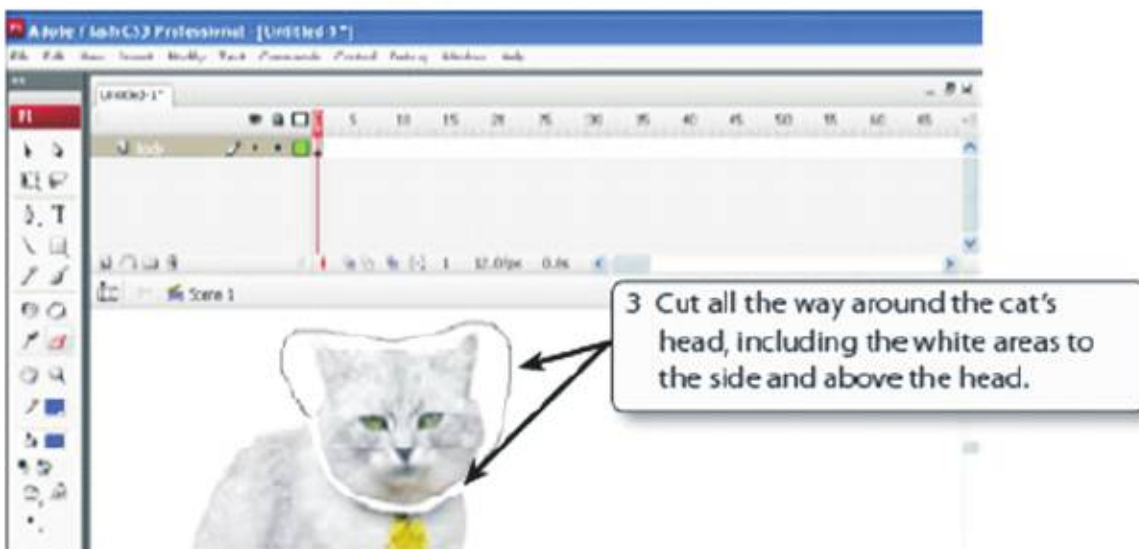
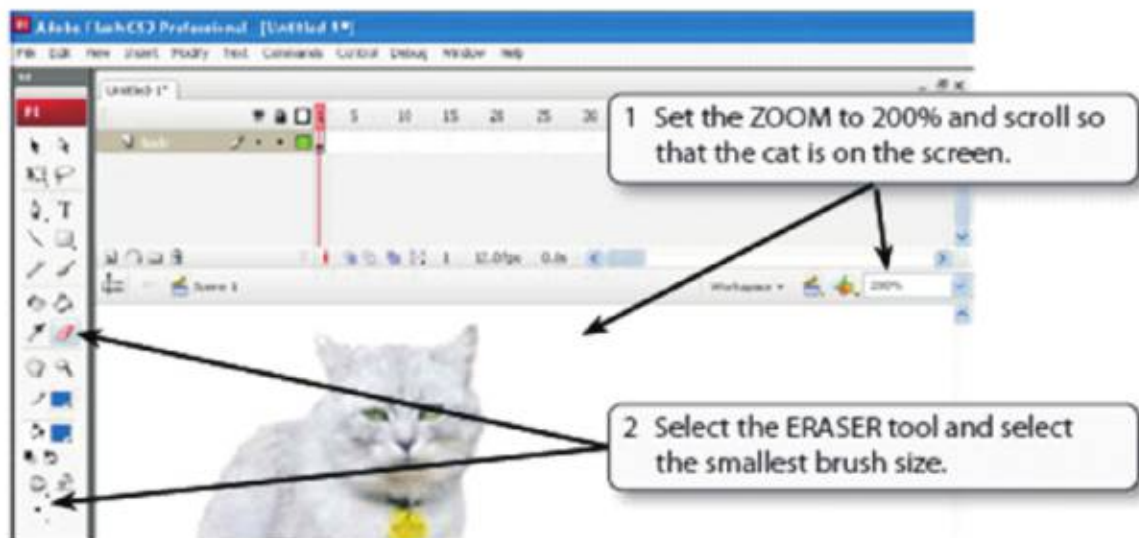


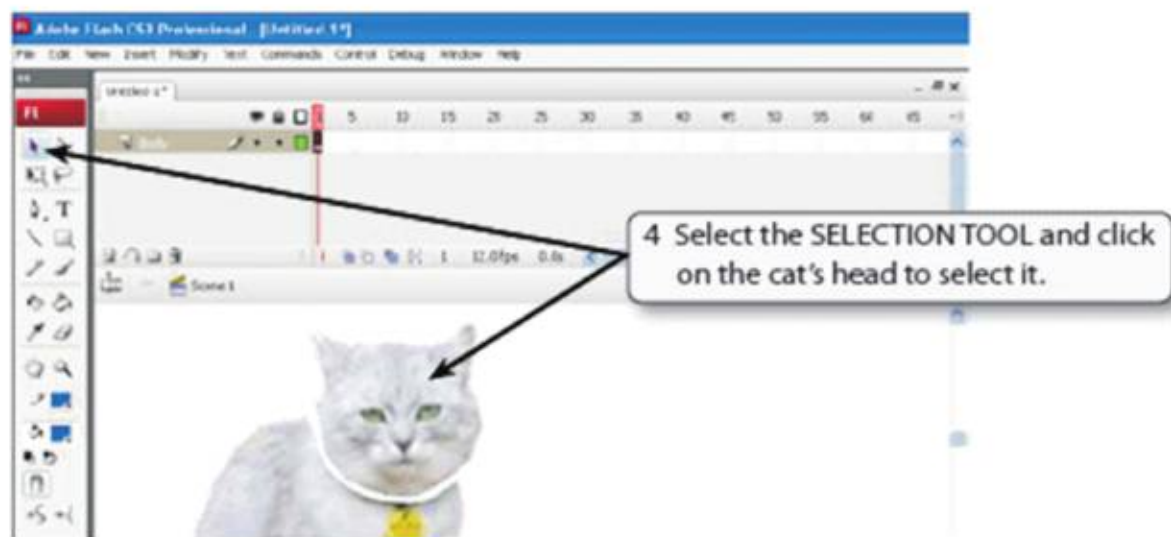


Separating the Head

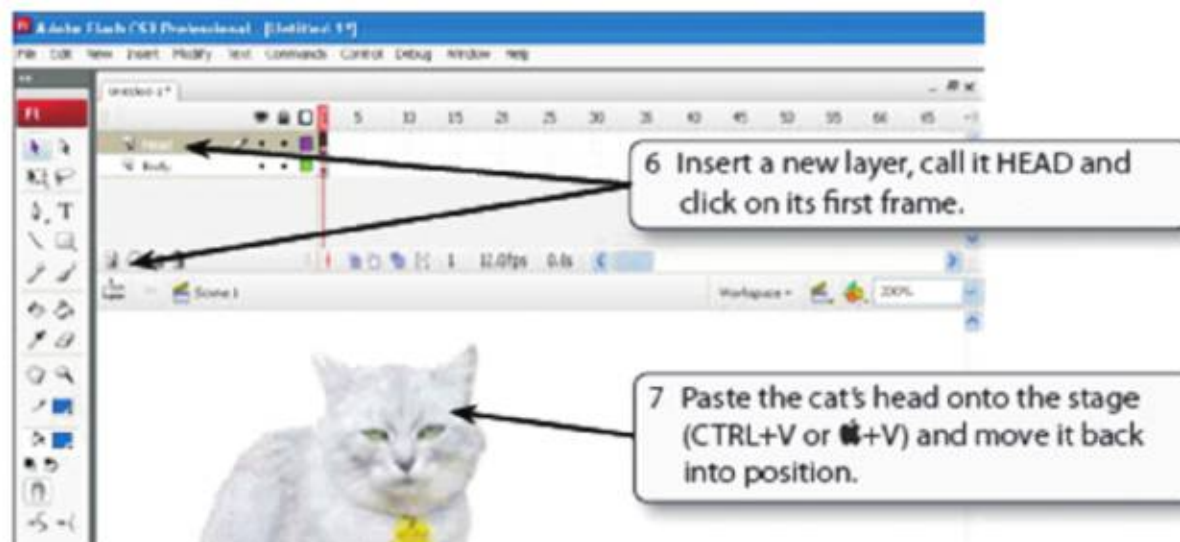
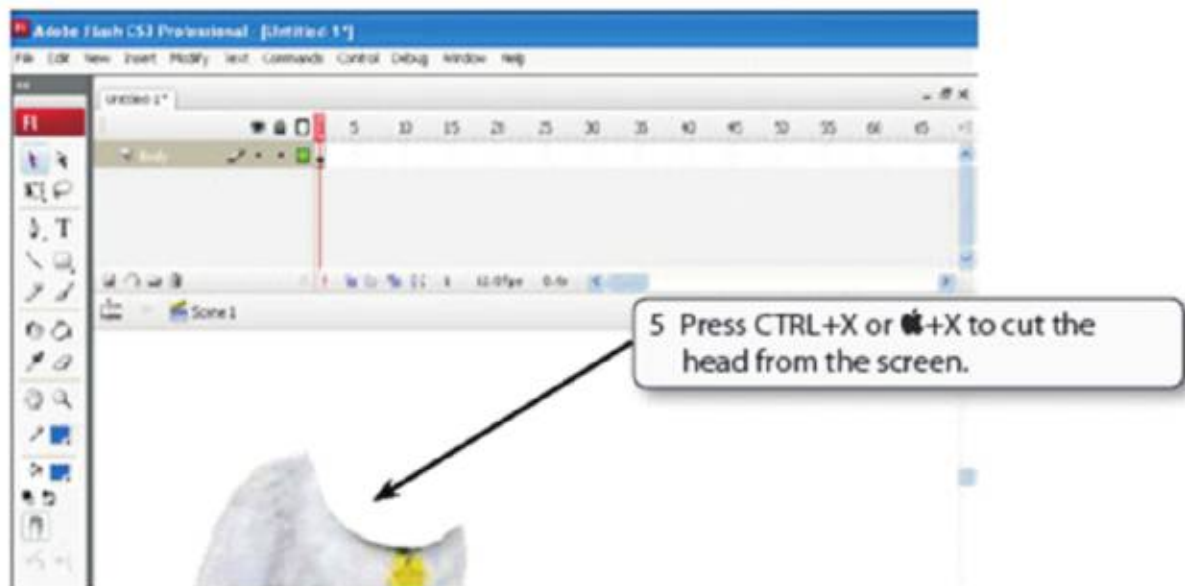
To animate the head it needs to be separated from the body and converted to a symbol.

It is also good practice to place the head in its own layer.





NOTE : If the whole cat's body is selected you need to erase around the cat's head again.

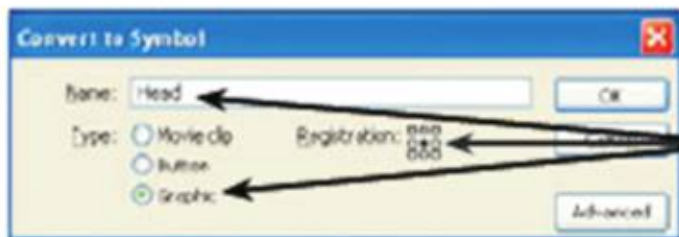


NOTE : The arrow keys on the keyboard can be used to nudge the head into position.

Converting the Head to a Symbol

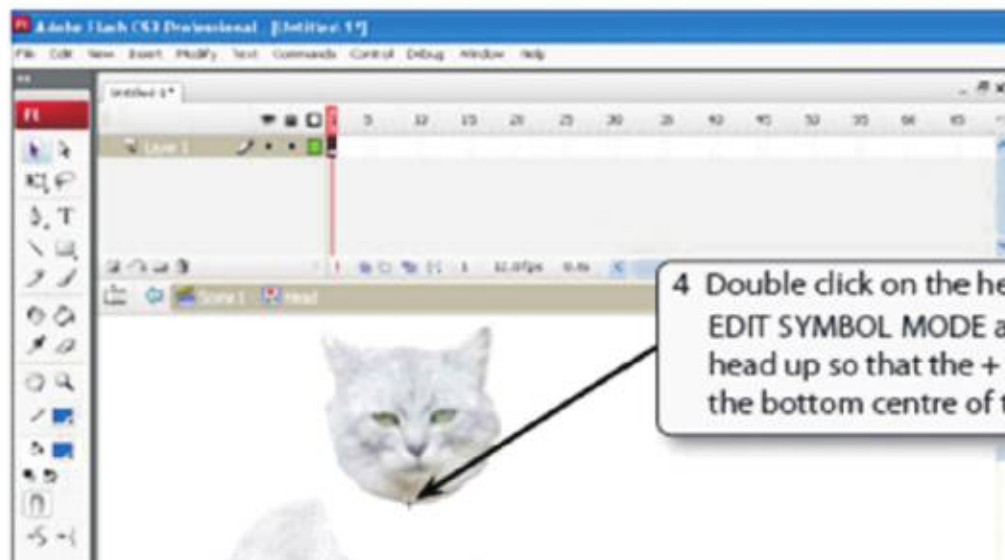
To animate the head using a motion tween it needs to be converted to a symbol and its reference (or rotation) point set to the base of the head.

1. With the head selected press F8 to convert it to a symbol.

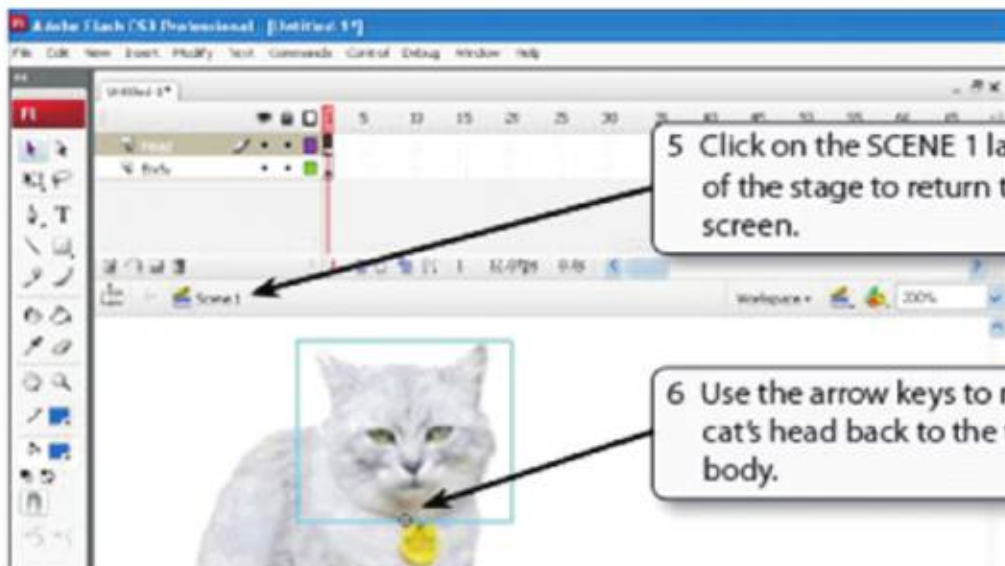


- 2 Call the symbol HEAD, set its TYPE to GRAPHIC, its REGISTRATION to CENTRE.

3. Select OK to create the symbol.



- 4 Double click on the head to open EDIT SYMBOL MODE and move the head up so that the + marker is at the bottom centre of the head.



- 5 Click on the SCENE 1 label at the top of the stage to return to the normal screen.

- 6 Use the arrow keys to nudge the cat's head back to the top of the body.



Work Station



A Tick (3) the correct option.

(NEP) Analytical Skills

- These are symbols that has its own independent timeline.
a. Pictures ☐ b. Clip art ☐ c. Word art ☐ d. Movie clips ☐
- This panel can be used as a zoom of the symbol as you are creating it.
a. Layer ☐ b. Mask ☐ c. Filter ☐ d. Library ☐
- We could have used a MOTION TWEEN or a SHAPE TWEEN as the:
a. Texture ☐ b. Masks ☐ c. Motion ☐ d. Animation ☐
- Some interesting and effective animations can be created using a layer to cover part of another layer.
a. Using layer ☐ b. Using motion ☐
c. Using masks ☐ d. Using clips ☐
- In order to edit the photo it must be broken apart into individual:
a. Pieces ☐ b. Pixels ☐ c. Parts ☐ d. Points ☐

B Fill in the blanks.

- The portion of layer being covered can only be seen though whatever shape is drawn in the mask layer is called _____.
- Movie clips do not have to be _____ animations.
- You can also _____ on the layer name and select MASK from the SHORTCUTS menu.
- You can use the shape tools such as the _____ and _____ to be the mask shape.
- The _____ is indented under the MASK layer to indicated that it is masked by the MASK layer.

C Write T for True or F for False.

- Image clips are an important feature of Flash. ☐
- You can create an animated movie clip and use it many times within other animations without affecting the size of the file. ☐
- Movie clips are created as a pixel and then used as required. ☐
- Life can be added to still photos by using Flash to animate parts of them. ☐
- You can drag a layer to the lower edge of a MASK layer to quickly set it to a MASKED layer. ☐

D Answer the following questions.

1. What is the use of creating an animated movie clip?
2. Write the use of LIBRARY panel.
3. What is ONIONSKIN?
4. What is motion tween?
5. Define Masking.

LAB ACTIVITY



A Flying rocket in the main timeline.

(NEP) Experimental Learning

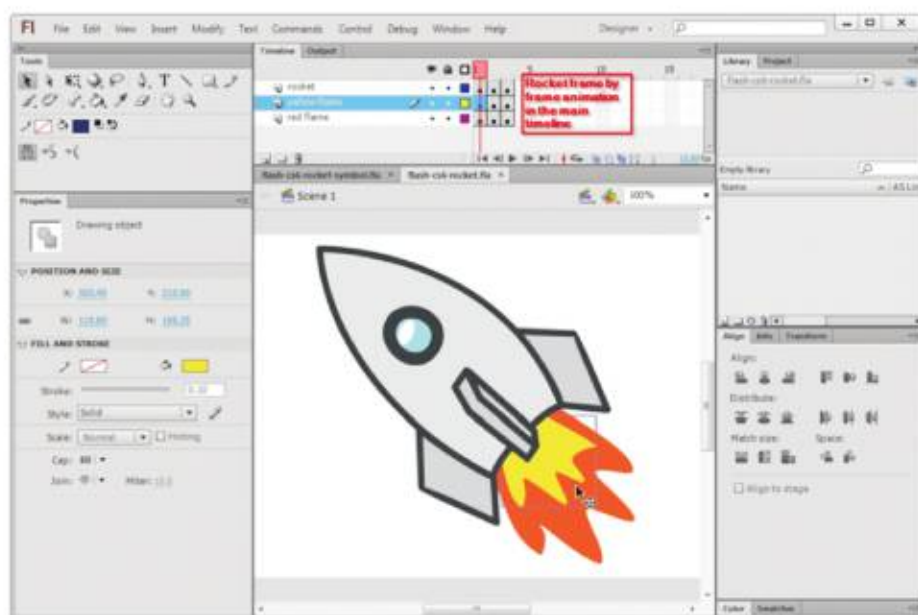
The following screenshot shows a rocket frame by frame animation made in the main timeline.

The animation only uses three frames, i.e. we only implement some minimalistic "flaming" animation.

- We use three layers for drawings, one for the rocket and two for the flames
- Yellow flaming is made with three different drawings in frames 1-2-3 (layer 2)

Red flaming is made with three different drawings in frames 1-2-3 (layer 3)

If you want to add a second, a third rocket and other flying objects you will have to duplicate the three layers. In the six new layers you will have to move the drawings to positions. This is awfully complicated and we will not show an example. Therefore, in most cases, do not create animations in the main timeline.





Tables & Links



Lesson at a Glance

HTML Tables
Cellpadding and Cellspacing
Table Caption

Basic Table Tags
Colspan and Rowspan
HTML Image Links

Table Borders
Table Background



HTML Tables

A table is a structured set of data made up of rows and columns (tabular data). A table allows you to quickly and easily look up values that indicate some kind of connection between types of data, for example a person and their age, or a day of the week or the time table for a local swimming pool.

Using tables to divide the page into different sections is an extremely powerful tool. Almost all major sites on the web are using invisible tables to layout the pages.

Basic Table Tags

Tables are defined with `<table>` tag. A table is divided into rows (with the `<tr>` tag), and each row is divided into data cells (with the `<td>` tag). TD stands for “table data”, and holds the content of a data cell. A `<td>` tag can contain text, links, images, lists forms, other tables etc.

The basic structure of a table consists of the following tags :

1. `<TABLE>` ... `<1 Table>`
2. `<TR>` ... `<1 TR>`
3. `<TD>` ... `<1 TD>`

Now our table has a single row and it looks like below :

```
<table>
```

```
<tr>
```

```
.....
```

```
.....
```


</tr>

<table>

An empty row has been created and now we need to have some table cells so that we can put something in those cells. There are two types of cell tags used in a table.

<th>:Marks a heading cell within a row. It is used to give a proper heading to the columns of the table. This tag is optional and you can create tables without using <th>tag.

<td>:It is a data cell tag and used for the body of the table. Again the closing tags </th> and </td> are optional.

The lines of code illustrate this concept:

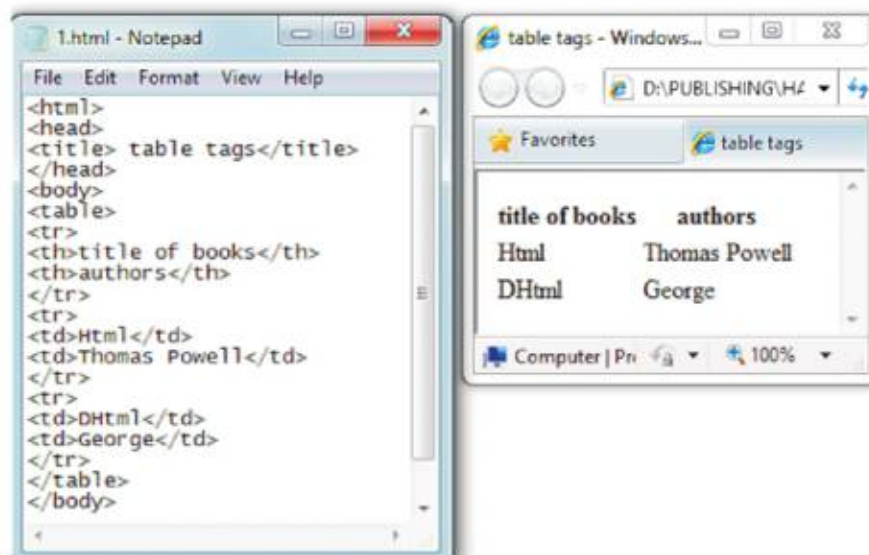
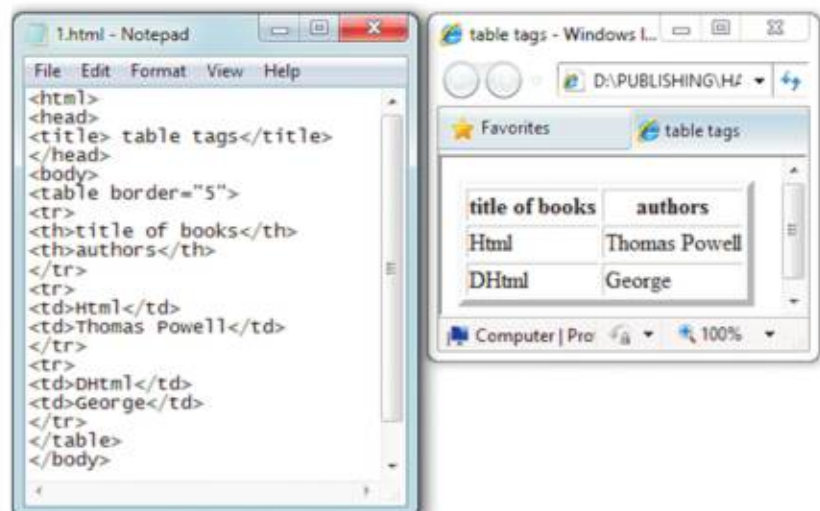
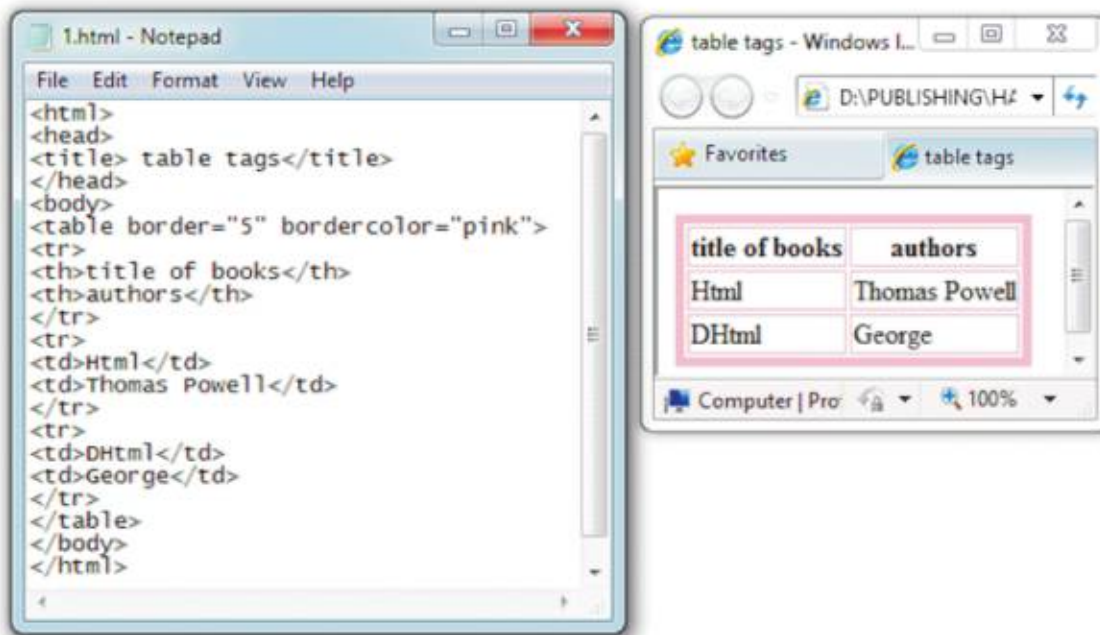


Table Borders

Tables with borders are much easier to read and are more attractive. You can specify a table border by using "border = number" attribute in the opening table tag. The number specifies the width of the border around the table in pixels. The default value is generally zero. The following code shows the use of border attribute in a table.



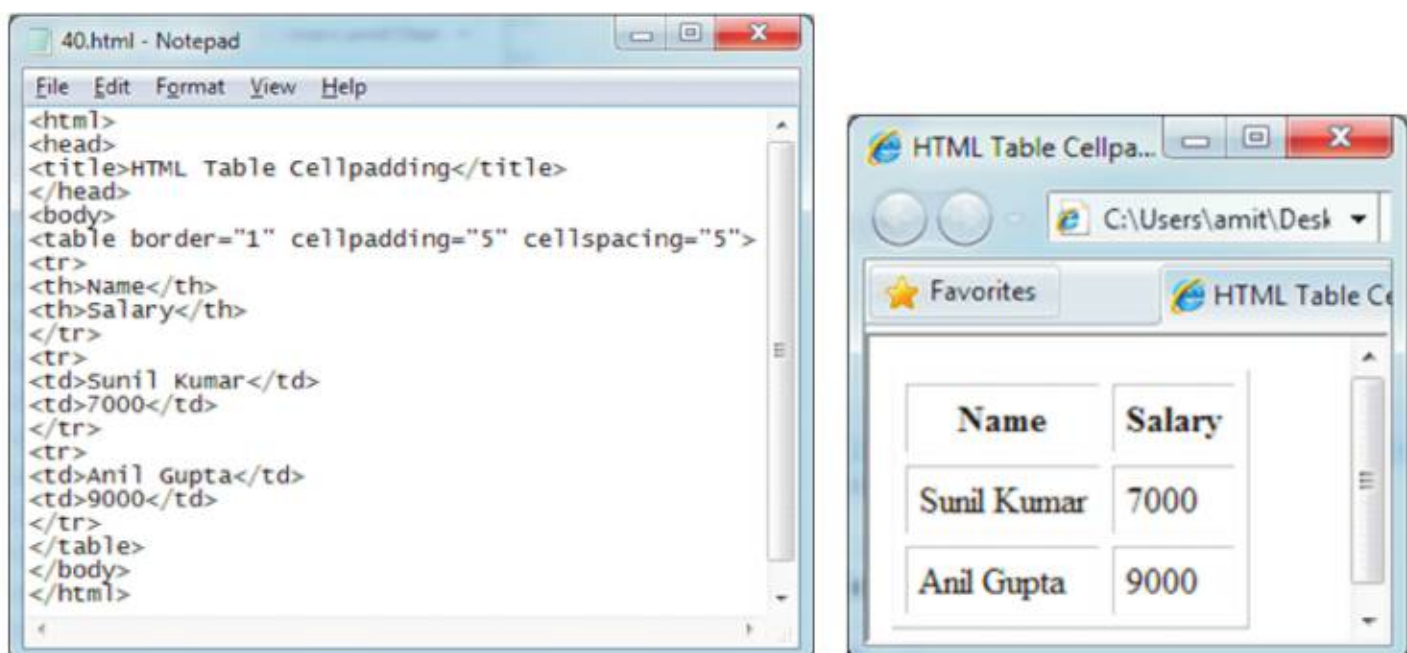
Also the color of the border can be set using the “border color” attributes in <table> tag. The following code shows the use of border color attribute in a table as:



Cellpadding and Cellspacing

There are two attributes called cellpadding and cellspacing which you will use to adjust the white space in your table cells. The cellspacing attribute defines the width of the border, while cellpadding represents the distance between cell borders and the content within a cell.

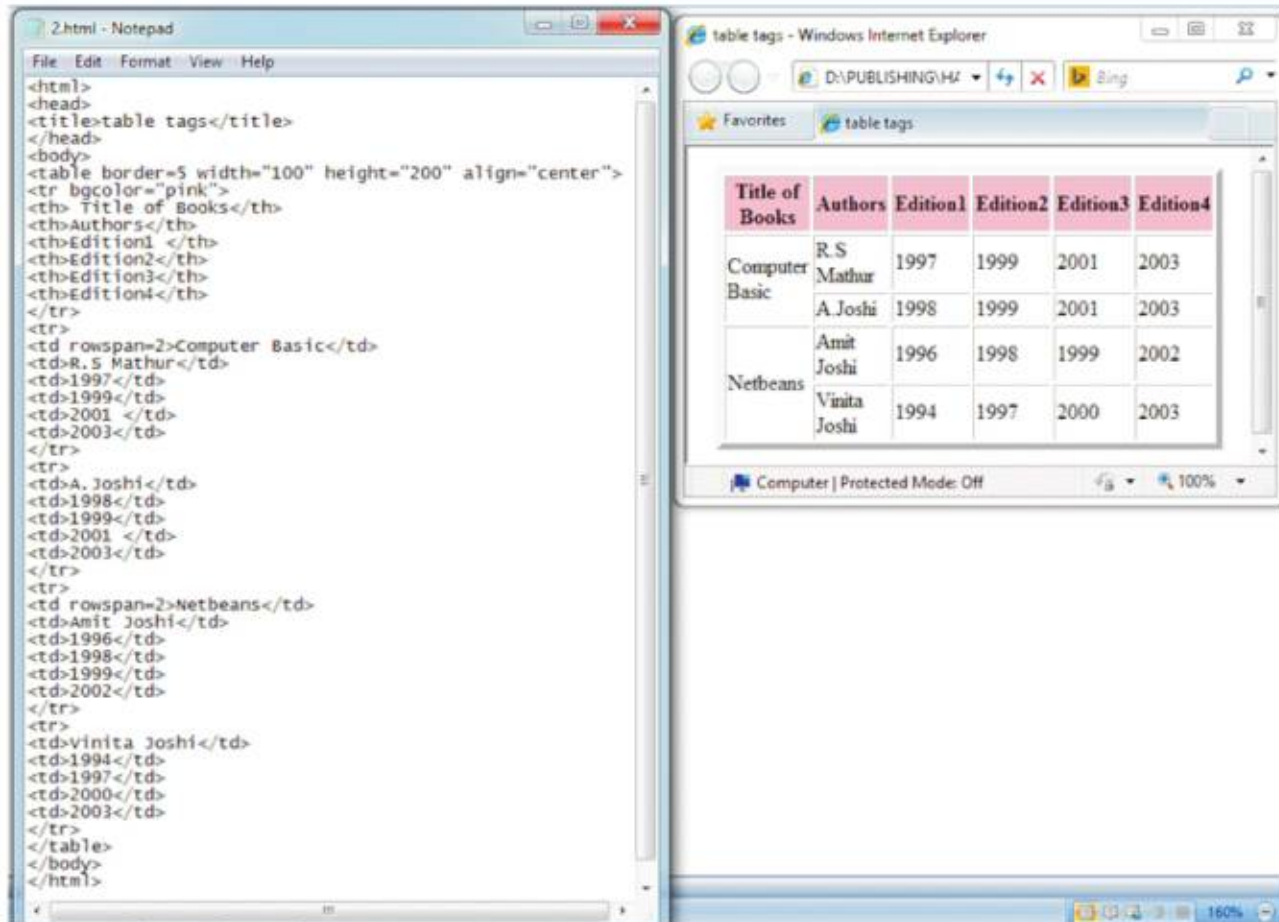
Example :



Colspan and Rowspan Attributes

Two attributes "rowspan" and "colspan" are used to combine adjoining cells into larger cells. These attributes are used in either <th> or <td> tags and accepts a number as value. For rowspan attribute this number indicates the number of rows a cell should take up and for colspan the number of columns the cell should span. In both the cases the present cell is included in counting.

The html code is as shown below :



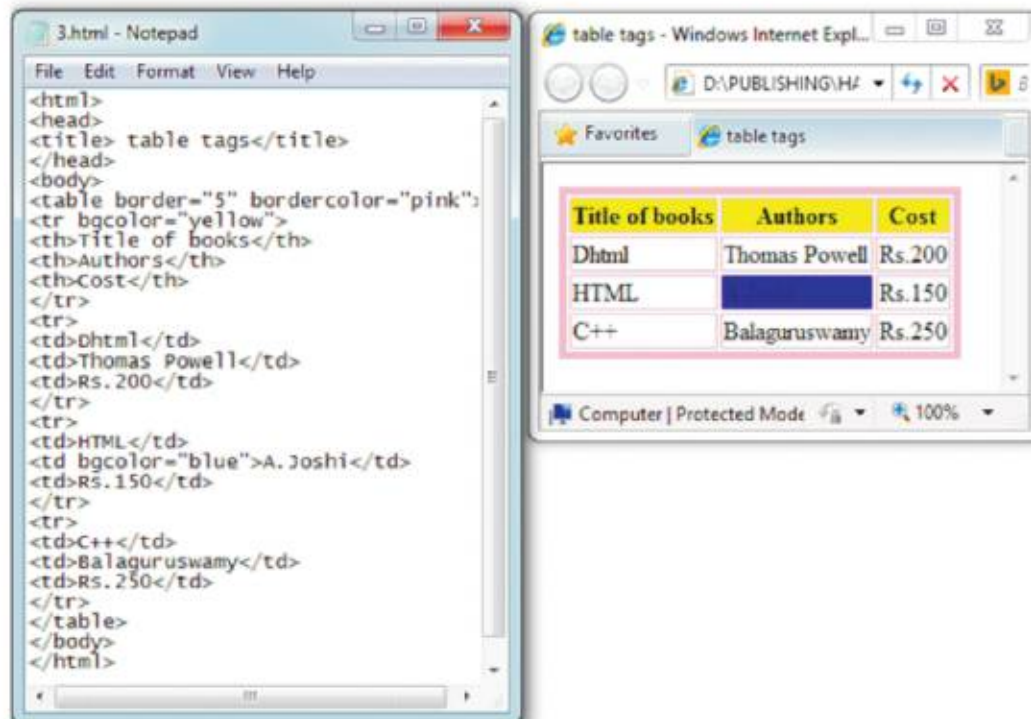
Tables Background

You can even set background color for the whole table, row or for a table cell (data or header). The background color can be set for the whole table by using bgcolor = "color" attribute in the <table> tag.

In order to set different background colors for a particular cell or a row, you can use the bgcolor attribute with <td> and <tr> tags.

The following code sets pink color for the first row and blue color for the cell containing text "award".

Example :

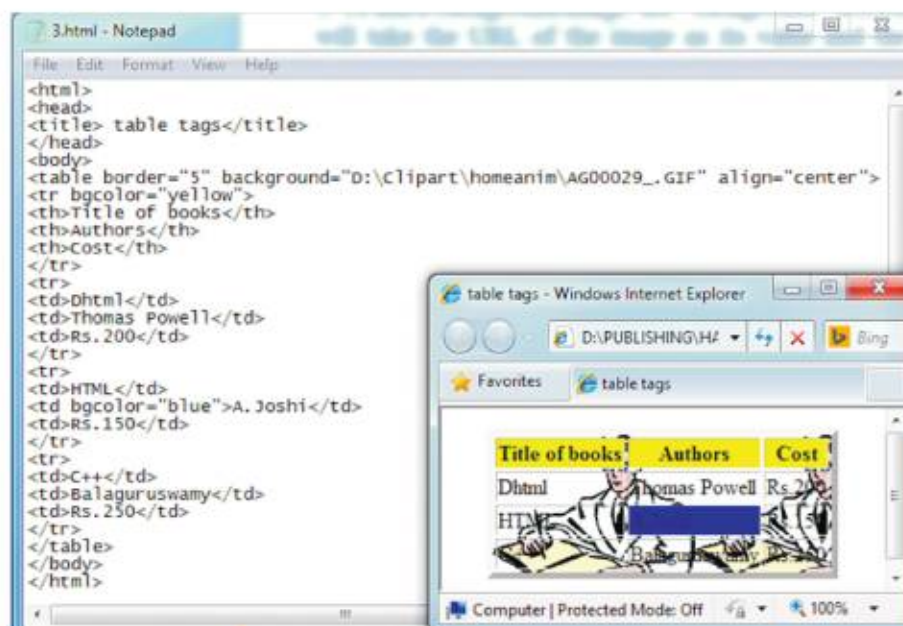


Including images within tables

An image can either be included in a table cell or can act as a background in a table :

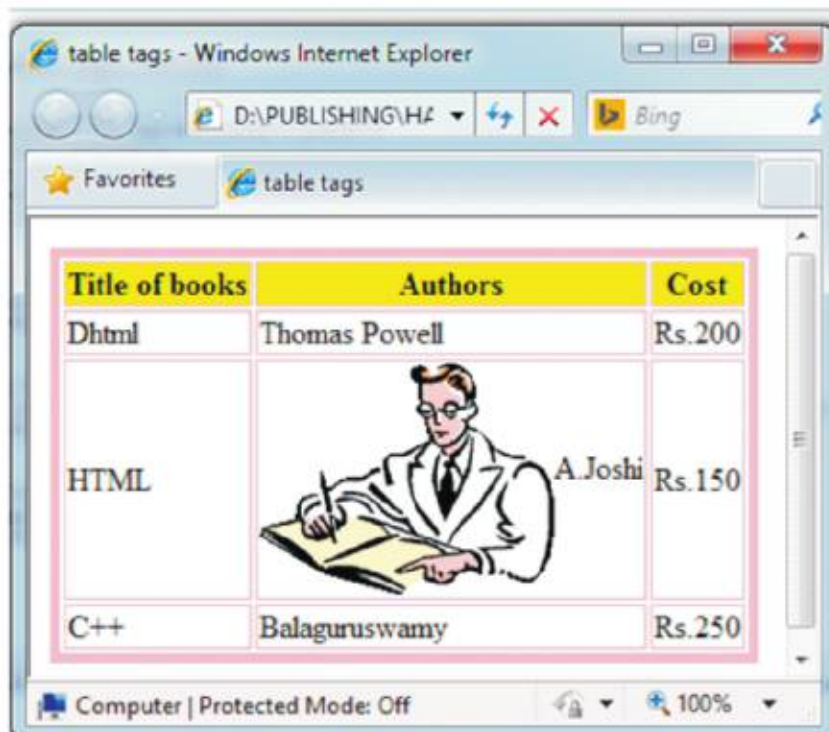
1. To add a background image: use "background" attribute in the opening <table>tag. It will take the URL of the image as its value and the background images are titled throughout the available table space.

Let's include image as a table background :



2. To include images within cells: images can be included within header or data cells. For that add `<img>` attribute between the `<td>` or `<th>` cell tags with proper URL specifying the source of the image. The following code shows the insertion of an image in a specific cell.

```
3.html - Notepad
File Edit Format View Help
<html>
<head>
<title>table tags</title>
</head>
<body>
<table border="5" bordercolor="pink">
<tr bgcolor="yellow">
<th>Title of books</th>
<th>Authors</th>
<th>Cost</th>
</tr>
<tr>
<td>Dhtml</td>
<td>Thomas Powell</td>
<td>Rs.200</td>
</tr>
<tr>
<td>HTML</td>
<td>A.Joshi</td>
<td>Rs.150</td>
</tr>
<tr>
<td>C++</td>
<td>Balaguruswamy</td>
<td>Rs.250</td>
</tr>
</table>
</body>
</html>
```



Specifying cell size and alignment

The width of a cell is specified as number of pixels or as a percentage of table width. This attribute can be used with <th> and <td> tags to adjust the width of cells. The following code will change the width of cells in the first rows as :

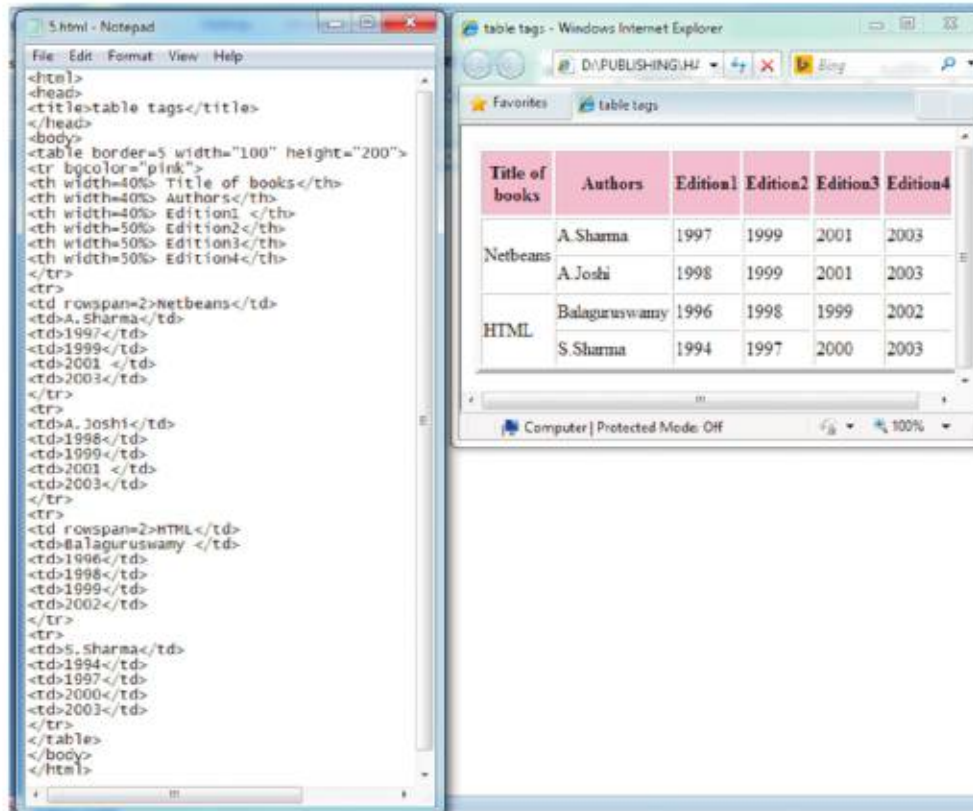
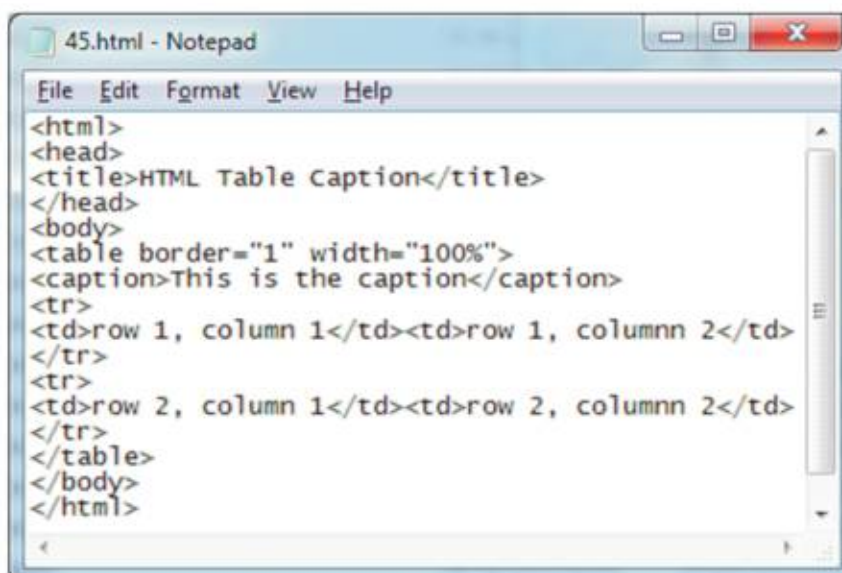


Table Caption

The caption tag will serve as a title or explanation for the table and it shows up at the top of the table.

Example :



This will produce following result :

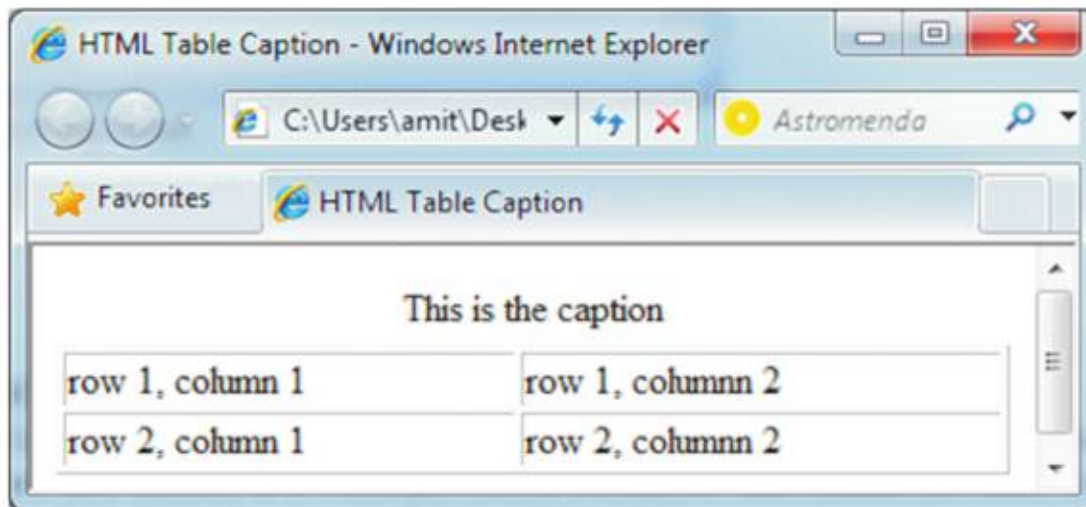


Table header, body, and footer

Tables can be divided into three portions: a header, a body, and a foot. The head and foot are rather similar to headers and footers in a word-processed document that remain the same for every page, while the body is the main content holder of the table.

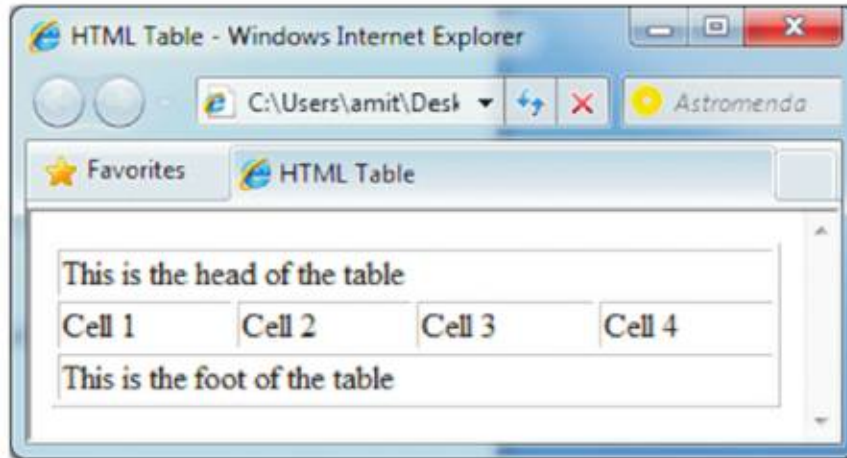
The three elements for separating the head, body, and foot of a table are:

- `<thead>` - to create a separate table header.
- `<tbody>` - to indicate the main body of the table.
- `<tfoot>` - to create a separate table footer.

Example :

```
46.html - Notepad
File Edit Format View Help
<html>
<head>
<title>HTML Table</title>
</head>
<body>
<table border="1" width="100%">
<thead>
<tr>
<td colspan="4">This is the head of the table</td>
</tr>
</thead>
<tfoot>
<tr>
<td colspan="4">This is the foot of the table</td>
</tr>
</tfoot>
<tbody>
<tr>
<td>Cell 1</td>
<td>Cell 2</td>
<td>Cell 3</td>
<td>Cell 4</td>
</tr>
</tbody>
</table>
</body>
</html>
```

This will produce following result :



Nested Tables

You can use one table inside another table. Not only tables you can use almost all the tags inside table data tag <td>.

Example : following is the example of using another table and other tags inside a table cell.

```
47.html - Notepad
File Edit Format View Help
<html>
<head>
<title>HTML Table</title></head>
<body>
<table border="1" width="100%">
<tr><td> <table border="1" width="100%">
<tr><th>Name</th> <th>Salary</th> </tr>
<tr><td>Ramesh Sharma</td> <td>6000</td> </tr>
<tr><td>Sameer Kumar</td> <td>9000</td> </tr>
</table></td></tr>
</table>
</body>
</html>
```

This will produce following result :

